Note: The following is a redacted version of the original report published October 13, 2026 [31 pgs]. All company mentions by GS analysts are illustrative and should not be interpreted as investment recommendations.

Goldman Sachs Research

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TOP *of* MIND

HIGHER BOND YIELDS: HOW CONCERNING?

Global bond yields have risen to multi-decade highs, yet growth and markets have remained remarkably resilient. But with some pressures beginning to emerge (e.g. in European sovereign bond markets), can this resilience last, or will more vulnerabilities be revealed? We ask the IMF's Tobias Adrian, Moody's Analytics' Mark Zandi, and GS economists. Adrian is relatively sanguine about the economy's resilience, though notes downside risks, while GS economists expect only a modest growth drag from higher rates, but a more consequential one if yields remain near current or higher levels. Zandi is generally more concerned, describing the US economy as vulnerable rather than resilient, and sees one-in-three odds of the economy going off the rails. Meanwhile, GS strategists and analysts are not especially concerned about banks or markets more broadly, but warn that further increases in yields or rate volatility could test risk assets. Key to watch: whether the increasingly interconnected and leveraged AI investment boom can continue to withstand higher rates.



INTERVIEWS *with...*

Tobias Adrian, Special Advisor and former Director of Monetary and Capital Markets, IMF

"The global economy has demonstrated considerable resilience in the face of multiple severe shocks, and I would not be surprised if it continues to shrug off higher yields, provided they do not keep rising sharply in a disorderly fashion. At the same time, I do worry about downside risks."

Mark Zandi, Chief Economist, Moody's Analytics

"Many continue to describe the US economy as resilient, which I find fascinating, bordering on frustrating. In my view, the right word is vulnerable... That shift in perspective completely changes how one evaluates the risks ahead."

ALSO *inside...*

Rising rates: key questions answered | William Marshall, GS Rates Strategy Research

A modest US growth drag | Pierfrancesco Mei, GS US Economics Research

Higher rates & DM growth: rules of thumb | Joseph Briggs and Johan Allen, GS Global Economics Research

Higher rates, greater US fiscal pressure | David Mericle and Pierfrancesco Mei, GS US Economics Research

Rates won't derail equities, for now | Ben Snider, GS US Portfolio Strategy Research

Rates risk in credit | Amanda Lynam, GS Credit Strategy Research

When to yield to bonds | Christian Mueller-Glissmann and Andrea Ferrario, GS Asset Allocation Research

Q&A: impact of higher rates on financials | Richard Ramsden and Alexander Blostein, GS Equity Research

Q&A on US Treasury buybacks | Isabella Rosenberg, GS Rates Strategy Research

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The Goldman Sachs Group, Inc.

Macro news and views

We provide a brief snapshot on the most important economies for the global markets

US

Latest GS proprietary datapoints/major changes in views

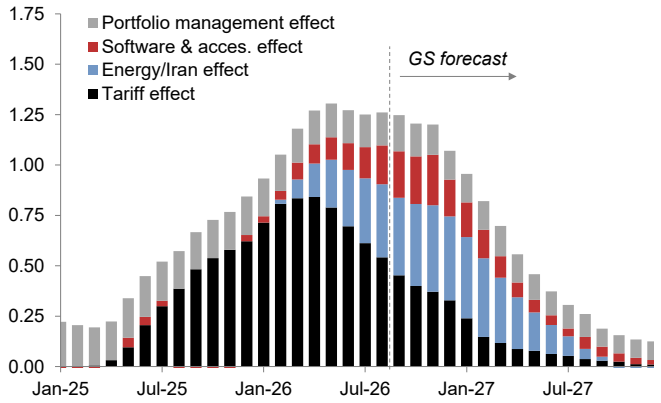
- We lowered our 2026 core PCE inflation forecast to 3.0% (Q4/Q4, from 3.2%) after larger-than-expected downward revisions to yoy core PCE growth, which partly led us to push back our forecast for the next Fed hike to Dec (from Oct).

Datapoints/trends we're focused on

- US economic growth, which has remained resilient this year despite higher rates and higher oil prices, though the economy is bifurcated, with strength among upper-income consumers and AI-exposed sectors.
- US midterm elections, where prediction markets imply relatively high odds of a Democratic sweep as voters remain focused on higher gasoline prices.

A fading US inflation impulse

Effects of temporary factors on yoy core PCE inflation, pp



Source: Bureau of Economic Analysis, Goldman Sachs GIR.

Japan

Latest GS proprietary datapoints/major changes in views

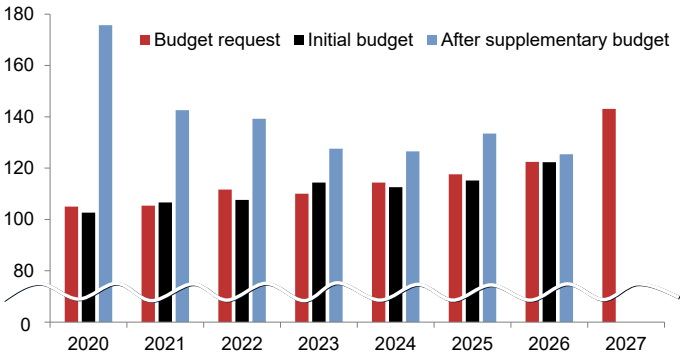
- No major changes in views.

Datapoints/trends we're focused on

- BoJ policy; we continue to expect the next rate hike in January 2027, though the timing could be brought forward in response to heightened inflation risks.
- FY2027 Ministry of Finance budget request, which increased significantly from FY2026 to a record ¥143tn and includes larger-than-usual JGB issuance.
- Japan real wages, which have maintained positive yoy growth since the beginning of the year.
- Japan consumer sentiment, which declined in September.

Japan's ever-increasing budget requests

General account budget requests, initial budgets, and supplementary budgets by fiscal year, ¥tn



Source: Ministry of Finance, Goldman Sachs GIR.

Europe

Latest GS proprietary datapoints/major changes in views

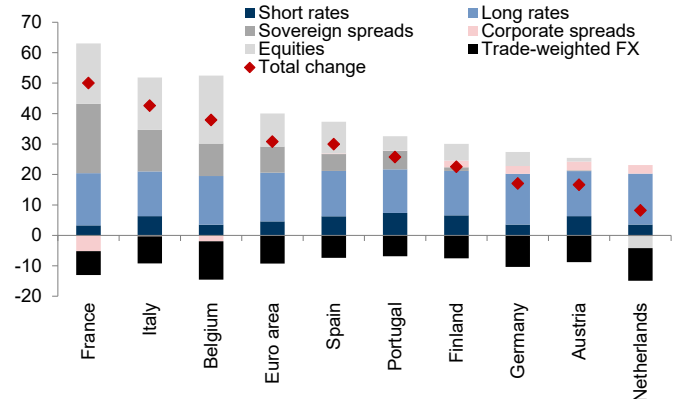
- No major changes in views.

Datapoints/trends we're focused on

- Positive growth spillovers from the global AI capex boom, which we think should continue into 2027.
- UK growth, which we expect to slow in the next two quarters.
- French elections; frontrunner Le Pen's proposal to address fiscal challenges is likely to face implementation challenges.
- Italy's 2027/2028 fiscal deficit, which was revised up significantly, weakening the country's debt outlook.
- Rising sovereign stress, which has tightened EA financial conditions, reducing the urgency for the ECB to hike rates beyond Dec, when we expect the last hike.

Tighter EA financial conditions, limited ECB hikes

Change in FCI since July 3, 2026, bp



Source: Goldman Sachs GIR.

Emerging Markets (EM)

Latest GS proprietary datapoints/major changes in views

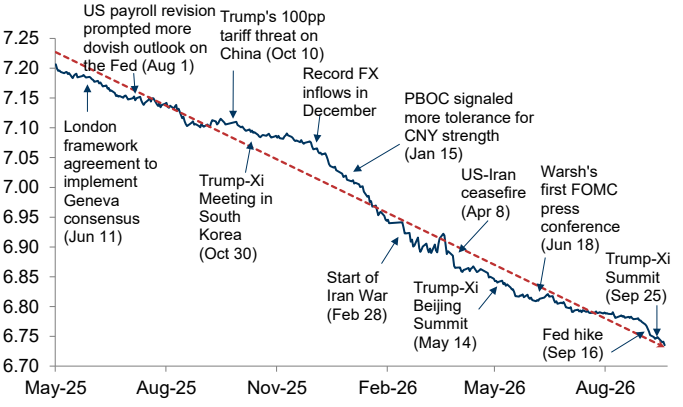
- No major changes in views.

Datapoints/trends we're focused on

- China's export boom, which we find still has room to run over the next few years but will eventually require stronger economic growth in importing countries.
- China FX policy; we see risk that RMB appreciation overshoots the gradual pace preferred by the PBOC, even as policymakers continue to lean against market expectations.
- Asia's AI dividend, which is broadening beyond exports.
- Brazilian presidential elections, which will be settled in a second-round runoff on October 25, with a market-friendly election outcome likely to lead to further BRL appreciation.

USD/CNY on the decline

USD/CNY fixing



Source: Bloomberg, Goldman Sachs GIR.

Higher bond yields: how concerning?

Recent acceleration in the rise in global bond yields to multi-decade highs has begun to raise questions about the implications for growth and markets. Both have remained remarkably resilient so far, with US growth in particular holding relatively firm and the S&P 500 and other equity markets near new all-time highs. But with pressure beginning to emerge—most notably in European sovereign bond markets—and yields looking unlikely to retreat soon, can that resilience last, or will more vulnerabilities be revealed?

GS Head of US Rates Strategy William Marshall first lays out how we arrived at higher rates, underscoring the striking breadth of the rise, which he attributes mainly to fundamentals—strong growth, inflation risks, hawkish policy shifts, the ongoing investment cycle, and long-run fiscal risks—but notes that technical factors likely amplified the recent move. And while he ultimately expects yields to fall (with YE 10-year US Treasury yields of 4.75%/4.4% in 2026/2027, respectively) as markets become more confident in easing inflation, he thinks elevated long-term yields should have some staying power, and sees continued upside risk from energy price and fiscal pressures.

So, just how much of a threat do higher rates pose to the macro outlook? For answers, we first turn to Tobias Adrian, Special Advisor to the Managing Director and former Financial Counsellor and Director of the Monetary and Capital Markets Department at the IMF, Mark Zandi, Chief Economist of Moody's Analytics, and GS US economists.

Adrian sees a global economy that has so far responded to the yield shock much as it has to the many other major shocks of the past seven years: resiliently. He notes that the rise in yields has not significantly tightened financial conditions or, more importantly, disrupted market functioning or forced damaging deleveraging. He believes that resilience could persist as long as the move in yields doesn't accelerate or become more disorderly. But he is concerned about vulnerabilities in the AI and technology complex, sovereign bond markets, and EMs—not to mention cyber risks—all of which leaves him "vigilant but cautiously optimistic."

GS US economists also characterize the US economy as resilient. Their expectation that the Fed will deliver just one more rate hike this year before cutting next year—consistent with Marshall's expectation of lower Treasury yields—leads Pierfrancesco Mei to expect only a modest 0.2pp growth drag in 2027 from higher rates. But he finds that the drag could rise to a more consequential 0.5pp if yields remain near current levels (see pg. 9 for GS global economists Joseph Briggs' and Johan Allen's rules of thumb for the impact of higher yields on rate-sensitive areas across major DMs.) David Mericle and Mei also worry about the implications for the already-concerning US fiscal trajectory, which they believe is unlikely to trigger a classic debt crisis but is too risky to ignore.

Zandi, for his part, is generally more concerned. While his baseline outlook is relatively sanguine, he describes the US economy as vulnerable rather than resilient and views the "distribution of possible outcomes as very flat and heavily skewed toward darker tail scenarios." He sees the clearest softness in the labor market, with weak hiring and falling real wage growth for many workers weighing on consumer spending. And he takes little comfort from US growth continuing to run around 2%—roughly its potential rate—noting that AI accounts for roughly one-third of that growth, leaving the non-AI economy growing at only around 1-1.5%, which he calls "a pretty punk economy." Zandi also worries that the US fiscal situation is "as bad as I have ever seen it" and that higher rates are already inflicting damage on real estate markets and squeezing existing debtors. As these stresses mount, he sees elevated risk of something breaking in the financial system. All told, he puts the odds of the current vulnerabilities pushing the economy completely off the rails at around one-in-three.

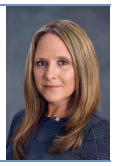
With banks historically at the center of the rate shocks and economic crises that Zandi is worried about, we then explore how much risk banks pose today. Adrian and GS Head of the Financials Equity Group Richard Ramsden both see little cause for alarm, explaining that higher rates should boost banks' net interest income and revenues given their asset-sensitive balance sheets. Ramsden also doesn't expect a significant deterioration in credit quality absent a broader economic downturn and says that most banks should have excess capital even after accounting for rate-driven portfolio losses.

But even if banks can weather higher rates, what about markets more broadly? GS strategists and analysts are not especially concerned, with Chief US Equity Strategist Ben Snider arguing that current rate levels shouldn't prevent continued strong US earnings growth and the S&P 500 from climbing higher, Chief Credit Strategist Amanda Lynam viewing the headwind to corporate credit from higher rates as manageable amid the solid growth backdrop, and US Financials Analyst Alexander Blostein expecting the recent slowdown in private market activity to rebound once rates stabilize. GS multi-asset strategists Christian Mueller-Glissmann and Andrea Ferrario add that higher yields have strengthened the strategic case for bonds in a multi-asset portfolio, though strong earnings continue to favor equities for now. But they all warn that if rate volatility remains elevated, and especially if yields continue to rise from here, risk assets could face mounting pressures.

All that said, with the AI boom powering the economy and markets, everything may come down to whether higher rates will derail it. Snider views this as unlikely given the strong balance sheets of the firms investing in AI and the expected profitability of these investments, with Lynam also seeing a high bar to derail AI-related debt issuance. Adrian also maintains a positive baseline for the AI economy but is concerned about the circularity of investment, increased leverage, and the sustainability of revenue growth in the AI complex. And Zandi again seems the most worried, warning that "AI could become the part of the economy that breaks" if rates remain at current levels for much longer.

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Interview with Mark Zandi

Mark Zandi is Chief Economist of Moody's Analytics. Below, he argues that the US economy is vulnerable rather than resilient and worries that high interest rates could deepen existing pockets of weakness and ultimately cause something to break in the financial system.

The views stated herein are those of the interviewee and do not necessarily reflect those of Goldman Sachs.



Allison Nathan: Prior to the September 2026 Fed rate hike, you argued that a hike would do little to address supply-driven inflation but could damage already weak parts of the US economy. Where are you seeing the most softness?

Mark Zandi: The clearest softness is in the labor market. Monthly payroll growth has averaged around 40,000 over the past year and has been essentially zero when excluding the healthcare sector. Hiring is moribund, hours worked are very low, and while the unemployment rate remains low, that is partly a reflection of declining labor force participation. Adjusting for that cyclical drop would put underlying unemployment in the mid-4% range—above the level consistent with full employment—and I expect the headline rate to move toward that level over the next 6-18 months. Real wage growth is, at best, flat, and for many Americans is actually declining, which is eroding purchasing power, particularly for lower- and middle-income households, who are increasingly drawing down savings to sustain spending. So, while the economy is still growing, the underlying labor market and household fundamentals point to very tenuous growth.

> "While the economy is still growing, the underlying labor market and household fundamentals point to very tenuous growth."

Allison Nathan: What is causing this softness?

Mark Zandi: Policy choices are at the root of the weakness. Tariffs, while more consequential last year, continue to weigh on growth and boost inflation. Heavy-handed immigration policy is also an underappreciated source of weakness. Many firms losing immigrant workers cannot afford to raise wages enough to attract native-born workers, so they are raising prices to align demand with the supply they can produce with a reduced labor force. But the most immediate and significant driver of the softness is the Iran war, which has pushed up energy prices, further fueling inflation and weighing on growth.

Allison Nathan: But US real GDP growth remains near 2%—roughly the economy's potential rate. Isn't that a relatively healthy growth pace?

Mark Zandi: Real GDP growth is indeed running at 2% yoy, but I estimate that AI-related activity accounts for roughly one-third of that, or around 0.6–0.7pp. Roughly half of that comes from AI capital spending, net of electronics imports for data centers, while the balance reflects wealth effects from the surge in AI-related stock prices that have contributed to an almost doubling in the S&P 500 since ChatGPT debuted four years ago, boosting net worth and consumer spending, especially among high-income households. So, the non-AI economy is growing at only around 1%–1.5%. That's a pretty punk economy, and a difficult place to be.

Allison Nathan: The September SEP indicated that most FOMC members expect one more rate hike this year, and the market is expecting the same. How damaging could further tightening be?

Mark Zandi: While I expect another hike this year, with the economy already barely growing at potential, additional hikes could push growth below potential. That would further strain the already weak non-AI economy and trigger layoffs, significantly raising the likelihood of a self-reinforcing negative spiral. So, further tightening would be a risky endeavor.

> "With the economy already barely growing at potential, additional hikes could push growth below potential. That would further strain the already weak non-AI economy and trigger layoffs, significantly raising the likelihood of a self-reinforcing negative spiral."

Higher rates are already inflicting damage on the economy's most rate-sensitive sectors, particularly housing and commercial real estate. Refinancing is in a deep freeze. Existing home sales have fallen to levels last seen during the teeth of the housing crash and the Global Financial Crisis. Home prices are moving sideways on average and outright declining in half the country. That could easily create broader problems for households, especially for those that bought homes in the last years after rates began to rise and have virtually no home equity. And while commercial real estate had begun to show signs of life, renewed rate increases are once again freezing activity and putting pressure on property values.

Higher rates are also squeezing existing debtors as borrowing costs on credit cards, home loans, auto loans, and corporate debt rise. I am particularly worried about highly leveraged, often private equity-backed companies that have managed the higher rate environment by extending maturities and limiting near-term interest payments. That strategy cannot continue indefinitely if borrowing costs remain elevated or continue to rise.

The equity market is another key area of concern. AI continues to propel the equity market, but it is not impervious to higher rates. Persistently elevated rates will eventually weigh on valuations, increasing the risk of a major correction that could quickly reverberate through the real economy as positive wealth effects reverse and consumer spending weakens. AI capex is also vulnerable as a higher cost of capital makes investment

more expensive and falling valuations make it harder to raise the capital necessary to sustain the current pace of the AI buildout. For now, AI is operating under its own momentum, supported by considerable optimism and euphoria. But if rates remain at these levels or higher for longer than a quarter or two, AI could become the part of the economy that breaks.

Last but not least, higher rates will worsen the US fiscal situation, which is itself contributing to elevated yields today. While we have navigated this fiscal risk for some time, the fiscal situation is as bad as I have ever seen it. The deficit exceeds 6% of GDP, the primary deficit is above 3%, and debt is roughly 100% of GDP and on track to reach 125% in a decade. Annual federal interest payments are already well above $1tn and will rise rapidly if rates remain elevated. All told, I see a roughly one-in-five chance of a fiscal cliff event over the coming year—unlikely, but large enough to warrant attention.

So, higher rates will weigh on the economy through several channels. Historically, rate increases of this magnitude have almost invariably caused something to break in the financial system. We cannot know exactly where the fault line will emerge, but the risk of these pockets of weakness becoming a broader problem is significant.

> "I view the distribution of possible outcomes as very flat and heavily skewed toward darker tail scenarios."

Allison Nathan: How do these risks translate into your forecasts for economic growth?

Mark Zandi: My baseline forecast is relatively sanguine based on the assumption that the Iran war begins to wind down, allowing oil prices to decline to around $80/bbl as we move into next year. That should ease inflation, reduce Fed tightening concerns, and bring the 10-year Treasury yield back toward 4.5% by early next year—a level the economy can digest.

Importantly, though, I view the distribution of possible outcomes as very flat and heavily skewed toward darker tail scenarios. The war may not fade into the background, keeping energy prices elevated and raising the risk of further inflation pressure. That could prompt additional Fed tightening, push yields even higher, and ultimately cause something to break in the financial system.

Meanwhile, the Treasury market has grown more fragile as traditional, price-insensitive buyers—including the Fed, banks, and foreign institutional investors—have stepped back, increasing the market's reliance on highly leveraged, price-sensitive hedge funds that could rush for the exits under stress. A debt-limit confrontation under a likely divided US government next year could provide the catalyst, particularly if lawmakers underestimate the consequences of a breach. In that scenario, an even sharper bond market selloff than we are currently experiencing could occur. While none of this is my baseline—and I remain inherently optimistic because avoiding the darkest scenarios is largely a policy choice—I put the odds of a severe bond market selloff at around one-in-four and of the economy going completely off the rails for any of the reasons we have discussed at around one-in-three. So, the downside risks are significant.

> "I put the odds of a severe bond market selloff at around one-in-four and of the economy going completely off the rails for any of the reasons we have discussed at around one-in-three. So, the downside risks are significant."

Allison Nathan: The Fed has considerable room to cut rates. Should that provide any comfort that the worst-case scenarios may be avoided?

Mark Zandi: That remains an open question. If inflation fades and allows the Fed to take its foot off the brake, rate cuts should bring yields down across the curve—though not back to pre-war levels. But in a darker scenario involving a debt limit crisis and even higher yields, short-term rate cuts would have little effect on the long end. The Fed would need to turn to alternative policy tools, including quantitative easing and bond purchases. That would amount to financial repression—a strategy similar to the one employed by the BoJ for many years—which could prove counterproductive by reducing the pressure on lawmakers to address the underlying problems while fueling even more persistent inflation.

Allison Nathan: Why do you think others are more sanguine about the risks that higher interest rates pose to the US economy?

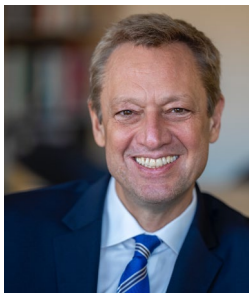
Mark Zandi: Some likely see greater underlying economic strength and expect AI to drive much stronger growth. Others may believe that the neutral rate is higher, perhaps for the same reasons, while some may be more concerned about persistently higher inflation.

But the more fundamental difference is the broader lens through which I currently view the economy. Many continue to describe the US economy as resilient, which I find fascinating, bordering on frustrating. In my view, the right word is vulnerable. Again, growth is pedestrian, the labor market is soft, which was further underscored by the much weaker-than-expected September jobs report, real wage growth is decelerating, inflation remains elevated, and interest rates are rising. That shift in perspective completely changes how one evaluates the risks ahead.

Interview with Tobias Adrian

Tobias Adrian is Special Advisor to the Managing Director and former Financial Counsellor and Director of the Monetary and Capital Markets Department at the IMF. Below, he argues that the global economy has been resilient despite higher bond yields and could very well remain so. But he sees several areas of concern in the economy and markets.

The views stated herein are those of the interviewee and do not necessarily reflect those of Goldman Sachs, the IMF, IMF Management, or the IMF Executive Board.



Jenny Grimberg: The IMF has cautioned that sharp increases in global bond yields could create strains across economies and markets. How would you assess the global economy's response to the recent rise in yields?

Tobias Adrian: The global economy has so far responded to the rise in yields much the same way as it has to the shocks of the past seven years: resiliently. If somebody had told you in late 2019 that the global economy would face a pandemic, multiple wars, deep geopolitical fragmentation, and trade and energy supply shocks, you would have been hard-pressed not to expect a global recession. The pandemic did, of course, trigger a contraction, but it was exceptionally short-lived despite the severe supply and demand shocks. And the global monetary tightening cycle that began in 2021 and accelerated sharply in 2022 in response to the pandemic-related inflation surge did not trigger a recession at all, which is unusual by historical standards. So, the global economy has proven deeply resilient.

Jenny Grimberg: What has that resilience owed to?

Tobias Adrian: Three factors explain this resilience: policy, the adaptability of the economy, and some luck related to technological innovation. Over the past seven years, fiscal and monetary policymakers acted aggressively to stabilize economic activity. Fiscal policy provided substantial support during the pandemic, while central banks cut interest rates, expanded asset purchases, and established emergency standing facilities. That fiscal support has of course come at a significant long-term cost as debt-to-GDP ratios in advanced economies have surged since 2019, and deficits in many countries remain large, contributing to the rise in long-term yields. But policy has undoubtedly been critical to the economy's resilience to date.

The economy itself has also become more adaptable. A dividend of the pandemic is that firms have developed a greater ability to rewire global value chains quickly in response to a range of disruptions, from shifts in energy and food prices to geopolitical developments, which has helped the global economy more effectively absorb shocks.

And luck has played a role in that the more recent challenges have coincided with the emergence and rapid adoption of generative AI technology, which has helped fuel expectations of a broad productivity boom and created massive demand for investment that is stimulating aggregate demand. If large language models (LLMs) had not come to the fore in November 2022, global growth in the more recent period certainly would not have been as strong.

Jenny Grimberg: As bond yields continue to climb, are you concerned that this resilience could falter?

Tobias Adrian: So far, the rise in bond yields has not triggered a substantial, broad-based tightening in financial conditions or, more importantly, disruptions to market functioning or forced deleveraging that could damage markets and the economy. But I see three main areas of concern: the AI and technology complex, sovereign bond markets, and emerging markets (EMs).

Three vulnerabilities stand out in the AI and technology complex. First, the interconnectivity and circularity of investment among many AI firms raises the risk that one firm's earnings hit could fairly quickly spread across the broader complex. Second, many AI firms have taken on significant leverage, and investor leverage has also grown through the use of leveraged ETFs and derivatives. Leverage upon leverage can be problematic when adverse shocks hit. Third, while relatively subdued valuations suggest that the market is not in a 1999 state of indiscriminate price appreciation, with investors continuing to differentiate across firms and sectors, the strength of the complex is predicated on strong earnings and high revenue growth, and it is not clear how long that will last. While my baseline outlook is positive, greater competition from open-source models, for example, could eat into AI firms' profitability and pressure the broader value chain.

Sovereign debt markets are another key concern. As I mentioned, global public debt is approaching 100% of GDP, and debt levels in major advanced economies exceed 120% of GDP. Those are staggering figures from a historical perspective. The investor base for sovereign bonds has also increasingly shifted to more leveraged investors who are more sensitive to macro developments. A key indicator to watch is auction tails—the difference between the yield at which a government sells bonds at auction and the yield the market expected beforehand. Looking at 30-year bond auctions across G4 sovereign bond markets, the tails have grown over time, suggesting increased market vulnerability to supply. 5-to-10-year forward term premia have risen to pre-Global Financial Crisis (GFC) levels. So, the quantitative easing period has now been fully priced out of the long end of the global yield curve. And I see no good reason for term premia to decline in the foreseeable future. If anything, both expected short rates and the term premium will probably remain somewhat elevated.

Finally, while EM economies have shown tremendous resilience through the shocks of the past seven years owing to stronger policy frameworks, resilient FX reserve buffers, and the development of local-currency bond markets that have reduced currency mismatches, vulnerabilities remain. Particularly striking is the K-shaped pattern of capital flows. Capital has flowed into EM sovereign bonds, while equity

capital has flowed out on net as domestic investors direct their savings into global—and, in particular, US—technology stocks. So, even though EM equities have rallied over the past two years, capital flows have moved in the other direction. That leaves these countries' external positions more leveraged, with less equity and more debt on net.

Jenny Grimberg: Higher yields played a key role in the 2023 regional banking crisis. Are you worried about banks today?

Tobias Adrian: Not particularly. Banks generally do well when interest rates are elevated, because financial intermediation is fundamentally about spreads. When rates rise, asset yields often reprice faster than deposit costs, supporting bank net revenues. That partly explained the banking system's overall resilience in 2023, even as a few outlier banks ran into trouble. Significant capital and liquidity buffers broadly absorbed mark-to-market losses on fixed-income securities, while higher spreads supported profitability.

Jenny Grimberg: What about residential and commercial real estate, where stresses were already visible?

Tobias Adrian: Real estate markets are faring differently across countries. London's housing market, for example, has cooled meaningfully, while other major financial centers like New York and Hong Kong continue to see strong home price appreciation. Real estate valuations are sensitive to long-term yields, and the pace of price appreciation will probably slow if yields remain at elevated levels. But with economic activity remaining relatively strong, immediate severe distress in either commercial or residential real estate seems unlikely.

Jenny Grimberg: When would you begin to worry that higher yields are creating a financial stability problem?

Tobias Adrian: The speed and magnitude of the yield move matter enormously. So far, even though bond yields have risen sharply, market functioning has remained orderly and no forced deleveraging appears to be occurring, as we discussed. I would be more concerned if the move in yields accelerated or became nonlinear, which could set off a damaging deleveraging spiral in which investors are forced to reduce leverage, adding to selling pressures and intensifying market stress. The question then becomes: who will catch the falling knife?

And leveraged investors may not be the only source of selling pressures. Before central banks launched massive asset purchase programs to stabilize markets in March 2020, open-ended funds were the sellers, while some hedge funds actually played a stabilizing role. Sharp selling pressure can even come from plain-vanilla mutual funds or ETFs. That said, it is reassuring that the major banks are well capitalized today, because it would be particularly worrisome if stress in the non-bank sector coincided with forced asset sales in the banking sector. We saw that dynamic during the GFC and the Long-Term Capital Management crisis.

Jenny Grimberg: What should investors watch for warning signs of systemic stress?

Tobias Adrian: I would watch repo market and money market functioning, because the early warning signs of stress typically show up in short-term funding markets. I would also look at a range of market-based indicators, including cross-currency and domestic basis measures, swap spreads, and convenience yields. I would also watch for a broader tightening of financial conditions, which can indicate that pressure is spreading beyond a particular market. And I would keep a close eye on implied volatility as a market-based assessment of risk that feeds directly into investors' risk-management models. When implied volatility rises sharply, investors may need to cut positions, which can reinforce the underlying market move.

Jenny Grimberg: If vulnerabilities were to become more acute, how well prepared are policymakers to respond?

Tobias Adrian: Central banks retain a powerful ability to inject liquidity through repo operations, outright asset purchases, and emergency standing facilities, and in recent years they have repeatedly demonstrated that they can act quickly and decisively when market functioning deteriorates. That is distinct from the broader stance of monetary policy. Ongoing inflationary pressures mean that monetary policy remains on a tightening bias in many countries. But central banks can hike rates and inject liquidity at the same time.

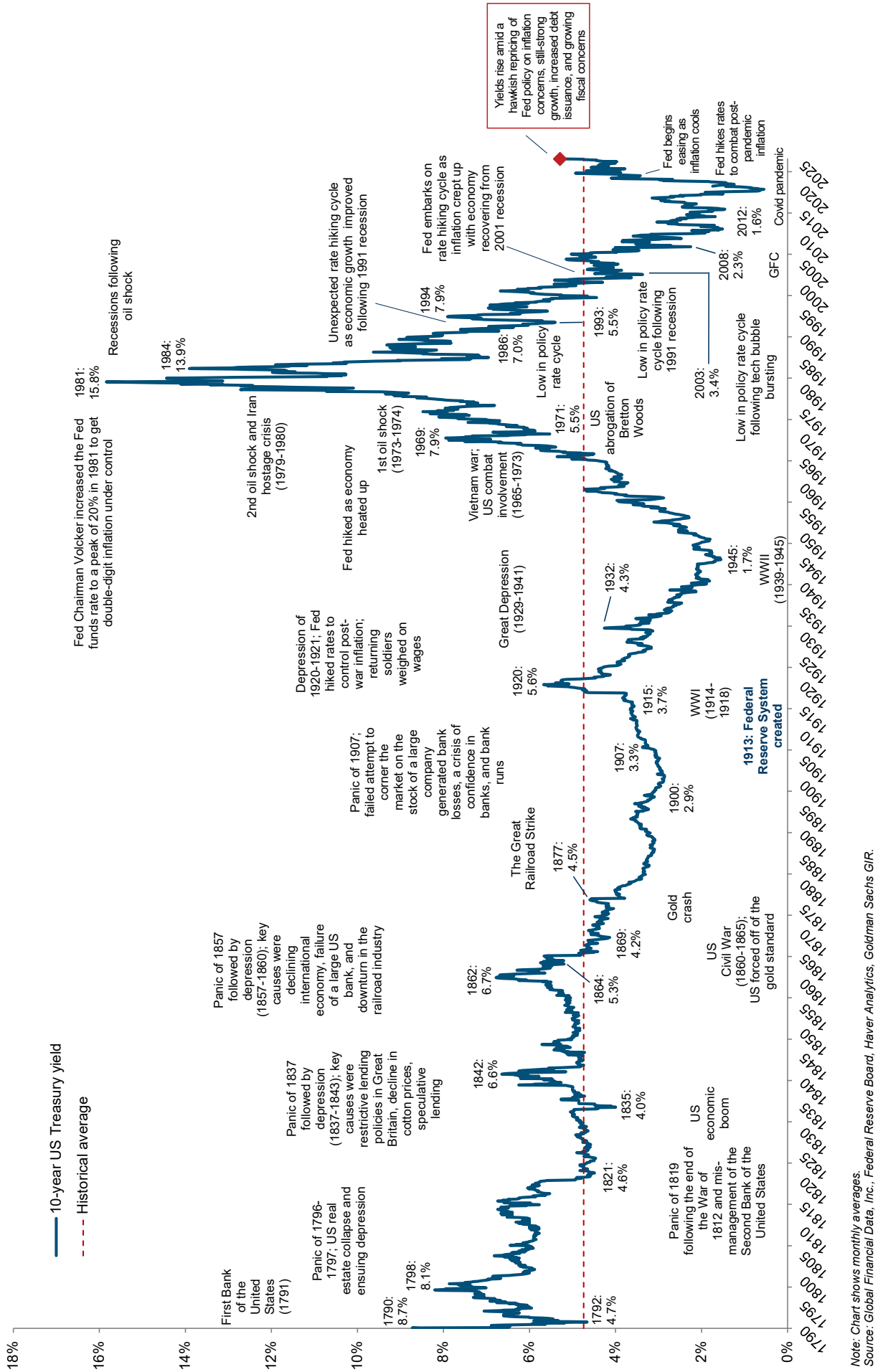
Fiscal policymakers are more constrained. Governments may still be willing to respond as forcefully as they did during the pandemic, but after taking such aggressive fiscal steps, less room exists to go big on fiscal policy today. Markets may also react to such support more negatively than in 2020 given already-elevated debt levels and ongoing concerns about fiscal sustainability. If monetary and fiscal authorities act together, as they did in 2020, that would temporarily mitigate market stress, but it would not resolve the longer-term fiscal problem.

Hopefully, the recent rise in bond yields will feed back into the politics of fiscal policy and prompt an adjustment. Bond market pressure has already triggered policy shifts in some countries. The question is how far yields must rise before adjustment occurs more broadly, and whether that adjustment can happen without uncomfortable volatility. But until it occurs, fiscal policymakers will remain relatively constrained.

Jenny Grimberg: Where does all of that leave you on the outlook for the global economy and markets?

Tobias Adrian: It is important to distinguish between the baseline and the downside. My baseline is fairly positive. The global economy has demonstrated considerable resilience in the face of multiple severe shocks, and I would not be surprised if it continues to shrug off higher yields, provided they do not keep rising sharply in a disorderly fashion. At the same time, I do worry about downside risks. Beyond the vulnerabilities we have discussed, cyber risk is an important concern. So far, cyber risks have not been macro-critical, but they certainly could become so. AI could strengthen cyber defenses, but it could also enhance offensive capabilities. And while sophisticated institutions may be resilient to cyberattacks, many firms may lack the capabilities and know-how to defend themselves effectively. So, plenty of room exists to worry about the downside, but the offset is policymaking, especially monetary policymaking, as we have discussed. All of that leaves me vigilant but cautiously optimistic.

A long history of US rates



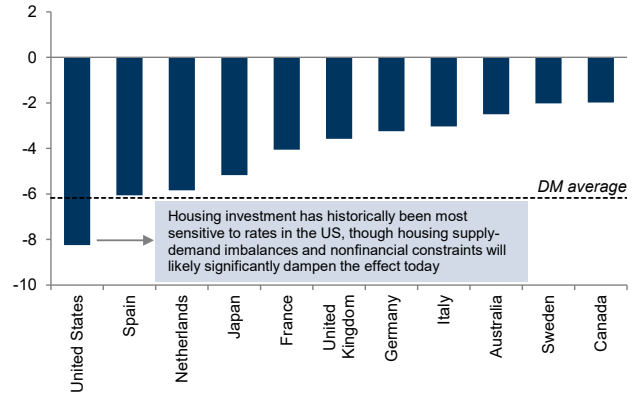
Note: Chart shows monthly averages.
Source: Global Financial Data, Inc., Federal Reserve Board, Haver Analytics, Goldman Sachs GIR.

Higher rates & DM growth: rules of thumb

GS global economists Joseph Briggs and Johan Allen develop rules of thumb to quantify the near-term rate sensitivity of major DM economies across four channels: (1) housing, (2) consumers, credit, and cashflow, (3) corporates and capex, and (4) debt and deficits.

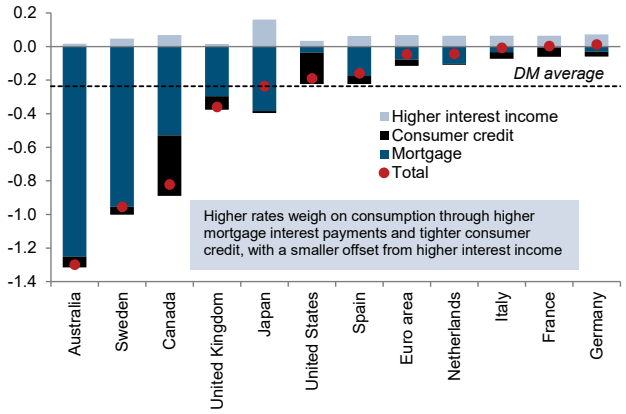
Each 100bp increase in rates lowers DM housing investment by 6% over the next 1-2 years on average...

Peak cumulative effect of a 100bp increase in bond yields on residential investment, %



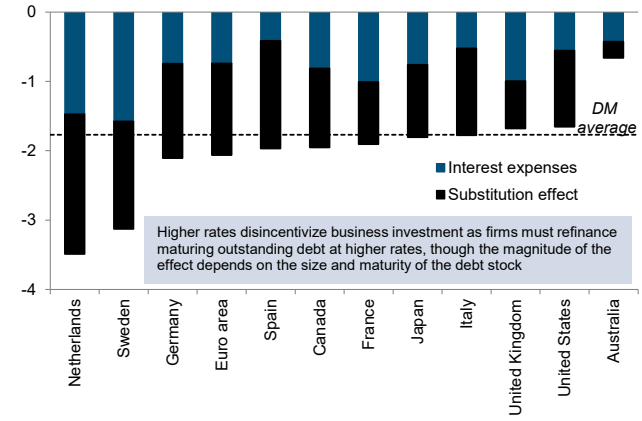
...DM consumer spending by 0.2% on average, with the largest effects in Australia, Sweden, and Canada...

Effect of 100bp higher interest rates on household consumption by credit type, %



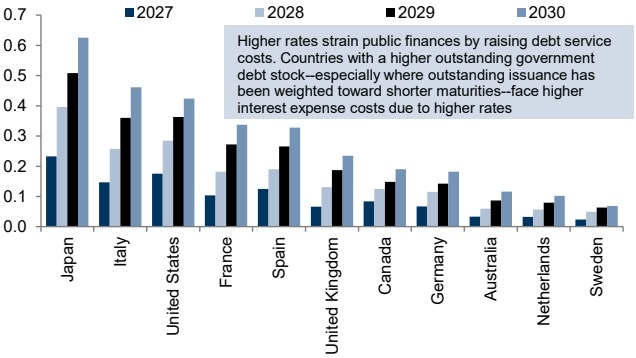
...and DM business investment by 1.8%, with larger effects in the Netherlands, Sweden, and Germany

Hit to corporate investment within next 12m, %



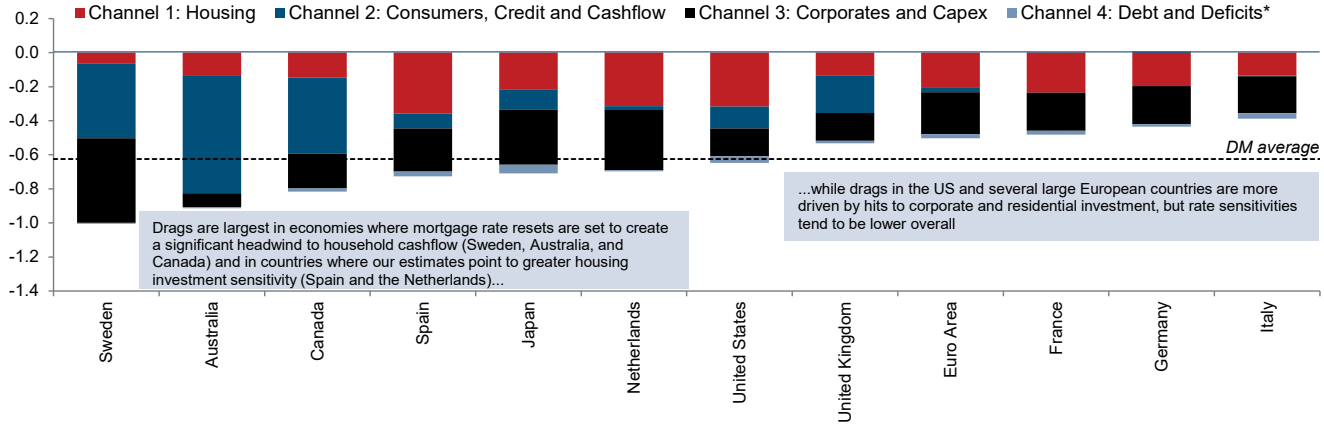
The current maturity structure of debt implies that a persistent 100bp increase in rates raises interest expenses by 0.0-0.2% of GDP over the next year and 0.1-0.6% of GDP by 2030, with the largest effects in Japan, Italy, and the US

Increase in interest expenses as a % of GDP assuming a 100bp increase in yields



The cumulative effect from a 100bp rise in rates across all four channels implies a 0.6% drag on DM GDP overall over the next year

Cumulative effect on overall GDP from a 100bp tightening, % of GDP



**We apply our estimate of the 2027 rise in interest expenses and assume that governments on average consolidate fiscal balance by 0.2pp for every 1pp increase in interest expenses, in line with Debrun and Kinda (2016).*

Source for all charts: Goldman Sachs GIR.

Rising rates: key questions answered

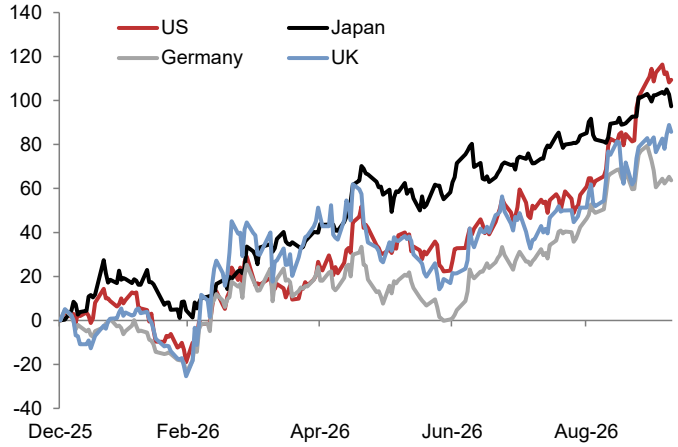
William Marshall explains the rise in global bond yields and where they may be headed

Q: What has driven the gradual rise in global bond yields over the past several months?

A: The most striking feature of the rise in yields has been its breadth. Yields in all major economies have risen, without a clear market leader or single source of the bond selloff, which has occurred across yield curves, with both front-end and long-end rates moving to fresh highs. Measures of inflation risk *and* real rates, as well as rate expectations *and* term premia, have played roles in the rise in nominal yields. In particular, estimates of term premium, which reflect the extra compensation investors demand for holding duration, have surged to the highest level in over a decade. And the move has not been isolated to sovereign bonds, occurring across all forms of duration. Until recently, the relative stability in interest rate volatility suggested that fundamentals, including stronger growth, higher inflation risks, hawkish policy, AI capex, and long-run fiscal risks, largely drove the rise in yields.

The rise in bond yields has not been unique to any market this year

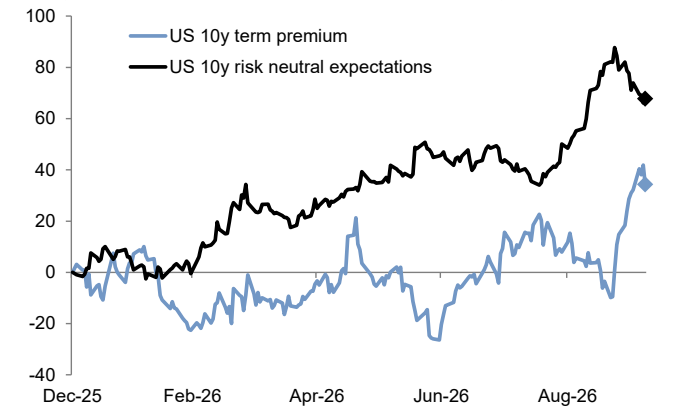
Year-to-date change in 10-year sovereign bond yields, bp



Source: Goldman Sachs FICC and Equities, Goldman Sachs GIR.

Both term premium and policy rate expectations have contributed

US 10-year term premium vs. risk neutral expectations, bp



Source: Goldman Sachs GIR.

Q: What role has AI capex and related issuance played?

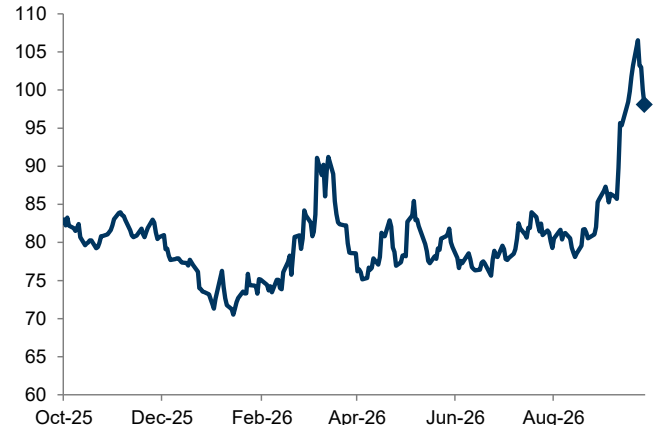
A: A swing of the investment/savings pendulum back toward investment creates a less supportive backdrop for bonds and requires higher yields to motivate investors to hold duration. The perception that investment spending (and the issuance to finance it) is relatively insensitive to higher yields is reinforcing that impact. While evidence that the absorption of duration supply is having a durable impact on longer-term yields is limited, the backdrop of price-insensitive issuance from both sovereigns and corporates is among the factors likely weighing on duration risk appetite. As long as investment and borrowing needs continue to grow, the skew of risks around where longer-term yields sit relative to macro drivers is likely to remain to the upside, with fundamental catalysts rather than valuations alone likely needed to reinvigorate duration demand.

Q: Beyond the steady march higher, bond yields have risen sharply in recent weeks. Are these moves fundamental or technical in nature?

A: The recent sharper rise partly reflects hawkish shifts in policy pricing, reinforced by the Fed, ECB, and BoJ all hiking in response to inflation pressure. Fundamental risks and central bank signals have modestly boosted front-end volatility. The rise in volatility toward the end of September points to these hawkish risks beginning to tighten financial conditions, as well as to technical factors related to hedging risks directly related to higher yields that may have helped to push yield levels higher than fundamentals would suggest.

Interest rate volatility has risen sharply recently

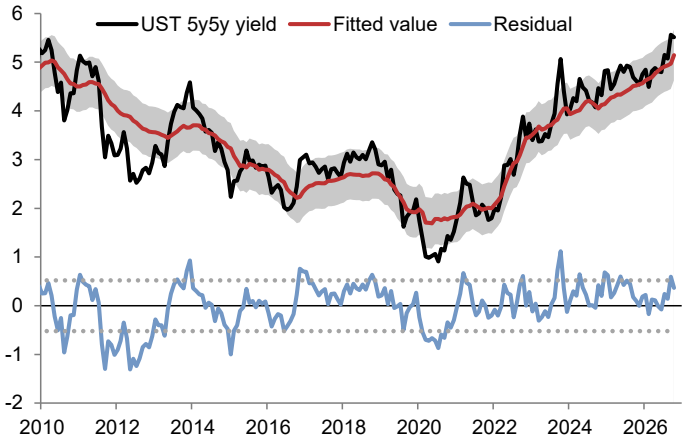
1y10y US implied volatility, annualized bp



Source: Goldman Sachs FICC and Equities, Goldman Sachs GIR.

The longer-term yield rise has lately de-coupled from fundamentals

UST 5y5y yield versus fitted value, %



Source: Bloomberg, Goldman Sachs GIR.

Q: Can the recent bond market operations of some Treasuries influence the level of bond yields?

A: Managing the reliance on longer-term issuance is a sensible response to the higher term premium environment. The combination of structural shifts in longer-term savings and higher yields is reducing the duration needs of liability-driven investors such as pension funds. However, while maturity adjustments that rebalance supply and demand across the yield curve can benefit long-end bonds at the margin, they are unlikely to meaningfully lower yields. Buybacks and increased front-end issuance do not change the deficit or the inflation trajectory (see pg. 24). Although these actions can potentially reduce interest costs on average, they risk raising interest cost volatility should front-end rates need to rise to combat higher inflation.

Q: On net, where do you expect global bond yields to go from here over the near, medium, and long term, and how do your forecasts compare to consensus?

A: We expect bond yields to fall across major markets, with 10y yields ending 2026 at 4.75% in the US, 3.25% in Germany, 5% in the UK, and 3% in Japan. Achieving this will require greater market confidence in a more benign underlying inflation trend. Our more dovish-than-priced outlook for policy rates across most markets argues for lower yields over the medium term, though for the US in particular we expect fiscal dynamics to exert steady upward pressure on longer-term yields over time.

Q: How is the balance of risks to your forecasts skewed, and what could cause yields to deviate meaningfully from these expectations?

A: Front-end rates are still tethered to inflation risks, especially from energy prices, and yields will remain skewed to the upside if energy volatility does not subside. This is especially true given that the growth data remain strong and the Fed recently delivered its first hike. Although we expect policy tightening to prove limited, visibility on the extent and duration of the hiking cycle is low and rates often struggle to price in the end of tightening in advance of the final hike. A credible reduction in inflation risk via a decline in energy volatility across natural gas, crude oil, and refined products, or progress in underlying inflation toward central bank targets would therefore likely be required to realize lower yields. Evidence of slowing AI investment or any revision of the economic potential of AI investment would also alleviate upward pressure on yields and may ultimately lead to much lower yields down the line, as would a return of cyclical concerns.

Q: What areas of the rates complex look vulnerable if government bond yields remain high?

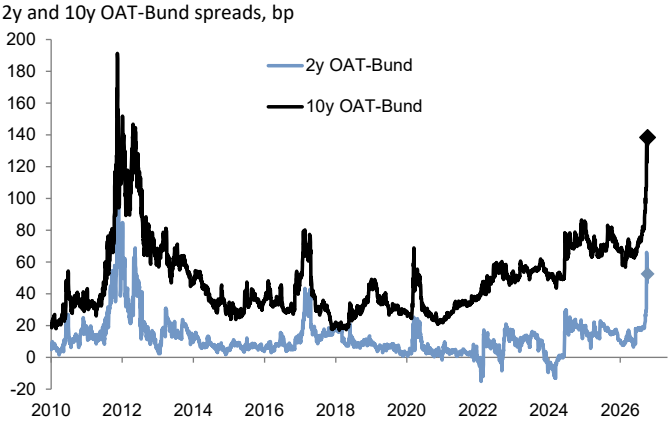
A: Aside from the effects on the real economy (see pgs. 9 and 12-13), the market is likely to focus on the credit implications of higher rates in both AI and non-AI-related sectors. In Europe, higher yields pose risks to sovereign credit spreads. While the growth backdrop remains supportive for now, and fiscal fundamentals in some countries such as Spain have improved, the pressure from higher German yields and ECB rate hikes will pressure markets like OATs (French bonds), which will price additional fiscal risks at higher refinancing rates.

Q: How concerned should we be about a sovereign bond crisis in France and contagion risk across Europe?

A: The relative rise in French versus German yields has been extreme by historical standards, particularly when benchmarked against other sovereign spreads. Domestic macro and fiscal fundamentals remain unfavorable for OATs, with ongoing budget negotiations and the presidential race increasing uncertainty. That said, French spreads have widened more than fundamental drivers would imply, suggesting that the latest leg of widening has been driven by uncertainty and risk premium rather than a clear shift in the outlook.

Stabilization likely requires inflation relief, signs of successful budget negotiations, or sufficient financial conditions tightening that shifts the path for ECB policy, with risk that elevated premium remains stickier. For Europe more broadly, spillovers have increased (particularly to Italian spreads), though risks still remain idiosyncratic to France. Similar to the case for France specifically, it will likely take a fundamental anchor to reduce risks of greater spillovers.

The rise in French OATs vs. German Bunds has been extreme by historical standards



Source: Goldman Sachs FICC and Equities, Goldman Sachs GIR.

Q: How can investors best navigate a world of sustained high bond yields?

A: The return of higher yields isn't necessarily bad news for investors, especially if it reflects higher nominal growth. A world of high term premium used to be a familiar one, and higher term premium is associated with positive excess returns. This should help to protect total returns for bond investors. But the competition with cash is likely to imply steeper curves than in past periods of high term premium/low volatility. The hedge value of bonds will therefore need to be recalibrated, but bonds should still offer strong protection against cyclical downturns.

William Marshall, Head of US Rates Strategy

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Tel: 212-357-0413

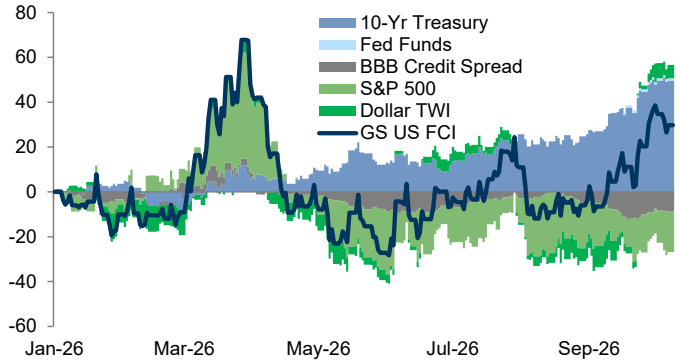
A modest US growth drag

Pierfrancesco Mei expects higher rates to only weigh modestly on US growth next year

The recent sharp rise in long-term interest rates has fueled a material tightening in US financial conditions, raising the question of how much this will affect US growth. Higher rates tend to weigh most heavily on residential investment, but also dampen consumer spending and capex, which can contribute meaningfully to the overall growth drag since they account for relatively large shares of GDP (~65% and 15%, respectively, vs. 5% for residential investment). Under our baseline forecast that sees the Fed delivering one more rate hike in December followed by three cuts starting in 2H27 and 10-year Treasury yields declining to 4.4% by end-2027, we expect only a modest 0.2pp drag on growth in 2027 from higher rates, though the drag could rise to 0.5pp if rates remain around current levels.

After moving sideways for a few months, financial conditions are now tightening amid the move higher in rates

Changes in GS US FCI since January 1, 2026, bp



Source: Goldman Sachs GIR.

Residential investment hardest hit...

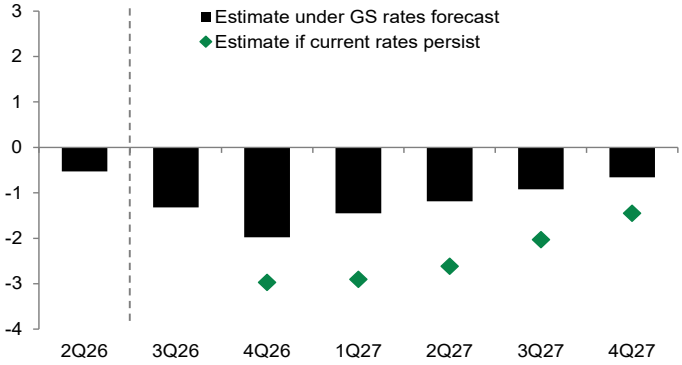
The sharp rise in 30-year mortgage rates to above 7% is likely weighing on residential investment, and will likely continue to do so by further cooling housing demand, which we already expected to remain tepid amid affordability concerns, slowing income growth, and a persistent drag on household formation from reduced immigration. Consistent with this, survey-based measures of purchase intentions (as expressed in the Conference Board and Michigan surveys) have declined since early summer.

Higher mortgage rates will also continue to weigh on housing turnover via the lock-in effect: because many borrowers took advantage of low rates in 2020 and 2021 to refinance, the cost of moving at current mortgage rates is significant. Indeed, with mortgage rates up more than 1pp since February, over 90% of borrowers now hold mortgages priced below market rates, leaving them with little incentive to move.

Although we noted during the last housing cycle that housing scarcity tended to dampen the hit to residential investment from rising rates, and such scarcity has eased, vacancy rates remain low by historical standards. On net, accounting for the prevailing vacancy rate, assuming mortgage rates evolve in line with our 10y Treasury yield forecast plus a 200bp spread, and measuring the counterfactual against our rates strategists' pre-Iran war forecasts, we expect higher rates to subtract around 2pp from annualized residential investment growth at the peak in 4Q26 and 1pp on average in 2027, with the drag likely to be roughly twice as large if rates remain at current levels.

We expect the drag from higher mortgage rates on residential investment growth to peak at ~2pp in 4Q26 and average ~1pp in 2027, but the drag would be twice as large if current rates persist

Effect of higher mortgage rates on QoQ annualized residential fixed investment growth, GS baseline vs. current market rates, pp



Source: Goldman Sachs GIR.

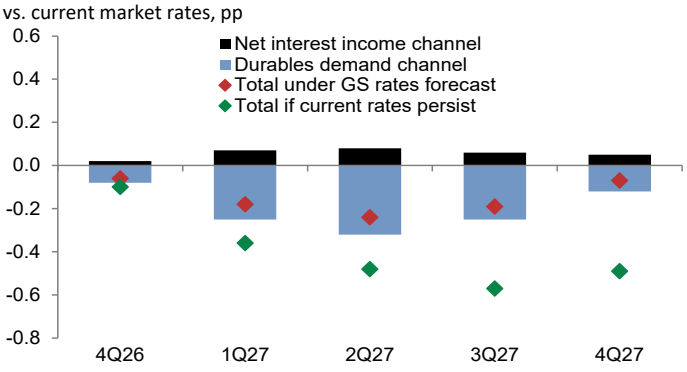
...but consumer spending will also take a hit...

Higher rates weigh on consumer spending mainly by reducing demand for durable goods. Using data since the 1960s, we find that a 50bp rise in the Fed funds rate—the cumulative increase we expect by December—has historically lowered consumer spending by just under 0.5%, with the decline concentrated in durable goods that rely heavily on credit financing and, in particular, recreational goods and motor vehicles.

That said, we think the impact could be smaller today as strong household balance sheets should help cushion the rise in interest expense and negative wealth effects should be limited given our equity strategists' baseline expectation that equity prices will continue to edge higher through 2027.

We expect the decline in durable goods spending to outweigh the boost to spending from higher net interest income, resulting in a ~0.2pp decline in consumption growth through 2027

Effect of higher interest rates on QoQ annualized real PCE growth, GS baseline vs. current market rates, pp



Source: Goldman Sachs GIR.

On net, focusing on reduced durables demand (given the higher incentive to save and higher financing costs) and an offsetting rise in households' net interest income[1] (since households are

[1] We isolate the direct substitution effect of higher rates on durables demand using estimates from economists Alisdair McKay and Johannes Wieland that strip out second-round effects from lower net wealth, and we estimate the effect on net interest income from the historical relationship between interest rates and net interest income and apply standard marginal propensities to consume to translate it into spending effects.

overall net lenders as they hold more interest-earning assets than interest-bearing debts), we estimate that higher rates will slow consumption growth by roughly 0.2pp through 2027. However, the drag would increase substantially if rates remain at current levels and stifle goods demand for longer.

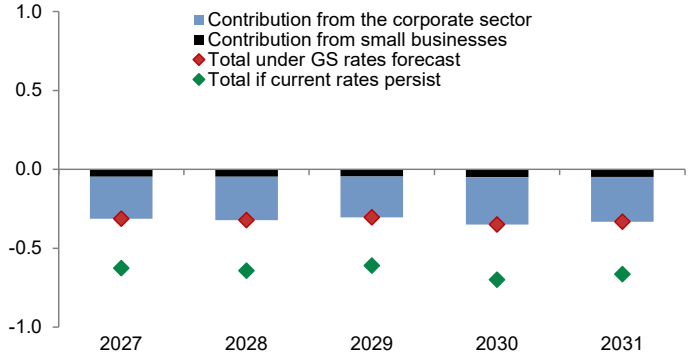
...as will capex

Higher rates also raise firms' marginal cost of funding, weighing on investment. A growing concern among investors is that higher rates could derail the AI-driven capex boom that has boosted investment growth in recent quarters, especially as companies have increasingly relied on debt and equity issuance to fund this investment. However, hyperscalers' capex—which accounts for over two-thirds of AI investment in the US—has so far proven largely resilient to higher rates and input costs, suggesting that these companies are unlikely to meaningfully pull back capex in the near term (see pgs. 18-19). Consistent with this, our portfolio strategists continue to expect a healthy pace of AI-related capex in 2027, and our credit strategists recently raised their forecasts for hyperscalers' debt issuance.

That said, non-AI companies will need to devote a larger share of revenue to interest expense as they refinance maturing debt at higher rates. In the corporate sector, around 15% of bonds are set to mature within the next two years—roughly in line with recent years but below the historical average—as the average maturity of corporate debt has roughly doubled over the last three decades. This should slow the passthrough from policy rate hikes to corporate borrowing costs, reducing the economy's near-term sensitivity to the funds rate. However, small businesses are more exposed since around half of their debt is variable-rate and reprices quickly.

We estimate that higher refinancing costs for businesses will dampen capex growth by ~0.3pp/year over the next five years

Effect of higher interest expenses from maturing debt on aggregate capex growth, GS baseline vs. current market rates, pp



Source: Goldman Sachs GIR.

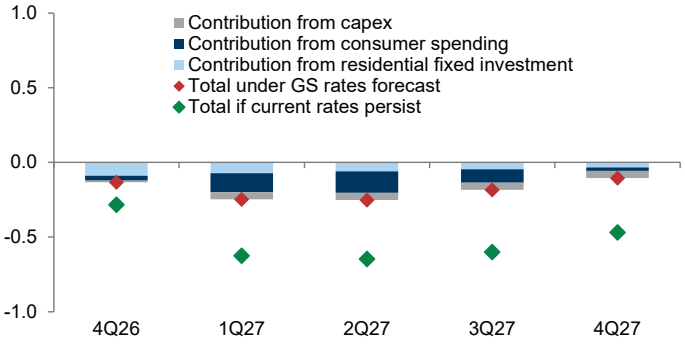
Taken together, we expect higher interest expense to weigh modestly on capex growth—by around 0.3pp in 2027 and subsequent years—assuming that borrowing rates reset in line with our strategists' forecasts for Treasury yields and credit spreads, though the impact would more than double if rates remain around current levels[2].

A modest growth drag, but larger if rates remain at current levels...

Until recently, higher equity prices, tighter credit spreads, and a weaker Dollar roughly offset the drag on financial conditions from higher rates, resulting in only a 0.1pp hit to GDP growth this year (Q4/Q4). Looking ahead, we expect the combined hit to residential investment, consumer spending, and capex from higher rates to subtract around 0.2pp from GDP growth in 2027 (Q4/Q4), which would still leave the economy growing close to our 2.3% potential growth estimate. That said, if rates were to remain at current levels, we estimate the drag would more than double, rising to just over 0.5pp.

We estimate that higher rates will subtract around 0.2pp from GDP growth in 2027, with the drag over twice as large if rates remain at current levels

Impact of higher rates on QoQ annualized real GDP growth, GS baseline vs. current market rates, pp



Source: Goldman Sachs GIR.

...and watch additional risks

Beyond the impact on the most rate-sensitive areas of growth, we see two additional risks if the current level of rates persists. First, equity prices could come under pressure. The most rate-sensitive segments of the equity market—unprofitable tech, utilities, REITs, and homebuilders—have already pulled back sharply from their highs, though the S&P 500 has been roughly flat since August. Our strategists think that current rate levels should not prevent the equity market from rising further (see pg. 16), but if equities remained roughly flat through 2027, we estimate that the missing boost to household net worth would lower consumer spending growth by just under 0.5pp.

Second, the fiscal sustainability outlook could deteriorate. Large primary deficits remain the main driver of the US' rising debt-to-GDP ratio, but the average interest rate on outstanding debt has remained below nominal GDP growth in recent years, allowing the government to issue new debt without mechanically pushing the ratio higher. This would remain the case under our baseline rate forecast, but not if current rate levels persist and maturing debts are refinanced at a higher cost (see pg. 15). In this scenario, interest expense and debt as a share of GDP would rise to an unprecedented 6% and 132% by 2035, respectively, increasing the urgency of deficit reduction, which could create a further drag on growth.

Pierfrancesco Mei, US Economist

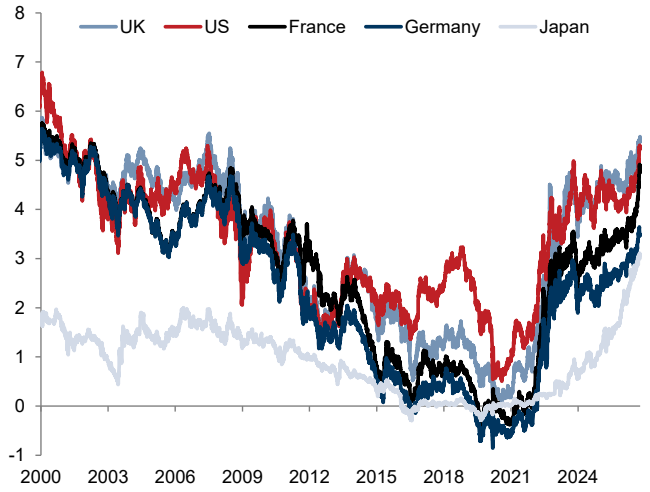
Email: pierfrancesco.mei@gs.com *Goldman Sachs & Co. LLC*
Tel: 212-902-8809

[2] We estimate the increase in interest expense separately for large and small businesses based on their financing needs over the next few years and then apply our standing estimates of capex sensitivity to interest payments.

A look at growing global fiscal risks

A sharp rise in sovereign bond yields around the world...

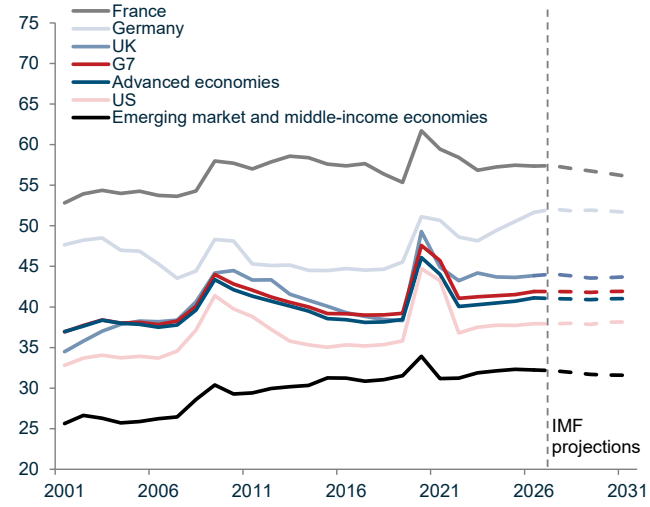
10-year sovereign bond yields, %



Source: Bloomberg, Goldman Sachs GIR.

...government spending is projected to remain high...

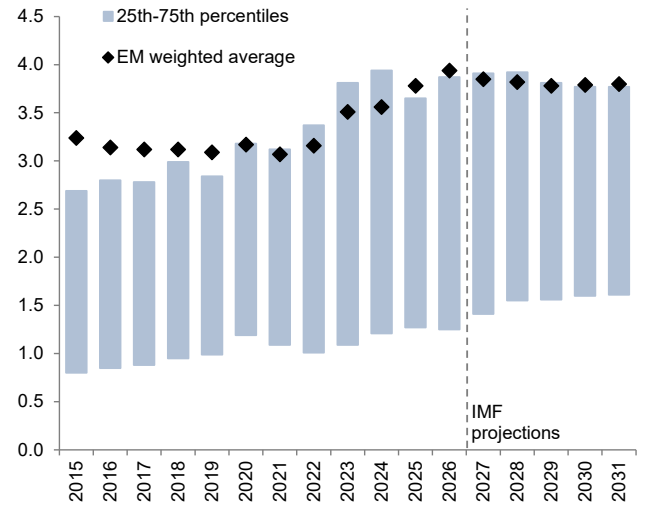
General government expenditure, % of GDP



Source: IMF (projections as of April 2026, found here), Goldman Sachs GIR.

...while debt servicing costs for the most vulnerable EMs are set to remain structurally elevated

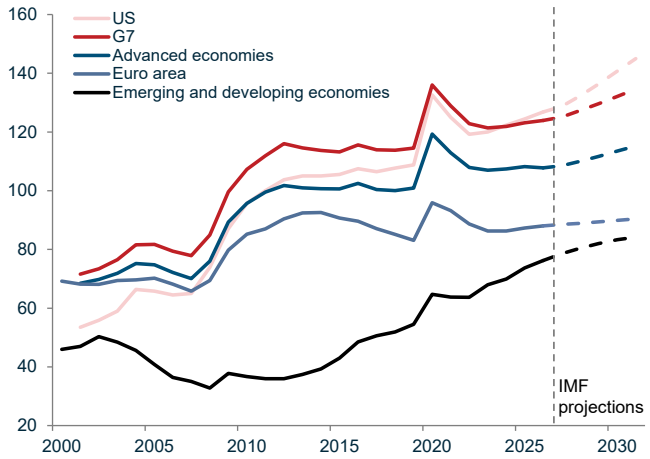
Interest expense in emerging markets (EM), % of GDP



Source: IMF (projections as of April 2026, found here), Goldman Sachs GIR.

...has both contributed to and reflected growing fiscal concerns, as already elevated debt-to-GDP ratios are set to rise further...

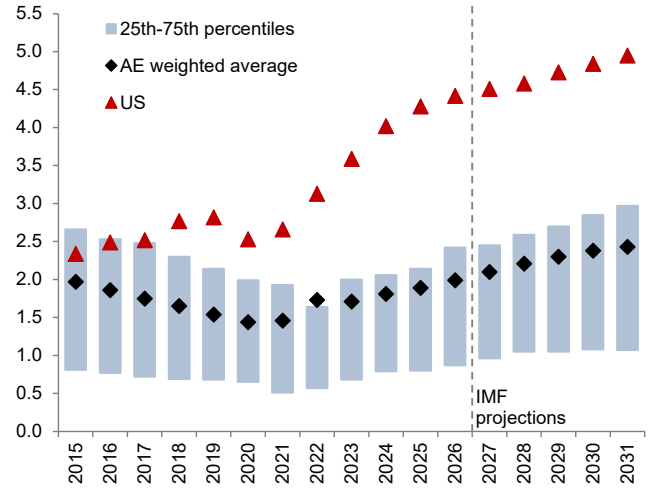
General government gross debt, % of GDP



Source: IMF (projections as of October 2026, found here), Goldman Sachs GIR.

...and interest expenses in advanced economies are expected to increase...

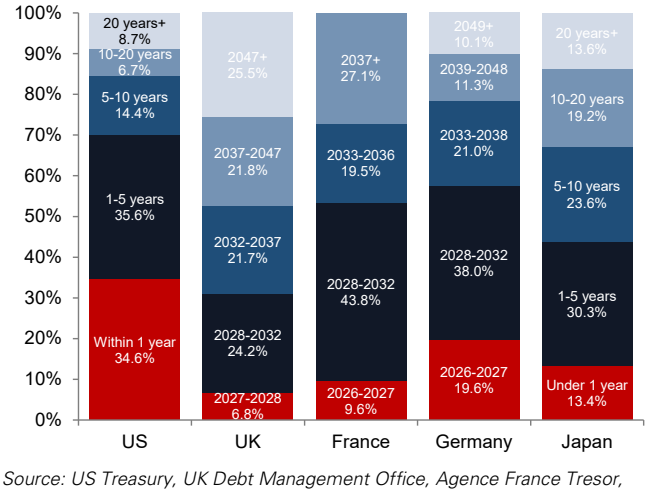
Interest expense in advanced economies (AE), % of GDP



Source: IMF (projections as of April 2026, found here), Goldman Sachs GIR.

Different debt maturity profiles across advanced economies mean that higher yields will impact the US relatively soon, while delaying the refinancing shock for longer-duration issuers like the UK

% of debt maturing within each timeframe, by country



Source: US Treasury, UK Debt Management Office, Agence France Tresor, Bundesrepublik Deutschland Finanzagentur, Japan Ministry of Finance, GS GIR.

Higher rates, greater US fiscal pressure

David Mericle and Pierfrancesco Mei assess the US fiscal outlook amid higher rates

While concerns about the US fiscal trajectory have grown steadily in recent years, driven by unusually large primary deficits and a debt-to-GDP ratio approaching post-WWII highs, the recent sharp rise in rates has intensified those concerns. Under our baseline forecast of nominal GDP growth close to 4.5%, an average primary deficit of around 3% of GDP, and an average interest rate on new government debt of just over 3.5%, we expect the debt-to-GDP ratio to rise from 100% today to 124% by 2035, and an even higher 132% if current rates persist—a problematic situation that is politically difficult to tackle, but too risky to ignore.

High debt is problematic…

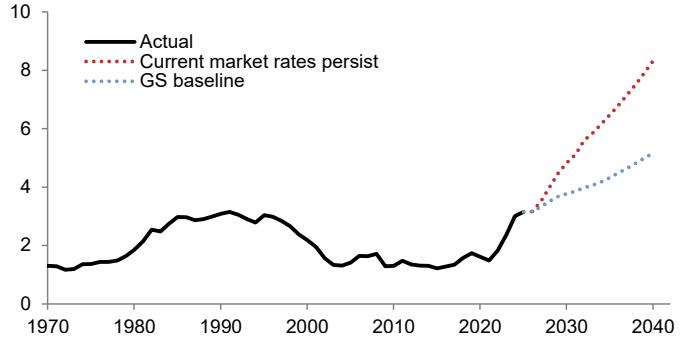
A large and growing debt is problematic because it puts upward pressure on nominal net interest expense as a share of GDP, which captures the cash outflows required to service the debt and is likely how policymakers think about the debt servicing burden. Nominal interest expense is already near its 1990s peak and is set to surpass it substantially in the coming years, especially if rates remain at current levels or rise even further.

A larger debt stock can also raise interest rates, which could further push up debt levels. The average interest rate on outstanding debt has been below nominal GDP growth in recent years, which has allowed the government to cover its interest expense by issuing new debt without contributing to an increase in debt-to-GDP (even though the large primary deficit has continued to push it higher). This would remain true under our baseline rates forecast, but would not if current interest rates persisted as maturing debt is refinanced.

To be clear, we don't think such a rise would likely trigger a classic debt crisis—where falling confidence in the government's ability or willingness to repay its debt causes a spike in interest rates that leads to financial market instability and/or recession—given that the US issues debt in dollars, sets its own monetary policy, and has a flexible exchange rate. But failing to address the rise is still risky and could prove costly. If the debt servicing burden or the necessary deficit reduction become large and painful enough, there could be pressure on the Fed to keep rates too low, resulting in high inflation. An unfavorable debt situation could also limit policymakers' ability and willingness to fight recessions with fiscal stimulus.

Nominal interest expense as a share of GDP will likely rise beyond its 1990s peak over the coming years…

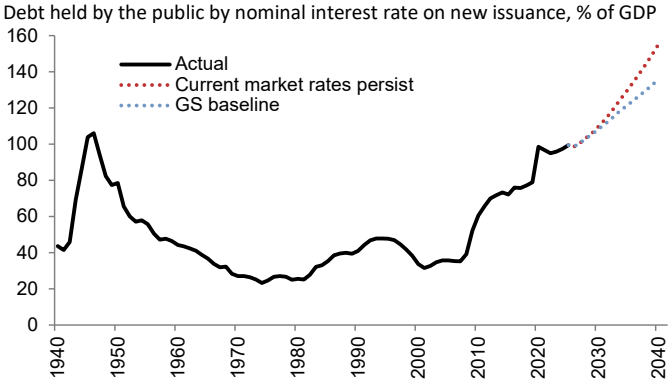
Federal nominal interest expense by nominal interest rate on new issuance, % of GDP



Source: Goldman Sachs GIR.

…and the debt-to-GDP ratio will likely reach 124% by 2035, and even higher if current rates persist

Debt held by the public by nominal interest rate on new issuance, % of GDP



Source: Goldman Sachs GIR.

…but politically difficult to tackle

The rise in the debt-to-GDP ratio that we expect over the next decade is driven entirely by a higher primary deficit rather than rising interest costs. So, balancing the primary budget would be sufficient to keep the ratio stable. The US has two advantages in achieving this adjustment. First, we expect growth to remain solid. This, together with substantial room for the Fed to ease and resilient private sector balance sheets, makes it more likely that deficit reduction would lower the debt-to-GDP ratio. Second, the US collects less revenue as a share of GDP than other G7 countries, giving it more room to raise additional revenue.

However, achieving a balanced primary budget would require a large deficit reduction given the current starting point—a tall order at a time when growing entitlement costs push in the opposite direction—and for now, deficit reduction looks politically unlikely.

Could AI solve the problem?

AI may be helpful in tackling the fiscal problem. A larger AI productivity boost to growth than we expect would slow the rise in the debt-to-GDP ratio, by boosting GDP and narrowing the primary deficit as revenues rise and cyclically sensitive outlays decline, much as the productivity boom that began in the late 1990s did. In a scenario in which AI lifts real GDP growth to 3% through 2035 (vs. ~2.5% in our baseline), we find that the debt-to-GDP ratio would rise to ~114% by 2035.

That said, AI may not be a magic bullet. The technology could displace workers, reducing government revenues and raising spending on income support, or it could shift national income from labor to capital, lowering the average tax rate and overall revenues. It could also improve healthcare and longevity, reducing some costs while potentially increasing overall entitlement spending. And it could boost the demand for capital, raising interest rates and government borrowing costs. This wide range of effects could dilute the improvement in government finances from higher productivity. So, AI is more likely to lower—not eliminate—the deficit reduction needed to tackle fiscal problems in the US and beyond.

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Rates won't derail equities, for now

Ben Snider argues that the current level of yields should not prevent the US equity market from rising further, though another sharp rise would likely pressure the rally

The S&P 500 is on pace for its fourth consecutive year of double-digit returns despite long-term Treasury yields surging to their highest levels in decades. For equity investors, the natural question is how much of a threat higher yields pose to stock prices, and at what level they might cause recent equity market strength to reverse. Ultimately, we think that the current level of yields should not prevent the equity market from climbing higher, though a further sharp rise would pressure the rally.

Higher rates weigh on valuations...

Rising rates most directly weigh on equity valuations. The value of an equity can be viewed as a discounted stream of future cash flows, and higher yields increase the discount rate applied to those future earnings. We estimate that over 75% of the S&P 500's current value reflects cash flows 10+ years in the future, the largest share in decades. In other words, the market is highly dependent on distant future profits, making valuations particularly sensitive to higher rates today.

Reflecting this pressure, equity multiples have declined substantially as bond yields have risen. The S&P 500 forward P/E multiple has fallen from 23x a year ago to 19x today, a nearly 20% decline. Although several factors have contributed to this decline—including AI-related uncertainty and skepticism about the durability of recent earnings strength—higher rates have played an important role.

However, while equity valuation multiples have fallen, valuations relative to bonds have remained surprisingly steady. The difference between the S&P 500 earnings yield (5%) and the real 10-year US Treasury yield (3%)—a simple proxy for the equity risk premium—currently stands at 230bp and has remained fairly stable over the past two years. This suggests that equity valuations have largely recalibrated to the rise in yields thus far. However, the steady recent relationship between equity valuations and interest rates also suggests that a further increase in yields from here would put additional pressure on stocks.

...but earnings matter more...

However, valuations are only part of the story. The continued appreciation of the equity market to new record highs this year despite rising rates and declining valuations is a testament to the predominance of earnings in driving stocks. S&P 500 earnings have grown by over 30% during the past 12 months, more than offsetting the rates drag on valuations. The more critical question for equity investors is therefore whether rising rates will derail the market's underlying earnings strength.

We see three reasons for optimism. First, US corporate balance sheets generally appear healthy today. Most S&P 500 company debt carries fixed rates and long maturities, and interest expenses remain small relative to strong profits. The S&P 500 interest coverage ratio exceeds 10x, close to the highest level on record, and interest coverage for the median S&P 500 stock ranks in the 70th percentile of the last 40 years.

Second, we estimate that AI investment spending is driving roughly half of S&P 500 earnings growth this year, and we don't expect the recent rise in interest rates to affect those spending plans. While debt is financing a growing share of AI spending, the largest companies driving this spending have particularly strong balance sheets and appear to be targeting returns well in excess of their current cost of capital.

Third, although the Fed recently hiked its policy rate, there is little evidence of economic overheating that would prompt policymakers to tighten aggressively to slow growth. And with the rates market already pricing over three additional 25bp hikes in the coming year, the bar for policy to surprise in a hawkish direction is high.

...and rate volatility matters too

Interest rate volatility also matters for equities. Over the past few decades, stocks have usually been able to digest gradually rising rates but have struggled to digest sharp rate increases, typically declining alongside increases in Treasury yields greater than two standard deviations. This relationship has been consistent across investment horizons and parts of the yield curve. Today, a two standard deviation move in the 10-year yield is roughly 50bp in a month, a "speed limit" only recently exceeded as yields surged in late September. So, further increases in interest rate volatility warrant close watch.

Some pockets of greater vulnerability

While the broad US equity market looks reasonably resilient at current yield levels, some pockets of the market are more vulnerable. Smaller companies typically have weaker balance sheets and more floating-rate debt than large-cap companies, leaving small-cap stocks more sensitive to rising rates. Real estate and housing stock earnings are also particularly tied to the interest rate environment. Recent rotations within the equity market have reflected these sensitivities. Homebuilder stocks, for example, have lagged the S&P 500 by more than 20pp since June, closely tracking moves in long-term yields.

This uneven rate sensitivity has pushed correlations across US stocks to near record lows. While concerns about rising rates have weighed on many stocks, the strong balance sheets and earnings growth of many companies exposed to the AI boom have kept the aggregate S&P 500 index near its record high. As a result, US equity market breadth has fallen to its lowest level since the Dot-Com Boom.

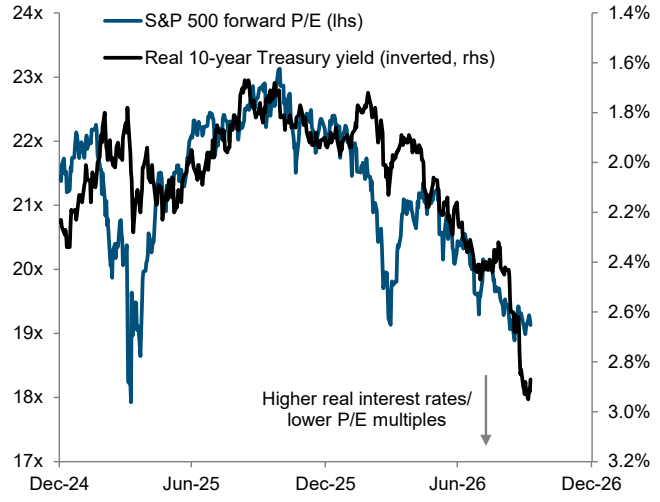
Not derailing equities, for now

Looking ahead, easing rate pressures would likely broaden the equity market rally and propel indices to new highs. However, even if yields remain at current levels, stocks should continue to climb gradually higher, as long as the earnings outlook remains intact and rate volatility remains contained.

Ben Snider, Chief US Equity Strategist

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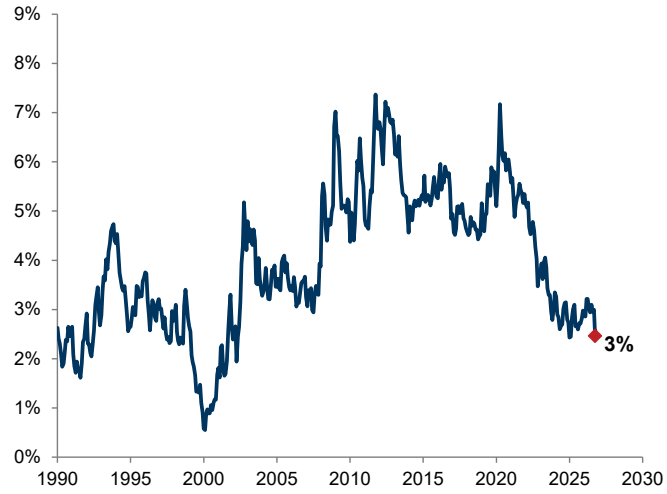
US equity multiples have declined this year as bond yields have risen...



Source: FactSet, Goldman Sachs GIR.

...though the market-implied equity risk premium has remained relatively stable in the past few years

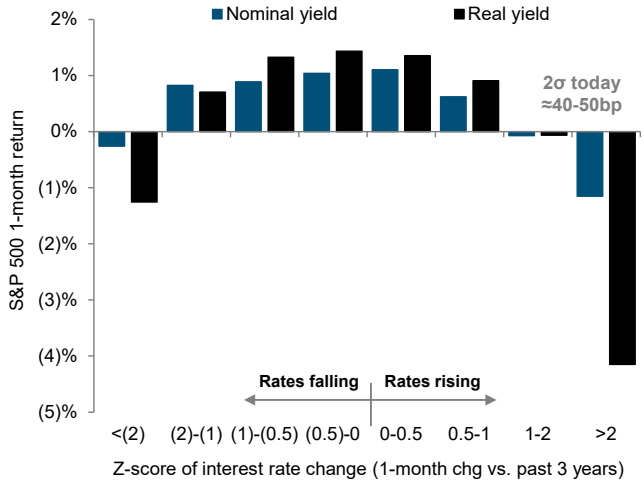
S&P 500 DDM-implied equity risk premium



Source: Goldman Sachs GIR.

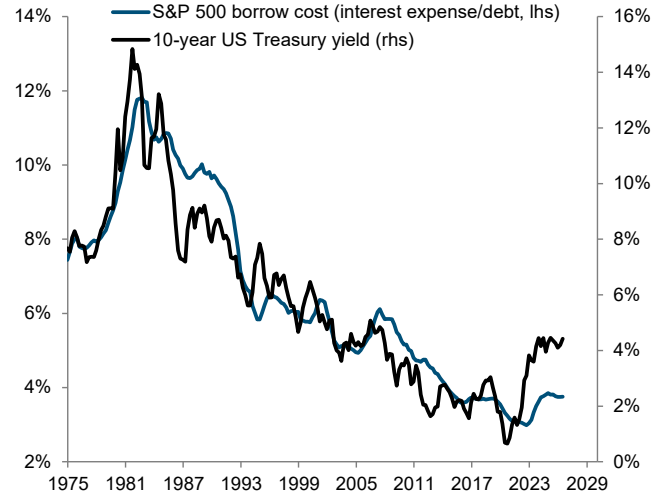
Equities typically struggle to digest sharp rises in bond yields...

Average monthly S&P 500 return vs. change in 10-year US Treasury yields



Source: Goldman Sachs GIR.

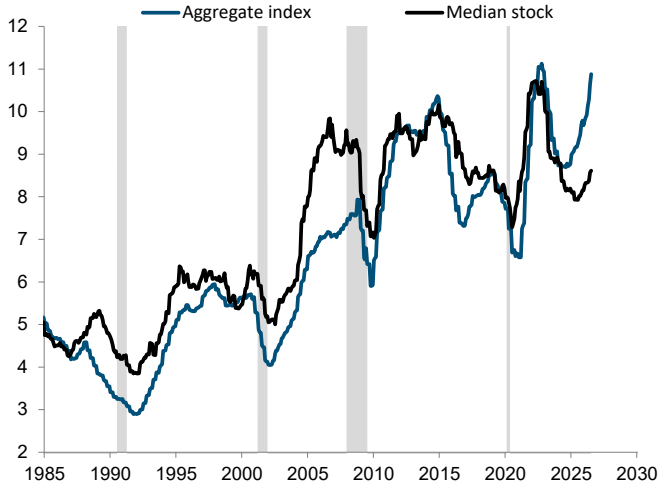
...though corporate balance sheets currently face limited fundamental risk from rising rates, as the increase in interest expense has been manageable...



Source: Compustat, Goldman Sachs GIR.

...and interest coverage ratios are historically high

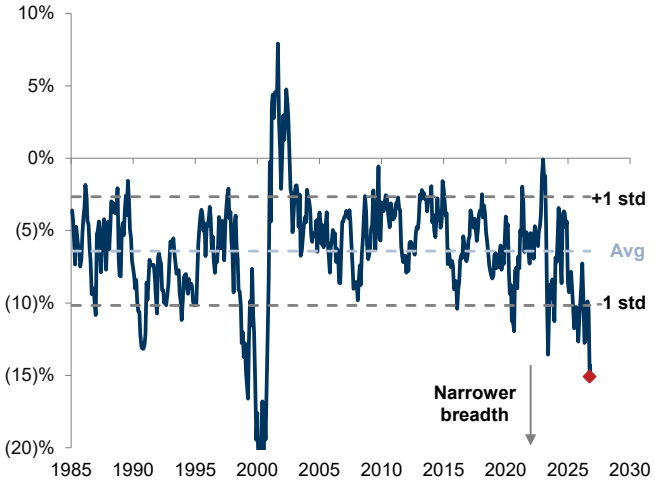
S&P 500 interest coverage (ex Financials)



Note: Aggregate excludes "other income" for GOOGL, AMZN, MSFT in recent quarters; grey bars denote recessions.
Source: Compustat, Goldman Sachs GIR.

But uneven rate sensitivity has pushed correlations across US stocks to near record lows, leading US equity market breadth to fall to its lowest level since the Dot-Com Boom

S&P 500 52-week market breadth



Source: Goldman Sachs GIR.

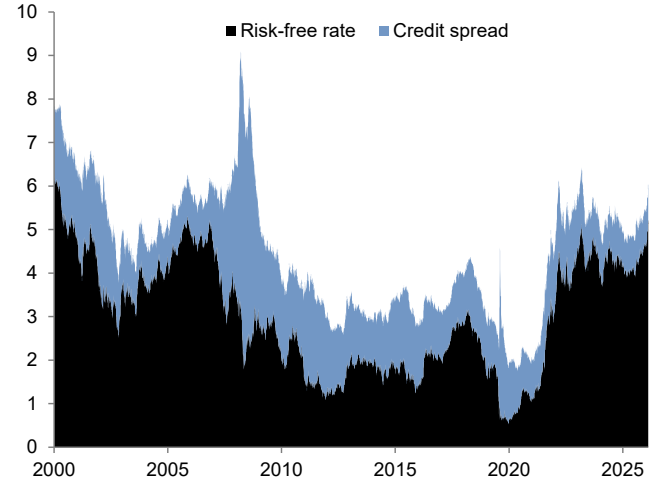
Rates risk in credit

Amanda Lynam sees growing risk that interest rates shift from a tailwind to a headwind for credit markets but views the headwind as manageable as long as growth holds up

For much of the past few years, elevated Treasury yields, or risk-free rates, have been a tailwind to corporate credit, sparking "yield-based" demand from institutional investors such as pensions, insurers, and foreign entities. The ability of these investors to achieve their all-in yield targets—even though much of the "heavy lifting" has been driven by the *risk-free rate rather than credit spreads*—supported demand for corporate credit and helped keep spreads anchored in a tight range. Credit spreads account for less than 15% of the all-in yield of the USD Investment Grade (IG) Corporate Index today, compared to 45% from 2019 to 2021, on average. This has effectively turned segments of the corporate credit market into a quasi-rates product. But with interest rate volatility now increasing, we are monitoring the risk that interest rates shift from a *tailwind* to a *headwind* for credit in two ways: by reducing investor demand and weighing on corporate fundamentals.

Credit spreads now account for less than 15% of the all-in yield of the USD IG Corporate Index

All-in yield for the Bloomberg USD IG Corporate Index, deconstructed between the spread and risk-free rate components, %



Note: Data as of September 28, 2026.
Source: Bloomberg, Goldman Sachs GIR.

Rate volatility poses a risk…

Increased rate volatility could reduce the confidence of yield-based investors to deploy capital. This would be especially negative for the IG market, as a large part of the yield-based demand was targeted toward that universe (due to insurers' rating-sensitive capital rules, for example). This flow is sizable, yet difficult to track. While weekly mutual fund and ETF flows are a closely watched, high-frequency data point for gauging credit market sentiment, they often *do not* capture flows from large-scale institutional allocators such as pensions, insurers, and sovereign wealth funds. The key signpost to watch for waning credit demand, in our view, is if rate volatility revisits the elevated levels of 2022-2023.

Higher rates could also weigh on corporate fundamentals through a higher cost of debt capital and/or pressure on interest-sensitive business models which rely heavily on financing, such as those tied to housing and autos. These fundamental concerns are most pronounced for leveraged finance borrowers, who tend to have thinner financial cushions relative to their larger and more diversified peers. These borrowers also tend to rely more heavily on floating rate debt, which experiences a headwind from higher base rates immediately. Fixed rate borrowers, by contrast, experience higher interest costs only if/when they refinance their lower-coupon debt.

We also believe the fundamental improvements visible over the past several quarters are likely to stall somewhat for issuers in the broadly syndicated leveraged loan and private credit markets. Higher base rates are an especially unwelcome development for issuers aiming to refinance upcoming Software loan maturities—especially those formed during the low interest rate backdrop of 2020-2021, before the potential for AI-related disruption to legacy Software business models became a key concern.

…though floating rate credit could benefit…

While higher rate volatility could weigh on credit demand, modestly higher base rates may actually support increased demand for floating rate credit, such as direct lending loans (within private credit) or broadly syndicated loans (either outright or as part of structured investments), on the margin.

The experience of the past few years in the broadly syndicated loan market shows how the linkage between higher base rates and demand can be nuanced. Notably, loan funds reported *outflows* when rates rose during the Federal Reserve's 2022-2023 hiking cycle, as recession concerns overshadowed higher loan yields. Given the magnitude and speed of the rate hiking cycle during 2022-2023, there was also a more pronounced focus on debt service and default risks.

Incrementally higher rates in the current market backdrop are unlikely to generate the same concerns, in our view. Indeed, inflows into loan funds have accelerated since April 2026 and the momentum has extended into late September. Similarly, private credit fundraising reached a new record in 2Q26, but we primarily attribute this to the broad-based improvement in sentiment toward the asset class.

…and growth is key

The drivers of any rate increases are also far more important than the absolute levels of rates. Higher rates reflecting resilient growth are more easily digested by credit fundamentals. Given the solid state of the economy and the relatively contained amount of rate hikes that our economists expect, we view the fundamental headwind as largely manageable and unlikely to spark a meaningful rise in defaults above our current forecasts.

A high bar to derail AI-related debt issuance

All that said, given the sharp increase in yields across the curve, market participants are increasingly questioning whether a higher cost of debt capital could impact higher-rated issuers—potentially derailing the pace of AI-related investment and, by extension, the AI-related debt issuance that we expect over the

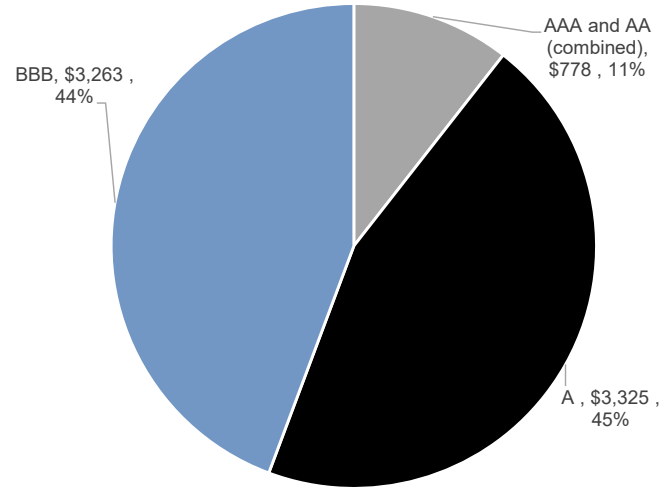
next few years. We view the bar as very high for such a scenario, for two key reasons.

First and foremost, we do not view today's all-in cost of debt as outsized by historical standards. For example, while the 10-year US Treasury yield (5.30%) is the highest since 2007, the USD IG index-level spread of 83bp still ranks toward the tight end of the historical range since 2000 (and below the average of 144bp). A similar takeaway is visible when isolating the spread of the USD IG Technology index. While the average spread of the sector has widened more than 20bp so far in 2026, the index-level spread measure of 91bp is below the average of 128bp since 2000. Moreover, while longer-term ROIC metrics on AI investment are still widely debated, even a relatively conservative assumption of a 15% ROIC leaves plenty of room for the after-tax cost of debt to increase, while still likely resulting in accretive debt-financed investment opportunities.

Second, highly rated Technology issuers have plenty of financial flexibility and debt capacity. For example, several AI-related firms—including MSFT, META, AMZN, GOOGL, NVDA, and TAISEM—have credit ratings in the range of AAA or AA. Such a high rating has become a relative rarity in the credit market, with only 11% of the market in that territory. Far more common these days is a rating in the A or BBB context—which allows for additional balance sheet leverage while still offering ample, consistent access to the USD IG debt capital markets.

Only 11% of the market value of the USD IG Corporate Index is rated AAA or AA

Market value by rating for the USD IG Bloomberg Corporate Index, $bn

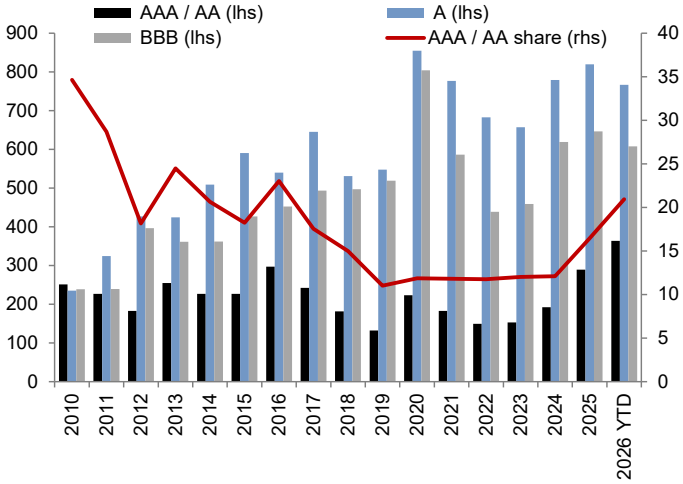


Note: Data as of 2Q26.
Source: Bloomberg, Goldman Sachs GIR.

The significant financial flexibility of the combined AAA/AA universe of issuers is visible when comparing key credit metrics side-by-side. The interest coverage ratio for the AAA/AA cohort is a striking 27.7x. This compares to 7.5x for the A-rated universe. Net leverage tells a similar story: 0.5x for the AAA/AAs, compared to 2.5x for As.

As Tech-related issuance has accelerated, the share of USD IG issuance rated AAA/AA has climbed to the highest since 2016—but A and BBB rated issuers still dominate the market

USD IG gross issuance by rating at launch (lhs, $bn) vs. the share of issuance that is rated AAA or AA (rhs, %)

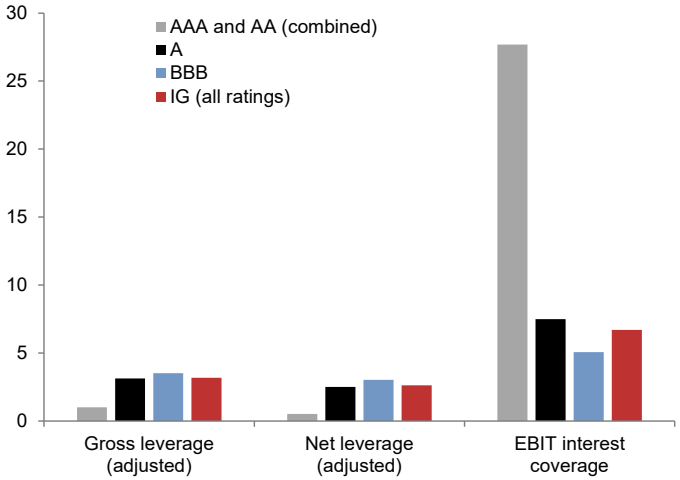


Note: Data as of September 28, 2026. Uses Dealogic Effective Rating at Launch.
Source: Dealogic (ION Analytics), Bloomberg, Goldman Sachs GIR.

We expect these highly rated, AI-focused firms to utilize the significant debt capacity available—irrespective of a higher cost of debt capital—as the risk of *under*investment in AI-related innovation appears greater than *over*investment.

AAA and AA rated firms have significant financial flexibility—and debt capacity—relative to the broader USD IG market

Trimmed mean aggregate credit metrics for the rating-specific cohorts of the USD IG Corporate Index, ratio (x)



Note: Data as of 2Q26. Gross leverage, net leverage, and interest coverage metrics use trimmed mean, which excludes the top 10% and bottom 10% of observations.
Source: Bloomberg, Goldman Sachs GIR.

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Q&A: impact of higher rates on financials

Banks

Q: What are the implications of the recent rise in rates across the curve for banks?



Richard Ramsden, Head of Financials Equity Research Group

A: Higher rates impact banks through three primary channels: 1) net interest income (NII), 2) credit quality, and 3) mark-to-market changes in bank securities' fair value.

Net interest income (NII): We expect higher rates to have an overall positive impact on US banks' NII given their asset-sensitive balance sheets, where asset yields reprice faster than liabilities. Higher long rates accelerate fixed-rate asset repricing as maturing loans and securities roll over into higher coupon assets. While higher short-term interest rates put upward pressure on funding costs, for most banks this is more than offset by the repricing of floating-rate loans and securities, as well as higher yields on cash held at central banks. Higher interest rates can also weigh on loan growth given that elevated borrowing costs adversely impact demand. However, the historical relationship between changes in rates and loan growth is relatively weak, with consumer spending and corporate confidence generally the larger drivers of loan demand.

Credit quality: Higher interest rates can increase delinquency and default risks, particularly in commercial real estate (CRE), where maturing debt is refinanced into higher coupon debt. However, a significant deterioration in unemployment or corporate bankruptcy rates has historically driven meaningful increases in nonperforming loans (NPLs), neither of which has occurred so far in this cycle. Accordingly, we do not expect the recent increase in rates to materially impact credit quality unless it is accompanied by broader economic deterioration.

Changes in securities values: Higher rates will also result in negative marks on banks' securities portfolios, decreasing their capital positions. However, we expect the impact of this shift to be manageable this quarter. We estimate that marks for available-for-sale (AFS) portfolios will decline by ~0.6%, resulting in a ~13bps reduction to CET1 ratios and ~8bps hit to tangible common equity-to-tangible assets (TCE/TA). Most banks should retain excess capital even after accounting for these losses. That said, the risk of further rate increases, alongside continued growth in bank balance sheets, could lead some banks to moderate buybacks in coming quarters to ensure that they maintain capital cushions. Higher rates also impact held-to-maturity (HTM) securities portfolios, which are larger than AFS portfolios. However, these losses do not impact a bank's capital position and pose limited risk to the banking system unless material deposit outflows force banks to realize losses in their HTM book. So far, deposit flows into the banking industry have remained positive (+6% YoY and 1% QoQ over the last 3 months) and HTM losses remain meaningfully below 2023 levels, reflecting both shorter securities duration across banks (the duration has fallen from 4.8 years to 3.7 years since 2023) and smaller overall HTM portfolios.

Q: Does the above calculus shift at all for smaller/regional banks?

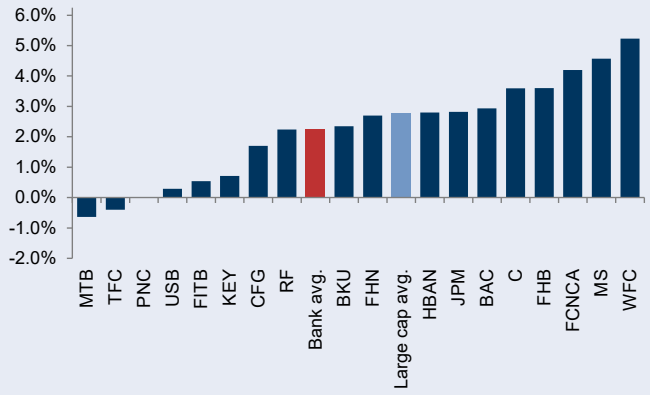
A: Regional banks are more sensitive to interest rate movements than large-cap banks because they generally have less diversified revenue streams and therefore rely more heavily on spread income. By contrast, large banks typically generate a greater share of their revenues from capital markets and wealth management businesses, reducing their rate sensitivity.

Q: Where do you see the most attractive investment opportunities if rates remain elevated?

A: In a higher rate environment, we prefer banks with high-quality deposit bases as those deposits are largely operational and not yield-seeking and therefore face limited upward pressure from higher rates. We also favor banks positioned to benefit from higher long-term rates as fixed-rate assets reprice.

Banks remain mostly asset sensitive

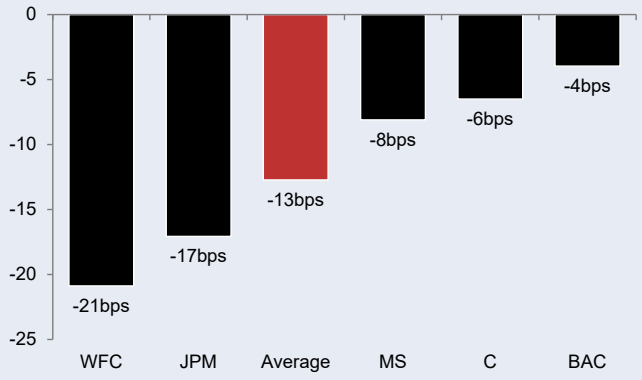
Estimated NII sensitivity to a +200bps parallel shift in rates



Source: Company data, Goldman Sachs GIR.

We estimate a ~13bps hit to CET1 ratios through Accumulated Other Comprehensive Income (AOCI) from AFS marks

Impact on CET1, bps



Source: Company data, Goldman Sachs GIR.

Private markets



Alexander Blostein, US Financials Equity Research Analyst

Q: What are the implications of the recent rise in rates across the curve for private markets?

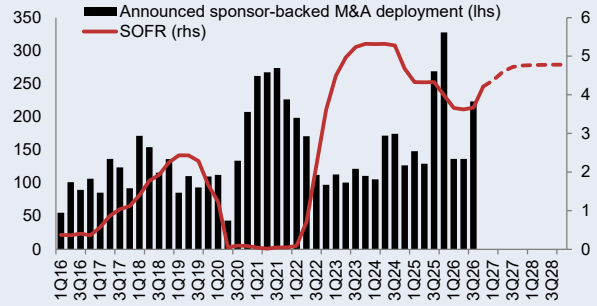
A: The rise in rates is generally negative for private markets, though the implications vary by asset class and, ultimately, the level of rates. At a high level, higher rates across the curve typically result in (1) lower levels of monetization and deployment activity in private equity, real estate, and the direct lending part of private credit (loans to sponsor-backed portfolio companies) and (2) potential headwinds to portfolio valuations, particularly in growth-oriented private equity assets and real estate.

Most private equity assets are funded with floating-rate loans, so the short-end of the curve tends to have a larger impact on financial sponsors' deployment and realization activity. When it comes to valuations, we believe much of this risk is hedged out, reducing the ultimate impact on portfolio companies' cash flows and investment performance, but higher-growth companies where most of the value is predicated on terminal value could be more negatively impacted.

In private real estate, the long end of the curve tends to matter more for both valuation and activity rates. Direct lending could also see slower activity rates as it is directly linked to private equity capital deployment. And higher borrowing costs from floating rates could weigh on cash flows and credit performance, though the absolute risk of a ~100bp move would be manageable.

We expect broad M&A deployment volumes to remain relatively range-bound amid higher rates

Announced sponsor-backed M&A deployment (lhs, $bn) vs. SOFR (rhs, %)



Source: Dealogic, Refinitiv Eikon, Goldman Sachs GIR.

Based on what we have seen so far across both the long and the short end of the curve (including market expectations for future rate hikes), we see the slowdown in private market activity as relatively temporary, with rate volatility, as opposed to the currently higher absolute level of rates, driving incremental uncertainty.

Q: If the rise in rates continues, at what point would you become more concerned about the outlook for private markets?

A: Rates rising +/- 100bps across the curve should not be a material headwind to private markets, and we expect activity to resume once the level of rates stabilizes. However, if rates rise significantly more, we expect more concerns around the valuation of longer-duration assets, such as real estate and growth-oriented private equity. A more prolonged hit to exit activity would also further delay the return of capital to investors (which has been running around two years behind typical timelines), leading to lower fundraising outlooks.

Q: What do you believe investors may be too concerned about, or underappreciating, about how higher yields could impact private markets?

A: Although higher rates are a net negative to private markets, the absolute level of rates matters. So far, the rise in yields of around +/-100bps has been relatively manageable, and activity rates are likely to resume once rates volatility stabilizes. Moreover, credit spreads remain tight by historical standards, and economic growth (a larger driver of value across private equity) remains on a strong footing. We think investors are likely overestimating the fundamental damage of higher rates on alternative managers, leading to stock price selloffs of >20% that represent a significant overshoot relative to the low-to-mid single-digit risks to 2027 earnings that we forecast in our base case. We note that the industry has spent the better part of the last three years operating in a higher rate environment, during which time fundraising, deployment, and realization activity have all experienced periods of pressure but continued to grow on a multi-year basis.

We think investors are also underappreciating the robust structural tailwinds that help offset macro headwinds, including (1) the earnings lever of the AI capex cycle across infrastructure, private credit, and syndication businesses; (2) the highly resilient fundraising trends; (3) the stabilization of retail alternative channels, where private credit redemption requests have peaked and are beginning to moderate alongside strong momentum in non-credit vehicles; (4) potentially stronger returns from newer vintages in private credit/direct lending amid higher base rates (since loans are floating rate); and (5) a potential acceleration in secondary market activity as LP liquidity is further pushed out by slower realizations.

Q: Where do you see the most attractive investment opportunities if rates remain near current levels or continue to rise?

A: If rates remain at current levels or move modestly higher, we would continue to see the most attractive opportunity set in infrastructure, private credit, and secondaries. In infrastructure and private credit, the secular investment cycle around AI-related power, energy, and digital infrastructure remains one of the most significant capital deployment opportunities in private markets today, supporting both deployment volumes and earnings growth for managers with relevant exposure. And in secondaries, depressed realizations and persistent pressure on distributed to paid-in capital (DPI) continue to drive a favorable supply-demand backdrop, creating opportunities to acquire assets at attractive discounts to net asset value.

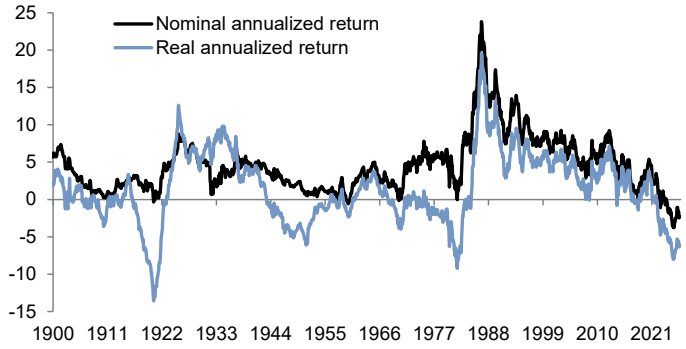
When to yield to bonds

Christian Mueller-Glissmann and Andrea Ferrario argue that higher yields strengthen the strategic case for bond allocations

Over the past five years, US 10-year bonds delivered the worst return in over a century—with real returns declining to levels seen only in the aftermath of WWI and WWII and in the 1970s—as global bond yields have risen sharply toward their long-run average from unusually low levels during the post-Global Financial Crisis (GFC) cycle (at 5.2%, the US 10-year yield is modestly above its 4.7% average over the past 250 years). These higher starting yields should improve carry, lift medium- to long-term total returns, and provide a larger portfolio buffer against further yield increases. This argues for a larger strategic allocation to bonds than over the past five years, but we do not expect an immediate return to the 60/40 portfolio, as strong earnings growth, sticky inflation, and capital competition still favor equities over bonds, at least for now.

US bonds recently had one of the worst returns in a century

5-year rolling returns of 10y Treasury bonds, %



Source: Haver Analytics, Robert Shiller, Goldman Sachs GIR.

Finding the right balance

The optimal equity/bond mix has varied materially over time, largely as a function of macro conditions. Since 1950, the 60/40 portfolio has been close to optimal on average, but the optimal mix has shifted significantly in recent decades. From the GFC to the Covid-19 pandemic, the optimal equity/bond allocation was closer to 40/60. But over the past five years, it has shifted to 100% equities, as most investors would have been better off pairing equities with cash than holding a balanced portfolio.

An improving tactical and strategic case for bonds…

However, higher yields are shifting the calculus for bonds, but their attractiveness depends on the investment horizon. In the near term, monetary policy and energy prices should remain the key drivers of equities and bonds. Further yield increases—particularly if driven by higher energy prices or more aggressive central bank tightening—would likely weigh more on equities than bonds, with increases above two standard deviations over three months historically triggering particularly large equity declines. Conversely, lower energy prices, continued core inflation normalization, and less hawkish central banks should help bonds stabilize into year-end and support equities.

Over medium- to long-term horizons, higher starting yields should improve carry and buy-and-hold returns while also giving bonds better asymmetry as the potential gains from rate relief now exceed the likely losses from a further rise in yields.

Accordingly, the case for higher strategic bond allocations is stronger now than in the past five years, and we see the optimal share of bonds in portfolios normalizing back toward the long-run optimal range of 35-55%.

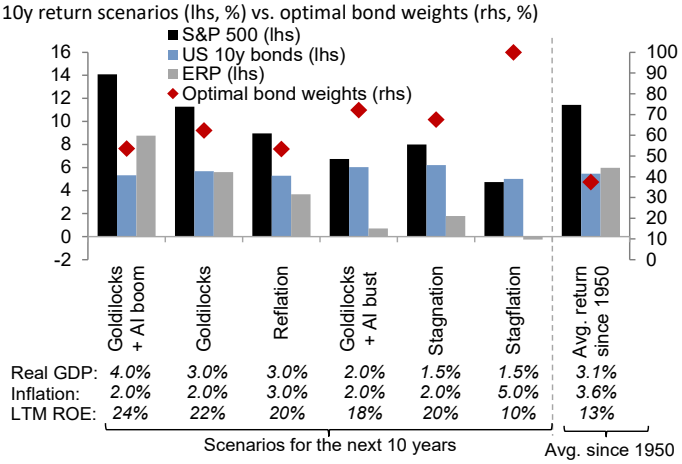
…but not quite a 60/40 balance yet

However, we do not expect a return to the traditional 60/40 portfolio anytime soon as strong earnings growth still favors equities over bonds in the near and medium term. Prospective global equity carry—the starting dividend yield plus subsequent change in earnings—has increased sharply this year, supported by the AI investment boom. Our equity strategists expect slower but still strong earnings growth next year, which should keep risk-adjusted equity returns attractive relative to bonds, even as relative returns moderate. That said, the case for higher bond allocations would strengthen on a 6-12m horizon if AI-driven earnings tailwinds fade or growth slows materially.

For now, bonds' diversification benefits are also likely to remain weaker than during the post-GFC period as equity/bond correlations should be less negative with less firmly anchored inflation. And even over a 10-year horizon, equities are likely to drive most of the variation across portfolio outcomes.

Structural regimes point to different long-term return forecasts

10y return scenarios (lhs, %) vs. optimal bond weights (rhs, %)



Source: Haver Analytics, Robert Shiller, Goldman Sachs GIR.

Beyond bonds for balanced portfolios

As the case for more "normal" strategic allocations to bonds grows, investors should favor the parts of the curve that offer the best carry vs. convexity balance, such as the belly of the curve. Rates receiver swaptions on shorter-dated rates can also provide a more convex hedge for recession risk as well as lower energy prices and rates relief.

Bond-like equities are also a valuable source of diversification. Correlations between the S&P 500 and both low-volatility and high-dividend stocks have fallen sharply, while the proportion of stocks with a negative beta to the market has risen, as it did around the Tech Bubble. Conversely, equity-like bonds such as convertibles can improve portfolio asymmetry, as embedded call option provides upside exposure to equities, while their relatively low duration limits sensitivity to rising bond yields.

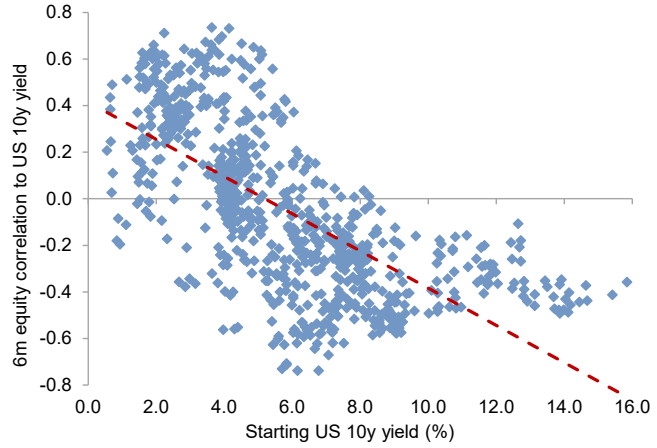
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Multi-asset implications of rising rates

US bond yields and equities tend to become more negatively correlated as bond yields rise...

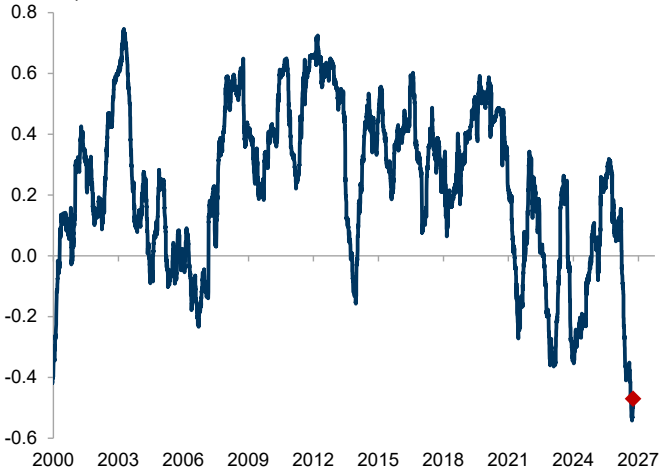
Starting US 10y yield vs. 6m correlation of S&P 500 index to US 10y Treasury yield



Note: Data since 1962.
Source: Bloomberg, Haver Analytics, Goldman Sachs GIR.

...with equity-bond yield correlations turning sharply negative amid the latest rise in yields...

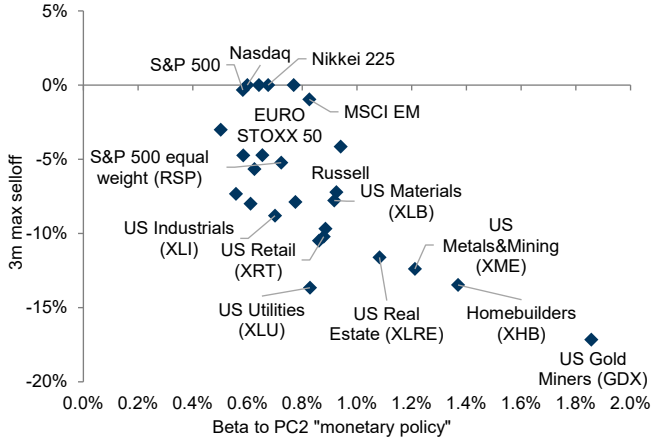
6m correlation of S&P 500 index and 10y Treasury yield (weekly returns)



Source: Datastream, Goldman Sachs GIR.

...and the most rate-sensitive equity market sectors selling off sharply in recent months

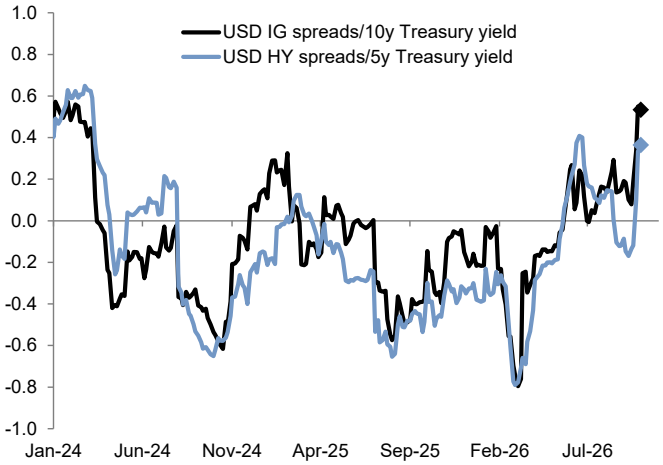
3y beta to "monetary policy" vs. 3m max selloff



Note: Beta is taken from a multivariate regression on both the PC1 "Global growth" and PC2 "Monetary policy" factors.
Source: Datastream, Bloomberg, Goldman Sachs GIR.

Rising yields have also pushed USD credit spreads wider as higher borrowing costs lead investors to demand a larger risk premium, with the correlation now firmly positive

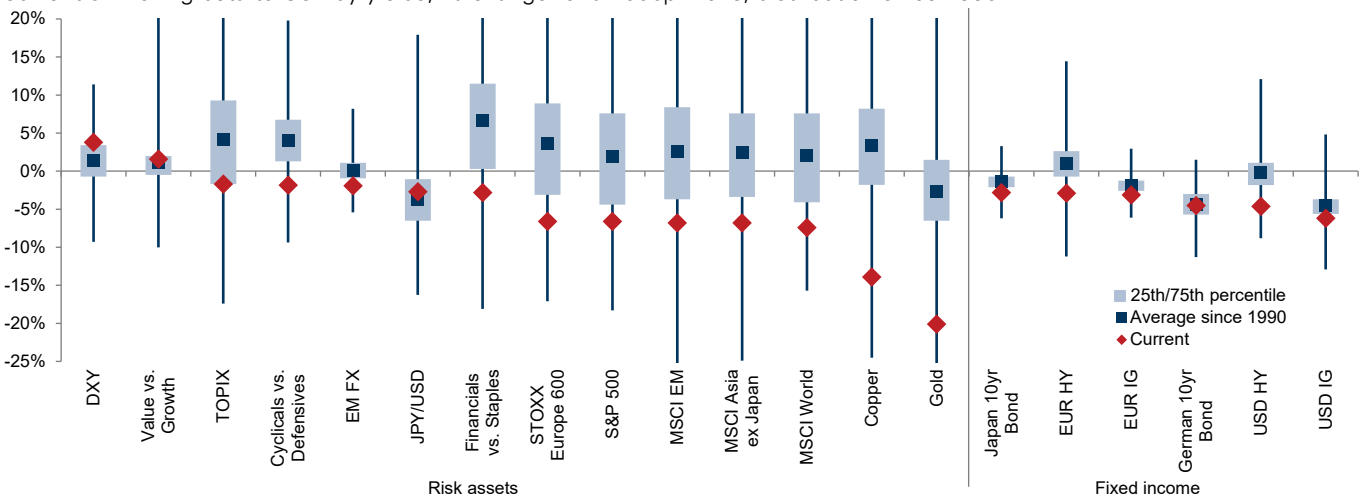
Rolling 3m correlation of USD credit spreads to bond yields



Source: Bloomberg, Goldman Sachs GIR.

More broadly, fixed income assets (unsurprisingly) tend to suffer when bond yields rise, while the impact on risk assets varies and can depend on the drivers of higher yields, though the latest increase has broadly weighed on risk assets

Current 3m rolling beta to US 10y yields, % change for a 100bp move, distribution since 1990



Source: Datastream, Goldman Sachs GIR.

Special thanks to GS Asset Allocation Strategist Andrea Ferrario and GS Global Credit Strategist Shamshad Ali for charts.

Q&A on US Treasury buybacks

Q: What are Treasury buyback operations and how do they work?

A: The Treasury conducts buyback operations generally once or twice a week by purchasing off-the-run Treasury securities (securities that are not the most recently issued) from primary dealers and other select market participants up to a pre-announced cap. While the Treasury is under no obligation to buy back any securities, it often purchases the maximum amount. The Treasury funds these purchases by issuing additional short-term debt, effectively removing longer-term debt from the market and replacing it with shorter-term debt.

Q: Why does the Treasury conduct such operations?

A: The purpose of buyback operations is to bolster market liquidity by establishing a "regular and predictable" opportunity for market participants to sell off-the-run securities. Bonds tend to become less liquid as they age, so by offering periodic opportunities for dealers to reduce inventories of these securities, the Treasury helps ensure sufficient liquidity and supports overall market functioning across the maturity spectrum.

Q: What did the US Treasury recently announce regarding its buyback operations?

A: On August 19, the Treasury announced that it would at least double the size of its buyback operations for longer-dated government debt (the 10- to 20-year and 20- to 30-year maturity buckets), from $2bn/operation to at least $4bn/operation. The announcement was notable given that the buyback schedule is typically set at each Quarterly Treasury Refunding—the Treasury's quarterly announcement of its borrowing strategy and upcoming debt auction sizes—making a mid-quarter change to operation sizes unusual.

Q: How meaningful is the announced increase in buybacks?

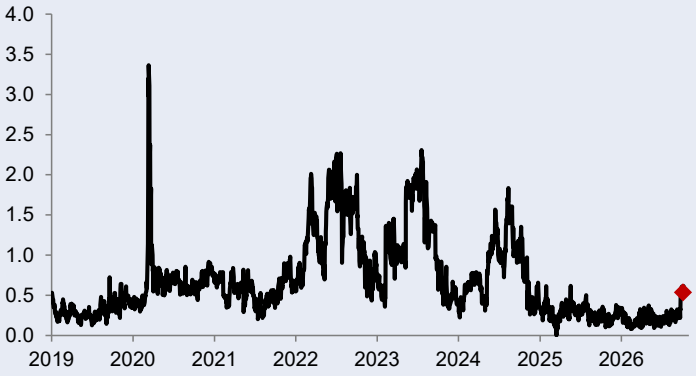
A: While the expanded operation is meaningful from a long-end supply perspective—at the new upsized amount, long-end buybacks should offset at least 30% of gross 10-30 year issuance annually, and 45% if the Treasury uses the entire $6bn cap it has announced so far for each buyback operation—it is small relative to overall Treasury supply. The stock of US Treasury debt with at least 10 years to maturity stands at around $4tn, or around 15% of privately held debt outstanding. While the larger buybacks should keep that share roughly stable over the coming years following several decades of steady increases, they will likely have a much more marginal impact on the total debt outstanding as a share of GDP.

Q: Why did the Treasury increase the operation size?

A: The Treasury cited liquidity support as the justification, pointing to the "significant volume of high-quality offers it routinely receives in longer-dated buyback operations" as evidence of strong dealer demand to sell older securities. That said, we see little evidence that yields have risen due to poor liquidity or an oversupply of long-end Treasuries, with macro fundamentals the main drivers of the recent bond selloff (see pgs. 10-11). The more likely motivation for the increase in buybacks looks to be a desire to temper the rise in longer-term bond yields. That said, the Treasury's purchase of over $4bn of long-dated bonds during each of the last three buyback operations has repeatedly fallen short of the $6bn cap—implying that it is not willing to purchase bonds regardless of price.

The dislocation of the Treasury curve vs. a modeled fitted curve remains relatively contained, signaling little evidence that higher yields have been due to poor liquidity or an oversupply of long-end Treasuries

US Treasury curve dislocations from model fitted values



Source: Goldman Sachs GIR.

Q: Has the increase in buybacks impacted the level of bond yields?

A: The larger buybacks have had very little impact on the level of yields so far, for a few key reasons. First, as mentioned, the shift in supply is relatively small compared to the overall size of the Treasury market. Our typical rule of thumb for the impact of duration removal on the level of yields argues for a relatively modest impact. We estimate that the Fed's 2011 Operation Twist program lowered 10-year yields by around 4bp per percent of GDP. Assuming a similar impact today, the buybacks would imply around a 1bp decline in 10-year yields levels. Second, as mentioned, the way in which the Treasury has conducted the operations suggests it remains a discerning—not indiscriminate—buyer. Third, and most importantly, removing long-end supply does little to address the macro fundamentals that have driven the selloff.

Q: What else can the Treasury do if it wants to lower yields?

A: The Treasury can potentially ease pressures on long-end bonds by reducing long-end auction sizes or doing away with 20-year issuance. Treasuries in other major economies, such as the UK and Japan, recently took similar actions, which had some impact—in Japan, it supported a reduction in long-end risk premium —but in both cases the efforts have failed to arrest the macro pressures that have taken long-end yields to cycle highs. Similarly, in the US, with the latest leg of the bond selloff having affected both long- and medium-term yields, any issuance adjustments at the long end may have only a limited impact.

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Summary of our key forecasts

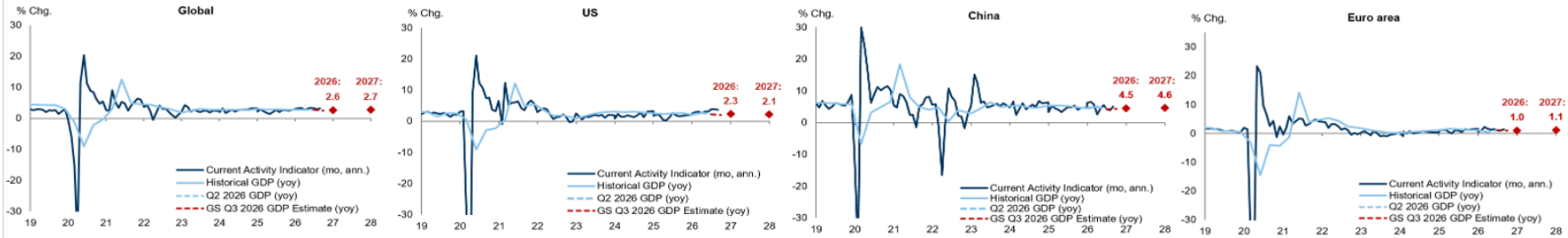
GS GIR: Macro at a glance

Watching

- **Globally,** we expect real GDP growth to slow to 2.6% yoy in 2026 amid headwinds from higher energy prices from the Middle East conflict. We expect global core inflation to end the year at 2.8%, reflecting a further normalization in shelter and wage inflation but a boost from energy price passthrough and lingering but fading tariff effects.
- **In the US,** we expect real GDP growth of 2.3% on a Q4/Q4 basis in 2026, largely reflecting the continued boost from the AI boom. We expect core PCE inflation to fall to 3.0% in 2026 on a Q4/Q4 basis as the effects of tariffs, energy price passthrough, and AI demand begin to fade, with it falling closer to 2% in 2027 as those effects subside further. We expect the unemployment rate to end 2026 at 4.2%.
- **We expect the Fed** to deliver one more 25bp hike in December to a peak policy rate range of 4.00-4.25%.
- **In the Euro area,** we expect real GDP growth of 1.0% on a Q4/Q4 basis in 2026, reflecting headwinds from elevated energy prices as well as subdued consumer confidence, though we expect growth to improve next year. We expect core inflation to rise to a peak of 2.7% yoy in 1Q27, reflecting energy price passthrough, before gradually declining to 2.0% by end-2028.
- **We expect the ECB** to deliver one more 25bp hike in December to a peak policy rate of 2.75%.
- **In China,** we expect real GDP growth of 4.5% yoy in 2026, reflecting broad-based export strength but headwinds from weak domestic demand. We expect CPI/PPI inflation to rise to 1.0%/2.0% yoy this year, largely owing to commodity price passthrough.
- **WATCH MIDDLE EAST CONFLICT.** The situation in the Middle East remains fluid and we will be closely watching how the region's energy exports as well as demand for its energy from key consumers like China continue to evolve.

Goldman Sachs Global Investment Research.

Growth



Source: Haver Analytics, Goldman Sachs Global Investment Research.
Note: GS CAI is a measure of current growth. For more information on the methodology of the CAI please see "Technical Updates to Our Global CAIs," Global Economics Comment, Sep. 01, 2025.

Forecasts

Economics							Markets								Equities								
GDP growth (%)	2026				2027		Interest rates 10Yr (%)	Last	E2026	E2027	FX		Last	3m	12m	S&P 500	E2026		E2027		Returns (%)	12m	YTD	E2026 P/E
	GS (Q4/Q4)	Cons. (Q4/Q4)	GS (CY)	Cons. (CY)	GS (CY)	Cons. (CY)											GS	Cons.	GS	Cons.				
Global	2.6	--	2.6	2.6	2.7	2.7	US	5.24	4.75	4.40	EUR/$		1.12	1.14	1.12	Price	8,000				S&P 500	11.4	14.0	22.2x
US	2.3	2.1	2.3	2.2	2.1	2.2	Germany	3.47	3.25	3.25	GBP/$		1.32	1.33	1.28	EPS	$375	$366	$415	$423	MXAPJ	28.8	20.4	12.3x
China	4.4	4.5	4.5	4.6	4.6	4.4	Japan	3.01	3.00	2.75	$/JPY		159	158	150	Growth	36%	33%	11%	15%	Topix	12.1	20.4	19.7x
Euro area	1.0	1.1	1.0	1.0	1.1	1.3	UK	5.26	5.00	4.85	$/CNY		6.68	6.70	6.40						STOXX 600	9.1	7.6	15x
Policy rates (%)	2026				2027		Commodities	Last	4Q26	1Q27	Credit (bp)		Last	4Q26	1Q27	Consumer	2026		2027		Wage Tracker 2026 (%)			
	GS	Mkt.			GS	Mkt.											CPI (%, yoy)	Unemp. Rate	CPI (%, yoy)	Unemp. Rate	Q1	Q2	Q3	Q4
US	4.13	4.14			3.63	4.68	Nat Gas, NYMEX ($/mmBtu)	3.17	2.90	2.95	USD	IG	81	82	80	US	3.4	4.2	2.4	4.1	3.5	3.6	3.5	--
Euro area	2.75	2.66			2.50	3.17	Nat Gas, TTF (EUR/MWh)	83.90	70	44		HY	308	295	290	Euro area	3.0	6.3	3.0	6.2	--	--	--	--
China	1.40	1.40			1.30	--	Copper ($/mt)	14,792	13,700	13,800	EUR	IG	108	88	86	China	1.0	--	1.0	--	--	--	--	--
Japan	1.25	1.43			1.75	2.15	Gold ($/troy oz)	4,172	4,605	4,765		HY	323	315	310									

Source: Bloomberg, Goldman Sachs Global Investment Research. For important disclosures, see the Disclosure Appendix or go to www.gs.com/research/hedge.html.

Market pricing as of October 12, 2026

Glossary of GS proprietary indices

Current Activity Indicator (CAI)

GS CAIs measure the growth signal in a broad range of weekly and monthly indicators, offering an alternative to Gross Domestic Product (GDP). GDP is an imperfect guide to current activity: In most countries, it is only available quarterly and is released with a substantial delay, and its initial estimates are often heavily revised. GDP also ignores important measures of real activity, such as employment and the purchasing managers' indexes (PMIs). All of these problems reduce the effectiveness of GDP for investment and policy decisions. Our CAIs aim to address GDP's shortcomings and provide a timelier read on the pace of growth.

For more, see our CAI page and Global Economics Comment: Technical Updates to Our Global CAIs.

Dynamic Equilibrium Exchange Rates (DEER)

The GSDEER framework establishes an equilibrium (or "fair") value of the real exchange rate based on relative productivity and terms-of-trade differentials.

For more, see our GSDEER page, Global Economics Paper No. 227: Finding Fair Value in EM FX, 26 January 2016, and Global Markets Analyst: A Look at Valuation Across G10 FX, 29 June 2017.

Financial Conditions Index (FCI)

GS FCIs gauge the "looseness" or "tightness" of financial conditions across the world's major economies, incorporating variables that directly affect spending on domestically produced goods and services. FCIs can provide valuable information about the economic growth outlook and the direct and indirect effects of monetary policy on real economic activity.

FCIs for the G10 economies are calculated as a weighted average of a policy rate, a long-term risk-free bond yield, a corporate credit spread, an equity price variable, and a trade-weighted exchange rate; the Euro area FCI also includes a sovereign credit spread. The weights mirror the effects of the financial variables on real GDP growth in our models over a one-year horizon. FCIs for emerging markets are calculated as a weighted average of a short-term interest rate, a long-term swap rate, a CDS spread, an equity price variable, a trade-weighted exchange rate, and—in economies with large foreign-currency-denominated debt stocks—a debt-weighted exchange rate index.

For more, see our FCI page, Global Economics Analyst: Our New G10 Financial Conditions Indices, 20 April 2017, and Global Economics Analyst: Tracking EM Financial Conditions – Our New FCIs, 6 October 2017.

Goldman Sachs Analyst Index (GSAI)

The US GSAI is based on a monthly survey of GS equity analysts to obtain their assessments of business conditions in the industries they follow. The results provide timely "bottom-up" information about US economic activity to supplement and cross-check our analysis of "top-down" data. Based on analysts' responses, we create a diffusion index for economic activity comparable to the ISM's indexes for activity in the manufacturing and nonmanufacturing sectors.

Macro-Data Assessment Platform (MAP)

GS MAP scores facilitate rapid interpretation of new data releases for economic indicators worldwide. MAP summarizes the importance of a specific data release (i.e., its historical correlation with GDP) and the degree of surprise relative to the consensus forecast. The sign on the degree of surprise characterizes underperformance with a negative number and outperformance with a positive number. Each of these two components is ranked on a scale from 0 to 5, with the MAP score being the product of the two, i.e., from -25 to +25. For example, a MAP score of +20 (5;+4) would indicate that the data has a very high correlation to GDP (5) and that it came out well above consensus expectations (+4), for a total MAP value of +20.

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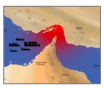
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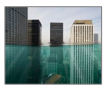
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Disclosure Appendix

Reg AC

We, Allison Nathan, Jenny Grimberg, Ashley Rhodes, Joseph Briggs, Amanda Lynam, William Marshall, David Mericle, Christian Mueller-Glissmann, Ben Snider, Andrea Ferrario, Isabella Rosenberg, Pierfrancesco Mei, and Johan Allen, hereby certify that all of the views expressed in this report accurately reflect our personal views, which have not been influenced by considerations of the firm's business or client relationships.

We, Richard Ramsden and Alexander Blostein, CFA hereby certify that all of the views expressed in this report accurately reflect our personal views about the subject company or companies and its or their securities. We also certify that no part of our compensation was, is, or will be, directly or indirectly, related to the specific recommendations or views expressed in this report.

Unless otherwise stated, the individuals listed on the cover page of this report are analysts in Goldman Sachs' Global Investment Research division.

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Goldman Sachs Research

TOP *of* MIND

ASSESSING A LESS TRANSPARENT FED

Fed Chairman Kevin Warsh is steering the Fed into a less transparent era. So, is that a positive or negative development, and what could it mean for markets and the economy? We speak with Former Fed governors Donald Kohn and Stephen Miran as well as GS' Jan Hatzius who agree that rigid "Odyssean" forward guidance that commits policymakers to a pre-set path risks policy mistakes and is therefore rarely appropriate. But they somewhat disagree about the benefits of greater transparency around the Fed's reaction function. Hatzius and Kohn argue that such transparency helps markets better anticipate and price in policy changes, accelerating monetary policy transmission. But Miran believes that the line between committing to a path and being transparent is clearer in theory than in practice. And while GS economists and strategists find that reduced transparency leads to increased market volatility, the question is whether this just increases the market's noise or also improves the market's signal. Hatzius argues largely for the former and Miran for the latter.



INTERVIEWS *with…*

Jan Hatzius, Chief Economist and Head of Global Investment Research, Goldman Sachs

"Markets price what they think the Fed will do, not what they think the Fed should do. That will remain true even if the Fed obscures its reaction function… market guesses about the Fed's next moves will simply become worse, leading to more volatility... and importantly, volatility that serves no constructive economic purpose."

Donald Kohn, former Vice Chairman of the Board of Governors of the Federal Reserve System, Robert V. Roosa Chair in International Economics and Senior Fellow, Brookings Institution

"I sympathize with Chairman Warsh to some extent. [But] the real problem with Warsh's approach is his reluctance to articulate how the FOMC sees the economy evolving… explaining the Fed's thinking about the economic outlook is essential in helping markets act as a stabilizing force and is a core component of accountability."

Stephen Miran, former Member of the Federal Reserve Board of Governors and Chairman of the Council of Economic Advisers, Senior Strategist, Hudson Bay Capital Management

"Reducing guidance generates more volatility but also more signal... I'd rather have that signal, even if it's messy, than throw the signal out by anchoring market expectations to the Fed's near-term projections."

ALSO *inside…*

Transparency: more benefits than costs | Joseph Briggs and Megan Peters, GS Global Economics Research

A less transparent Fed and US rates | William Marshall, GS Rates Strategy Research

Fed communication and FX volatility | Michael Cahill and Lexi Kanter, GS FX Strategy Research

Fed: more transparent than most | Joseph Briggs and Megan Peters, GS Global Economics Research

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Macro news and views

We provide a brief snapshot on the most important economies for the global markets

US

Latest GS proprietary datapoints/major changes in views

- No major changes in views.

Datapoints/trends we're focused on

- US consumer spending, which we expect to slow, with risks skewed to the downside as the Strait of Hormuz remains closed and further rises in gas prices would hurt consumers.
- US labor market; following weak July payroll growth, we estimate that the underlying pace of job growth now stands at 5k, well below our estimate of breakeven job growth.
- US inflation, which has improved meaningfully in the past two months, leaving us comfortable with our view that the Fed will remain on hold through the end of the year.

Japan

Latest GS proprietary datapoints/major changes in views

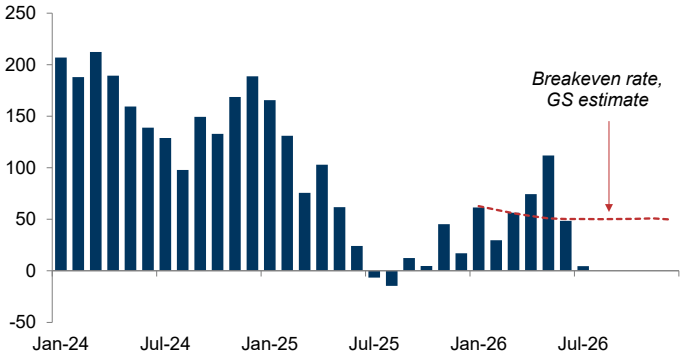
- No major changes in views.

Datapoints/trends we're focused on

- Recent US-Japan coordinated Yen intervention, which we think is unlikely to sustainably strengthen the Yen.
- BoJ policy; we expect the BoJ to continue hiking gradually, with the next hike in January 2027, but the risks are skewed toward an earlier rate hike, potentially as early as October.
- Japan consumption tax cut; while we think the tax cut will have an only limited impact on growth, it could materially worsen Japan's longer-term debt outlook.

US job growth: well below breakeven

Underlying US trend job growth, GS estimate*, thousands/month

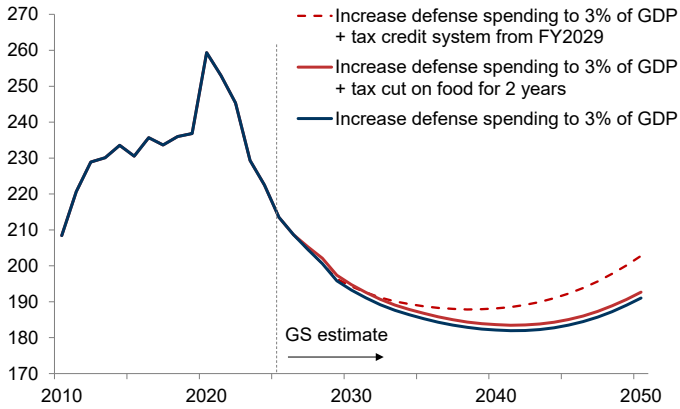


*We estimate underlying trend job growth as 0.75*3-month average payroll growth + 0.25*9-month average household employment growth.*

Source: US Bureau of Labor Statistics, Haver Analytics, Goldman Sachs GIR.

Japan's debt-to-GDP: an upward sloping trajectory

Japan public debt-to-GDP ratio, % of GDP



Source: Cabinet Office, Goldman Sachs GIR.

Europe

Latest GS proprietary datapoints/major changes in views

- No major changes in views.

Datapoints/trends we're focused on

- ECB policy; with core inflation only modestly above target, we maintain that the ECB will hike by 25bp in September but that the next move will be a cut (in mid-2027).
- 2027 European elections, including in France, Italy, Spain, and Greece, which risk producing hung parliaments.
- European defense spending, which has risen significantly but unevenly, with Germany driving much of the increase.
- BoE policy mandate; we think adding economic activity to the BoE's mandate would have dovish implications.

Emerging Markets (EM)

Latest GS proprietary datapoints/major changes in views

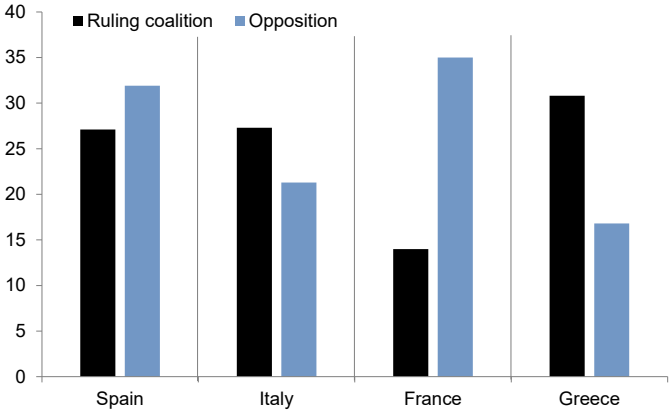
- No major changes in views.

Datapoints/trends we're focused on

- The AI boom, which is driving large trade surpluses in North Asia that are spilling over into stronger domestic activity.
- China's property downturn; we find the housing drag on the economy has diminished and should mostly fade by 2028.
- A historic opportunity for RMB bond internationalization, though we think sustaining gains will require broader structural changes.
- India's macro conditions, which are showing signs of cyclical improvement after a string of adverse shocks.

Europe: ruling parties losing share

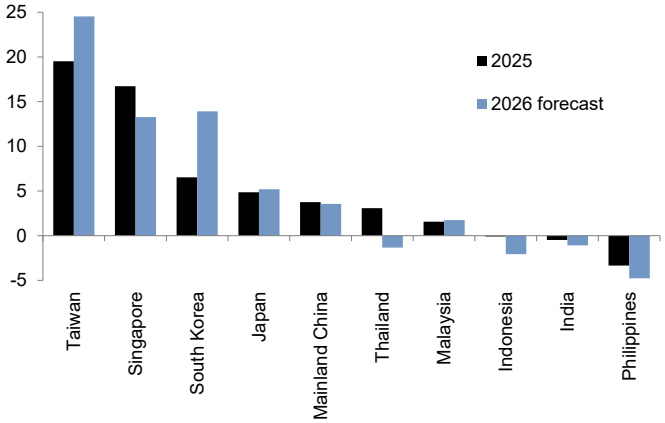
Main parties in latest opinion polls by country (as of July 2026), %



Source: Politico, Goldman Sachs GIR.

Super surpluses in North Asian tech exporters

Asia current account balances, % of GDP



Source: Goldman Sachs GIR.

Assessing less Fed transparency

New Fed Chairman Kevin Warsh is steering the Fed into a less transparent era characterized by shorter post-meeting statements, less forward guidance, and a more limited role for projections, marking a meaningful break from the transparency revolution that has defined central banking over the last quarter century. So, what could a less transparent Fed mean for markets and the economy?

For answers, we turn to former Fed governors Donald Kohn, now Robert V. Roosa Chair in International Economics and senior fellow at the Brookings Institution, and Stephen Miran, now senior strategist at Hudson Bay Capital Management, as well as Jan Hatzius, GS' Chief Economist and Head of Global Investment Research.

We first ask whether scaling back Fed communication is a positive or negative development. Hatzius says it's likely negative, though he draws a clear line between transparency and forward guidance. He argues that clarity about the Fed's reaction function helps markets better anticipate policy changes and quickly incorporate new information into financial conditions, thereby accelerating monetary policy transmission to the economy. So, he believes that unwinding that transparency would be a mistake. (Note: GS Global Economists Joseph Briggs and Megan Peters lay out the evidence backing this argument on pgs. 14-15.)

But Hatzius takes a more nuanced view of forward guidance, arguing that rigid "Odyssean" guidance that commits policymakers to a pre-set path regardless of new information only makes sense at the effective lower bound for the policy rate, while conditional "Delphic" guidance tied to the economic outlook is generally more defensible. Although markets could confuse the two, he believes that most sophisticated investors understand the difference and is therefore not particularly worried that markets have become overly reliant on the policy dots published in the Fed's Summary of Economic Projections (SEP). Instead, he considers them a "useful source of additional information about the Fed's reaction function" (except for the long-run policy dot). If anything, Hatzius would prefer the dots move in an even more informative direction—by publishing each policymaker's combined projections for growth, inflation, unemployment, and the policy rate on an anonymized basis.

Kohn agrees that clarity about the Fed's "narrative" and reaction function is vital. Explaining how the Fed sees the economy evolving and how policy would respond, he says, is "essential in helping markets act as a stabilizing force and is a core component of accountability." Still, he sympathizes somewhat with Chairman Warsh's view that the dot plot is "not helpful" (see pg. 9 for Warsh's own words on this), arguing that the market's hyper-focus on the SEP's median policy dot "illustrates the risk of excessive precision and false certainty," as small shifts in it can trigger unwarranted market moves. So, he has long favored omitting the median.

Miran, though, is more supportive of Warsh's push for a less transparent Fed. Like Hatzius and Kohn, he's concerned that excessive forward guidance that locks the Fed into a policy path raises the risk of policy errors and also argues that such guidance can impair market pricing of risk. But he's less convinced about the benefits of increased transparency around the Fed's reaction function, making the case that the distinction between the Fed committing to a pre-set policy path and being transparent about its reaction function is clear in theory, but less so in practice. So, he thinks "the policy dots absolutely have to go" but feels less strongly about omitting the SEP's economic projections.

But would less transparency allow markets to provide the Fed with more "direct and unfiltered" information about the economy, as Warsh has argued? The answer seems to boil down to the implications for market volatility, and whether any resulting volatility would simply increase the market's noise or also improve its signal.

As for the volatility implications, the evidence is fairly clear. Briggs and Peters find that increased central bank transparency has led to a reduction in market volatility over the last several decades. This is consistent with GS Head of US Rates Strategy William Marshall's recent observation of the reverse: that the US rates market has already priced substantial uncertainty associated with a less transparent Fed, with the July FOMC representing the largest non-cut meeting day surprise in almost three decades.

But Marshall argues that whether such meeting-day surprises lead to structurally higher rates volatility depends (again) on whether the Fed's reaction function is well understood. If it's not, he would expect higher rates volatility to persist. GS FX Strategists Michael Cahill and Lexi Kanter find a similar pattern in FX: that central bank communication shifts can change which events and data drive FX volatility without necessarily raising it, but less transparency could increase Dollar volatility—all else equal (though global macro conditions remain its biggest driver).

But the key debate seems to be whether the increased volatility resulting from less Fed transparency will improve or worsen the market's signal. Hatzius is clearly in the "worsen" camp. He argues that markets price what they think the Fed *will* do—not what it *should* do—and less transparency won't change that; it would simply produce worse guesses. So, he says that while volatility in and of itself is not necessarily bad, volatility driven by uncertainty about the Fed's reaction function would serve no constructive economic purpose.

Kohn agrees that the lack of a clear Fed narrative could create damaging uncertainty. But he argues that removing overly specific guidance could help free markets to process new information more independently and isn't too worried about modestly higher volatility as a result. And Miran seems most convinced that reducing transparency would improve the markets' signal, with volatility an unavoidable but acceptable cost: "The first best world of signal without volatility doesn't exist. I'd rather have that signal, even if it's messy, than throw the signal out by anchoring market expectations to the Fed's near-term projections."

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Interview with Jan Hatzius

Jan Hatzius is Chief Economist and Head of Global Investment Research at Goldman Sachs. Below, he argues that a shift toward a less transparent Fed is likely a negative development.



Ashley Rhodes: Fed Chairman Kevin Warsh appears to be ushering the Fed into a less transparent era. Is that shift a positive or negative development?

Jan Hatzius: It's likely a negative development, although I would distinguish between transparency and forward guidance. The transparency revolution in central banking over the last 30 years has been extremely successful. When a central bank provides meaningful information about its reaction function—how it's likely to respond to different economic outcomes—markets can better anticipate policy changes. So, new information gets incorporated quickly into financial conditions, accelerating the transmission of monetary policy. The central bank can later decide whether to ratify that move and always retains the option not to. That framework has worked well for the Fed over the past several decades. Unwinding transparency about the reaction function risks delaying the transmission of monetary policy and would be a mistake, in my view.

Forward guidance about the policy rate path is a separate and more nuanced issue. In assessing the pros and cons, it's important to distinguish between "Odyssean" and "Delphic" forward guidance. With Odyssean guidance, like Odysseus tying himself to the mast, the central bank commits to a policy path even if subsequent data point in a different direction. That only makes sense at the effective lower bound for the policy rate, when conventional monetary easing is no longer an option. Even then, the hurdle for Odyssean guidance should be high because there is a real risk that policymakers lock themselves into an inappropriate policy stance. By contrast, Delphic guidance, named for the Oracle of Delphi, is a conditional description of what policymakers expect to do given their economic outlook that will change with new information. Delphic guidance is more defensible, even in normal times.

Ashley Rhodes: Let's take these in turn. Do you agree with the argument that Odyssean forward guidance increases the risk of policy errors?

Jan Hatzius: That argument is a serious one. In terms of practical examples, it is heavily centered on the 2021–2022 episode, when the Fed tightened policy much too late because they made a major forecasting error on inflation at a time of rigid Odyssean forward guidance. That was a bad combination. If the forecast is wrong, policy will be set incorrectly under any communication or transparency framework. And Odyssean forward guidance compounds that problem because it delays the inevitable course correction. This is why Odyssean forward guidance should be used sparingly, if at all.

That said, I am not sure how big a difference to the inflation outcome or to economic performance more broadly it would have made if the Fed had started to tighten in, say, September 2021 rather than March 2022. By early 2022, markets had already recognized that the funds rate would need to rise, and financial conditions tightened substantially on the back of higher rate expectations well before the Fed delivered those hikes. Also, the error was ultimately not disastrous economically; inflation expectations did not de-anchor and the Fed course-corrected without causing a recession.

Ashley Rhodes: Even if the Fed just provides Delphic guidance, are concerns that the market might confuse it with Odyssean guidance warranted, especially given how much weight markets put on the dot plot?

Jan Hatzius: Such confusion would be problematic, but I think it doesn't happen frequently in practice. Most sophisticated observers understand that the policy dots included in the Summary of Economic Projections (SEP) are conditional on data evolving in line with the economic projections. And the dots only provide information about the individual and anonymized views of FOMC participants, including non-voters. So, they aren't strong forward guidance, not even strong Delphic forward guidance. And market expectations have often diverged meaningfully from them, undermining the argument that investors view them as a commitment.

Instead, the dots are a useful source of additional information about the Fed's reaction function. The SEP could be made even more informative by publishing each policymaker's combined projections for growth, inflation, unemployment, and the Fed funds rate on an anonymous basis, which we have long argued for. Doing so would provide a clearer picture of how policymakers' economic forecasts relate to their anticipated policy paths. But that seems unlikely given Warsh's apparent preference for moving in the opposite direction.

Concerns about market confusion are more understandable when it comes to the long-run policy dot, where the Fed has less information to convey. Nobody knows where R-star is—as Chair Powell liked to say, "it is seen by its works"—but markets seem to assume that the Fed has special knowledge and therefore place too much weight on the long-term dot. That can lock down longer-term rate expectations too firmly. That said, removing the long-run dot could complicate the Fed's communication in its current form, as it has become an important benchmark for assessing the stance of monetary policy. I would prefer to eliminate the long-run dot and communicate the stance of policy with greater emphasis on financial conditions. Overall, though, the dot plot works well, and, despite the controversy surrounding it, I remain a fan.

Ashley Rhodes: Aren't the Fed's inflation and employment targets sufficient for the market to infer the Fed's reaction?

Jan Hatzius: No, because getting from point A to point B requires more information than just a stated objective. Take the inflation target, for example. The Fed's goal is 2% inflation, but how do you assess whether inflation is moving sustainably toward that target? How do you determine the current underlying rate of inflation? Should you look at headline PCE, core PCE, trimmed mean inflation, or something else entirely? Do you project future inflation changes based on the

unemployment rate via a Phillips curve relationship, the monetary aggregates, or something else? To anticipate the Fed's moves—and thereby push financial conditions in a stabilizing direction—markets need insight into how the Fed assesses these questions. Those considerations go well beyond stating a broad policy goal.

Ashley Rhodes: Is there *any* merit to communicating less given that Fed officials have, at times, inadvertently made market-moving remarks?

Jan Hatzius: There is certainly an optimal level and frequency of communication. The Fed should not provide running commentary on every data release, especially as markets tend to overemphasize individual data points relative to the longer-term trend. The Reserve Bank presidents speak a lot more frequently than the governors and their comments are almost by definition less informative about the policy path than those of the Fed leadership. While that can complicate the public's understanding of the policy outlook, it can also be informative for market participants who nevertheless understand that the views of the troika—the Fed chairman, Fed vice chair, and New York Fed president—carry the greatest weight. On net, it's therefore unclear whether noise from frequent policymaker comments has had big practical costs. More broadly, communication mistakes are inevitable, and policymakers will say things they later wish they hadn't. But the question is whether these occasional mistakes outweigh the benefits from transparent communication. In my view, they don't.

Ashley Rhodes: Warsh believes that providing less guidance about the Fed's reaction function will allow policymakers to draw more "direct and unfiltered" information from markets. So, you don't agree?

Jan Hatzius: I don't. Markets price what they think the Fed *will* do, not what they think the Fed *should* do. That will remain true even if the Fed obscures its reaction function. So, if the Fed chooses to do so, market guesses about the Fed's next moves will simply become worse, leading to more volatility in rates markets and in financial conditions—and importantly, volatility that serves no constructive economic purpose. Volatility is not bad in and of itself. For example, if new data shows increased inflation risks, financial conditions should tighten as markets price in rate hikes. This accelerates the transmission of monetary policy. But volatility driven by uncertainty about the Fed's reaction function is more random and doesn't necessarily push financial conditions in a stabilizing direction.

Also, policymakers can already extract valuable information from market indicators. For example, a sharp rise in breakeven inflation above target is a statement that markets view the expected policy path as inappropriately loose. So, the information is already available without the Fed saying less.

Ashley Rhodes: Is there a risk of a "hall of mirrors" effect if the Fed uses market prices to not only glean information about the economy but also to guide what it should do?

Jan Hatzius: Yes, but I am not particularly concerned about this risk. While a "hall of mirrors" effect—whereby markets watch the Fed for clues, but the Fed watches the markets for clues—could push the policy stance far away from desirable levels, the likelihood of that scenario materializing is low. Warsh has said that he does not want to "be constrained or take verbatim from what the market's doing." And Fed officials, including the Chairman, are unlikely to take that logic to its extreme conclusion and follow the market simply because it has priced in a particular policy move.

Ashley Rhodes: Warsh has also said that financial markets "are probably the most important source of information to guide central bankers." Do you agree?

Jan Hatzius: Not entirely, as I would rank the economic data first. That said, financial market prices are very important, and central banks have become increasingly sophisticated in extracting information from them. There's always room for improvement, but again, the information is already available under the current regime. And communicating less transparently wouldn't help policymakers evaluate financial market signals more appropriately.

Ashley Rhodes: How durable is a less transparent Fed?

Jan Hatzius: That remains an open question. As Chairman, Warsh has significant control over the Fed's communications framework and can choose to provide less information at press conferences, as he has done. In that sense, a less transparent approach could prove durable, though its consequences remain to be seen. Other communication changes, such as eliminating the dot plot, would probably in practice require FOMC support, and my sense is that other Fed officials would see that as too great a step away from transparency.

A large reduction in Fed communication, defined by the number of speeches, is less likely to endure. Reserve Bank presidents have strong incentives to continue talking, in part because the influence and relevance of their institutions depend on it. This has been a perennial issue—Chairman Alan Greenspan was often very frustrated with the Bank presidents. Warsh could try to rein them in. But even Greenspan in his heyday failed to do so. If regional bank presidents continue speaking frequently while centralized communication declines, the cacophony could actually worsen.

Ashley Rhodes: More broadly, while the Fed held rates steady in July, three FOMC participants dissented in favor of a hike, fueling debate about whether further tightening is on the horizon. Where do you stand?

Jan Hatzius: We don't expect the Fed to hike, largely because we think much of the recent strength in core PCE inflation reflects temporary factors: higher oil prices related to the US-Iran war, roughly 70bp of lingering tariff pass-through, and distortions in software and accessories prices. As these effects fade, we expect core inflation to move close to 2% in 2027. That assessment is consistent with the ongoing deceleration in other measures of underlying inflation such as trimmed-mean PCE and core CPI.

The recent data are also broadly consistent with that view. The inflation data improved significantly in June and July, and the labor market reports for both months were somewhat weaker on balance. If those trends continue, policymakers will likely conclude that additional tightening is unnecessary, especially with the funds rate already around 3.5-3.75%—a level the Fed likely views as roughly neutral or even slightly restrictive.

Interview with Donald Kohn

Donald Kohn is former Vice Chairman of the Board of Governors of the Federal Reserve System and current Robert V. Roosa Chair in International Economics and senior fellow at the Brookings Institution. Below, he sympathizes with Warsh's desire to scale back forward guidance but sees a bigger issue with his reluctance to explain the Fed's reaction function.

The views stated herein are those of the interviewee and do not necessarily reflect those of Goldman Sachs.



Jenny Grimberg: You saw firsthand the Fed's evolution from a relatively opaque institution into one that communicates through statements, press conferences, projections, and public remarks. What drove this shift toward greater transparency?

Donald Kohn: A key motivation was to get financial markets on the Fed's side. While the Fed controls the overnight rate, which acts as the anchor for the short end of the yield curve, monetary policy transmission to the real economy works through broad financial conditions, including equity prices, the Dollar, and long-term yields. The Fed believed that the more the market understands its policy framework and objectives, the more it will work with, rather than against, the Fed—as new data arrives, markets will adjust in a stabilizing direction. The second motivation was accountability. The Fed benefits from being an independent institution largely insulated from the political process, but that independence comes with a responsibility to provide the public and Congress a clear explanation of its actions.

Jenny Grimberg: Why is it so important for markets to work with the Fed?

Donald Kohn: Markets work better and reinforce central bank policies more effectively when they understand how policymakers expect the economy to evolve and how policy will likely respond. Historically, a lack of explicit communication has sometimes caused costly misunderstandings. Prior to 1994, the Fed didn't announce its policy decisions; markets had to infer them from the New York Fed's open market operations. An infamous misunderstanding known as the "Thanksgiving turkey" occurred in November 1989, when markets misconstrued a routine reserve injection as policy easing. The resulting volatility prompted an emergency FOMC phone call where policymakers had to decide whether to validate the market's mistaken interpretation by cutting rates or maintain the existing policy stance, ultimately choosing the latter.

In 1994, the Fed introduced explicit policy announcements, which grew more complex over time by, for example, including a policy "tilt" to signal whether policy was more likely to tighten or loosen. This toolkit proved vital in the early 2000s, when markets prematurely priced in rate hikes, causing financial conditions to tighten faster than was consistent with the FOMC's objectives; the Fed was able to explain its thinking more clearly, successfully realigning expectations.

Jenny Grimberg: Which communication innovation has made the greatest practical difference?

Donald Kohn: The innovations we've discussed, as well as the move to increase the frequency of the publication of the Fed's economic forecasts from semi-annually to quarterly, were all important. But if I had to choose the single most important innovation, it would be the post-FOMC meeting press conference, which Chairman Ben Bernanke, who had a strong predilection for increasing transparency, introduced in 2011. The press conference provides the chair with a direct, unfiltered opportunity to articulate the FOMC's thinking and explain the rationale behind its decisions. That's very powerful.

Jenny Grimberg: Conversely, has history also provided lessons about the risks of greater transparency?

Donald Kohn: The most serious risk is that forward guidance—guidance about the likely future path of policy—can constrain both the FOMC's and the market's reaction to incoming data. A prominent example occurred during the Covid recovery, when the Fed committed to holding rates at zero until the economy reached full employment, almost regardless of the inflation rate. This "Odyssean" commitment effectively constrained the Fed's response to emerging inflationary pressures.

I've received blowback on this point from Fed officials who argue that the real problem was bad forecasts. It's true that most forecasters, including the Fed, failed to anticipate the persistence of the price pressures. But the forecasts, the Fed's Flexible Average Inflation Targeting Framework, and the resulting forward guidance were all born out of a 15-year mindset of combatting too-low inflation. So, the guidance was too constraining, even if the forecasts were also at fault.

The market's hyper-focus on the median forecast in the Summary of Economic Projections (SEP) also illustrates the risk of excessive precision and false certainty. This median is a very weak indicator of the FOMC's collective stance; a move by just one or two participants can shift it, triggering unwarranted market reactions, which is partly why I've long advocated for stopping its publication.

Jenny Grimberg: So, is Chairman Warsh's move to reduce transparency a positive or negative shift?

Donald Kohn: I sympathize with Chairman Warsh to some extent. Forward guidance with commitment is an important tool at the zero lower bound, even if the Fed's pandemic-era Odyssean guidance was a poor choice. But once policy rates are well above zero, the need for it diminishes. At that point, the Fed has ample conventional policy space to adjust rates as economic conditions evolve. And, as we've discussed, projecting a specific number of rate hikes or cuts as in the policy dot plot is pretending to know more than policymakers actually do. So, I would be a reluctant user of forward guidance away from the zero lower bound—and with the Fed funds rate currently at 3.5-3.75%, we're clearly far from it.

The real problem with Warsh's approach is his reluctance to articulate how the FOMC sees the economy evolving,

seemingly out of fear that the market will interpret what he says as a policy hint. But explaining the Fed's thinking about the economic outlook is essential in helping markets act as a stabilizing force and is a core component of accountability. Central banks need a narrative. Chairman Alan Greenspan, whom Warsh has expressed a desire to emulate, always had a story—even if it wasn't always well understood—that was backed by deep analysis, and when economic developments diverged from expectations, Greenspan could investigate why the data weren't aligning. So, a narrative is crucial.

That said, I see merit in some of Warsh's other ideas, like reducing the number of FOMC meetings. While he hasn't specified a target, a move from eight to six meetings a year would allow policymakers to evaluate two months of economic data, encouraging the FOMC and markets to focus on trends rather than individual data points. So, I would support such a move, provided it doesn't disrupt the flow of information from the Fed staff to policymakers or increase intermeeting actions.

Jenny Grimberg: Warsh seems to want markets to "play the ball, not the referee," enabling the Fed to obtain more "direct and unfiltered" information about the economy from markets. How viable is such a feedback loop?

Donald Kohn: Markets will always react to new information based partly on how they expect the Fed to react. While Warsh has urged market participants to focus on the ball rather than the referee, the Fed is *not* the referee—it's the 800-pound gorilla in the rates markets. So, the market will always interpret incoming data through the lens of expected Fed policy.

But a golden mean exists. Overly specific forward guidance can potentially damp the market's reaction to incoming data, so backing off from such specificity could help free markets to process new information more independently. But the Fed must still provide its narrative—explaining how it's thinking and what it's focused on—to help the market figure out the appropriate rate path to achieve the Fed's objectives.

Jenny Grimberg: Curtailing guidance would likely increase short-term market volatility. Does that concern you?

Donald Kohn: Not necessarily. If forward guidance has indeed suppressed market reactions, removing it should improve the quality of the signal. So, a modest increase in short-term volatility is acceptable, provided the Fed equips market participants with enough information to react intelligently to incoming data. But an important distinction exists between productive volatility and harmful uncertainty. Creating uncertainty for its own sake is counterproductive and can dampen investment, and failing to articulate a clear story generates this damaging uncertainty. If the Chairman doesn't tell that story, dissenting policymakers may step in to fill the void—as occurred after the recent FOMC meeting—confusing the market and obscuring the actual policy consensus.

Jenny Grimberg: Warsh has also said that financial markets "are probably the most important source of information to guide central bankers." Do you agree?

Donald Kohn: I'm not sure they're the most important, but they're certainly very important. When I was at the Fed, I looked carefully at what the markets were saying about what they thought we should do, what they were reacting to, and whether that reaction was appropriate. But the Fed must interpret markets relative to what markets expect the Fed to do. Otherwise, policymakers risk the "monkey in the mirror" problem—watching the reflection of their own actions and adjusting policy to match owing to the mistaken belief that the market is providing independent validation. Navigating this issue requires a delicate balance. The more accurate markets' reactions are, the more they'll behave in a stabilizing manner, which, as we've discussed, was a key driver behind the Fed's shift toward greater transparency. But Warsh is also correct that forward guidance can become too prescriptive, resulting in market pricing simply reflecting the FOMC's past guidance and obscuring the informational value of the signal. So, while market prices are an invaluable source of information for central bankers, interpreting them is undoubtedly a tricky game.

Jenny Grimberg: Ultimately, how durable is the quieter communication regime that Warsh has ushered in?

Donald Kohn: Warsh's level of quietness is likely unsustainable. The market reaction to his July press conference—declining short-term rates and rising long-term rates—was undoubtedly not the outcome he desired. I sat next to him for nearly four years at the Fed, and I know him to be a highly capable policymaker. So, I'm confident he'll take this lesson on board and move in the direction of filling the current communication void. I'm also confident that the communications task force he's established will help move him in that direction, especially with Mervyn King co-leading it. King revolutionized communication at the BoE by introducing inflation projections and outlining the "key judgements" behind the BoE's forecasts, and in his book *Radical Uncertainty* emphasized that central banks need a narrative built on rigorous analysis that can be validated or contradicted by incoming data.

Warsh has also talked about less discussion from the Reserve Bank presidents. These policymakers are a vital liaison between the Fed and local communities and so shouldn't be silenced. But the public would benefit from a different form of communication from the Reserve Bank presidents—their public remarks should explain the FOMC's collective decisions and broader economic trends rather than offer instant analysis of individual data points or telegraph policy paths. So, the objective shouldn't be to restrict communication, but to foster more substantive and less reactive dialogue.

Jenny Grimberg: Investors are also focused on the Fed's near-term policy path. What do you expect the Fed's next move will be?

Donald Kohn: I wouldn't have voted to raise rates in July given that the June CPI print was favorable and long-term inflation expectations seemed anchored. But if forced to say whether the Fed's next move is more likely to be a hike or a cut, I would lean toward a hike. I am worried that inflation could be stuck at 2.5%, and because it has run above target for some time, longer-term inflation expectations may be less firmly anchored than usual. So, as a matter of risk management, buying insurance against inflation running too high for too long seems prudent. But if the favorable June, and now July, CPI prints were followed by another favorable one in August, hiking in September probably wouldn't be appropriate.

Fed Chairman's task forces, explained

Task force	Leaders	Goal	What changes should we expect?
Communications	• Mervyn King, Former Governor, Bank of England • Peter Fisher, Professor of Practice, Foster School of Business, University of Washington • Arminio Fraga, Chairman, Gávea Investimentos; former President, Central Bank of Brazil	Review how the Federal Reserve conveys policy deliberations and decisions amid uncertainty.	The FOMC would probably see getting rid of the Summary of Economic Projections (SEP) entirely as too large a step away from transparency, but it could modify it by, for example, no longer publishing the median projection to avoid seeming to endorse it. While Chairman Warsh has criticized forward guidance, some implicit soft forward guidance without commitment is likely inevitable as long as Fed officials share their views on the risks to inflation and employment and on the neutral rate.
Inflation Frameworks	• Greg Mankiw, Professor of Economics, Harvard University; former Chairman, CEA • Thomas Sargent, Professor of Economics, New York University; Nobel laureate • William White, Senior Fellow, C.D. Howe Institute; former economic adviser, BIS	Revisit how the Federal Reserve understands and responds to the drivers of inflation.	The task force leaders will likely take a mainstream view of inflation drivers but might advocate taking another look at monetary aggregates. They could also suggest new ways to measure the underlying inflation trend; one approach is to consider an average of several different measures that might each be optimal under different circumstances. The more immediate question is how to respond to inflation caused by supply shocks. While all agree that the Fed must ensure supply-driven inflation doesn't broaden out or compromise inflation expectations, policymakers still face the difficult task of judging how long supply shocks will last, for which no easy answer exists.
Data Sources	• Raj Chetty, Professor of Economics, Harvard University • Doug McMillon, Former President and CEO, Walmart Inc. • Kevin Murphy, Professor of Economics, University of Chicago	Improve the quality and timeliness of real economic signals that inform the Federal Reserve's policy judgments.	While broad support exists for incorporating alternative data sources, they often fail to meet three requirements for high-quality economic statistics: representativeness, accurate seasonal adjustment, and continuous availability. As a result, alternative data sources are unlikely to fully replace the official data but could be used to shed light on a subset of the economy.
Balance Sheet Policy	• Karen Dynan, Professor of Economics, Harvard University • Raghuram Rajan, Professor, Chicago Booth; former Governor, Reserve Bank of India • Jeremy Stein, Professor of Economics, Harvard University; former Governor, Fed Board	Review the benefits and risks of the current ample-reserves regime and the composition of the Fed's balance sheet and assess alternative frameworks for the conduct and operation of monetary policy.	We see little support for abandoning the ample-reserves framework and consequently see only limited room for regulatory and supervisory changes to shrink banks' demand for reserve balances and the size of the balance sheet. But whether the Fed's asset holdings should be proportional to Treasury debt outstanding or consist mainly of bills to reduce volatility in Fed profits remains an open question.
Productivity and Jobs	• Marc Andreessen, Co-founder and General Partner, Andreessen Horowitz • Charles I. Jones, Professor of Economics, Stanford University (on leave at Anthropic) • Asha Sharma, Executive Vice President and Xbox CEO, Microsoft Corp.	Assess the economic impact of new general-purpose technologies, including artificial intelligence (AI), to inform the Federal Reserve's policy judgments.	Warsh might invoke the Greenspan years to argue that the FOMC should not raise rates solely in response to strong GDP growth if it is accompanied by strong productivity growth driven by AI—an argument other Fed officials would likely find uncontroversial. But most Fed officials will likely be skeptical that monetary policy decisions today should be influenced by a forecast of how AI will affect productivity growth in the future as productivity growth has historically proven very hard to forecast.

Source: Federal Reserve, Goldman Sachs GIR.

Special thanks to US economist Pierfrancesco Mei for the table, which originally appeared in US Economics Analyst: The Fed Task Forces.

Fed guidance, in Warsh's own words

"I think financial markets perform best when they react to incoming data. I think they—the financial markets work less efficiently when they ask a question: How will the Federal Reserve react to that incoming information? The more that markets are paying attention to what's happening in the real economy, deciding what's good data and what's less good data, the more financial markets can price what they believe is the most likely and what are the tail risks. Financial market prices are probably the most important source of information to guide central bankers."

- June 2026 FOMC press conference

"This afternoon, you also received the usual Summary of Economic Projections. It's been the practice of this Committee for participants to submit these projections, and I have encouraged my colleagues to continue to do so. I, however, have refrained from offering any projections of my own, consistent with my long-held views on the SEP, at least as currently structured... For me, it's not helpful in the conduct of policy."

- June 2026 FOMC press conference

"Financial markets and the real economy work best when you look at what's happening in the real economy, you make your own judgments. There has been a tendency, and I take plenty of blame [for] this, the '08 crisis where we we're trying to suppress volatility, where we thought we needed to spoon feed markets to get out of that. That was the right policy for a crisis. It's not the right policy for the time that we have now. And so sometimes unlearning is harder than learning and I'm going to keep at it."

- ECB Forum on Central Banking, July 2026

"I think part of the reason why, after making a mistake in 2021 and 2022, the mistake was compounded, is the Fed gives its forward guidance. The Fed tells the whole world what their dots are going to be, what their forecasts are going to be. Well, the Fed's human. Then they hold on to those forecasts longer than they should. I think if the Fed were to wait until it gets into a meeting before making a decision, that incremental deliberation can keep the central bank from compounding its errors. I think these are big changes that are needed."

- Warsh's Senate confirmation hearing, April 2026

"I understand the desire for rolling forecasts and commentary from this Committee. But for our part, we need to observe market reaction to developments, direct and unfiltered... We're trying not to interfere with that market signal. That's part of the reason why we've been somewhat spare on our words, when we pulled back from forward guidance."

- July 2026 FOMC press conference

"Market participants are learning to play the ball, not the referee — and market prices will continue to respond in the direction and magnitude they see fit. This is, in my view, a change for the better—and we are just getting started... By not spoon-feeding markets, by not previewing our decisions, by not sort of giving nudges and leans, my colleagues and I have found in the intermeeting period what we're getting is the views from a very accomplished economist. That's the internals of financial markets... We're not going to be constrained or take verbatim from what the market's doing. But I think it's useful... to understand that markets can be a very good source of information."

- July 2026 FOMC press conference

"In the monthly window between FOMC meetings, policy makers provide the minutes of the preceding meeting and make dozens of statements, speeches, interviews, forecasts, and other predictions. They pay great attention to the latest payroll data and inflation prints and their immediate effect on the next decision. There is, however, little to be learned from the latest Fed forecast, little insight to be gleaned from the last piece of government data. I worry that Fed policy makers find themselves settling scores with one another while Fed watchers score count hawks and doves. All the while, the big questions for the economy go unanswered. And the big risks over the horizon go unaddressed... We should not encourage the financial markets to be the handmaiden of the central bank. We should allow asset prices to be an independent source of economic insight and discipline."

- From a chapter in "The Structural Foundations of Monetary Policy", 2018

"[The Fed's] "forward guidance"... offers ambiguity in the name of clarity. It licenses a cacophony of communications in the name of transparency."

- WSJ op-ed, August 2016

Source: Federal Reserve, ECB, Congress.gov, The Structural Foundations of Monetary Policy, WSJ, compiled by Goldman Sachs GIR.

Interview with Stephen Miran

Stephen Miran is former Member of the Federal Reserve Board of Governors and Chairman of the Council of Economic Advisers. He is a Senior Strategist at Hudson Bay Capital Management. Below, he argues that less forward guidance would reduce the risk of policy errors and improve the quality of market signals.

The views stated herein are those of the interviewee and do not necessarily reflect those of Goldman Sachs or Hudson Bay Capital.



Allison Nathan: New Fed Chairman Kevin Warsh appears to be ushering the Fed into a less transparent era. Is this shift positive or negative?

Stephen Miran: It's positive. Forward guidance tends to dampen market volatility in the short run because telling the market what the Fed plans to do reduces the market's sensitivity to incoming data. But, over the long run, forward guidance tends to increase volatility for two reasons. First, it raises the risk of the Fed making a policy mistake because it leaves the Fed slower to adapt to changing economic conditions. For example, at the end of the post-pandemic period, the Fed was still buying mortgage bonds even though home prices were up 20%, in part because the Fed had provided calendar-based guidance committing it to doing so. But this ignored the fact that the way official shelter inflation data is constructed results in a long lag between changing market conditions and the pass-through to inflation, generating real persistence in inflation. The Fed continued to inject credit into the housing sector long after it was appropriate to do so.

Second, forward guidance can impair market pricing of risk. When the government provides explicit guidance about the future path of policy, many people naturally believe it's likely to happen. The experience of Silicon Valley Bank (SVB) illustrates the danger. SVB's management certainly made huge risk-management mistakes by extending duration and taking on excessive interest-rate risk. But part of the reason they felt comfortable doing so was because they took the Fed's guidance of an extended period of zero rates at face value. Over the longer run, forward guidance creates an environment in which these kinds of bad outcomes occur more frequently.

Allison Nathan: But we've also lived through periods of less Fed transparency that led to massive volatility, such as the Greenspan era. Is that a world we want to return to?

Stephen Miran: My sense is that Warsh does want to move in the direction of the Greenspan era. The question is, does that reduce the chance of categorical policy errors like those we discussed? Again, my answer is yes. Unlike large-scale inflation, I don't blame asset market bubbles and associated volatility on the Fed; they're an indelible part of capitalism. We don't like them, but they're part of life.

Allison Nathan: Isn't there an important distinction between the Fed committing to a policy path and being transparent about its reaction function?

Stephen Miran: In theory, yes. In practice, the difference is muddled because economic data often require interpretation. Inflation, for example, is the change in the general price level, which is not a directly observable metric like you get in physics or chemistry, but a constructed abstraction that incorporates many components, some of which may not accurately capture underlying price pressures. Case in point: the measured contribution to inflation of portfolio management fees, which are included in PCE inflation but not CPI inflation, is closely tied to stock market performance. This very positively contributed to PCE inflation in recent years as the stock market rose even though fees have declined steadily over the past two decades. GS' own economists estimate this one line contributed more than half of core PCE inflation in July. The Fed should clearly ignore this, and it needs the flexibility to do so. But, if there's too much transparency about the reaction function, the Fed then runs the risk of appearing to violate it or move the goalposts, denting the credibility of that reaction function. The Fed must have the flexibility to carefully interpret inflation data and separate true price pressures from measurement error when these are evolving in real time. So, the merits of reaction function transparency aren't as clear as they may first seem.

Allison Nathan: But wouldn't the Fed communicating more about how it's viewing and interpreting data—effectively the opposite of Warsh's intentions—avoid this issue?

Stephen Miran: I agree that an important distinction exists between guidance that tells people what you're going to do and guidance that tells people what you care about. And I feel more strongly about avoiding the former than the latter.

My view of the Summary of Economic Projections provides a useful illustration of this. The policy dots absolutely need to go because, as I explained, forward guidance on the Fed's policy path increases the likelihood of large risk events, and the policy dots play an important role in that, enabling markets to take cues from the Fed rather than forming their own expectations about the future. While the Fed emphasizes that the policy dots aren't a commitment or even a forecast, but a "projection," that's ultimately not credible because the Fed understands that if it violates its policy "projections" too much, people will start ignoring them. Forward guidance is only useful if people believe it. And it's only believable if the Fed follows it. So, the Fed is motivated to follow through on its projections. In contrast, the economy and inflation projections, which provide insight into what policymakers care about and their view of macro conditions, aren't as problematic. So, I wouldn't die on the hill for keeping or getting rid of them. I would be comfortable with narrowing the scope of guidance without removing it entirely.

Allison Nathan: Warsh believes that markets themselves should determine the appropriate policy path from macro conditions rather than rely on Fed guidance. But given that short-term interest rates price what markets expect the Fed *will* do—not what it *should* do—would less guidance simply make policy expectations harder to form?

Stephen Miran: A "hall of mirrors" problem will always exist at the front end of the interest rate curve, where the Fed is looking at the market to tell it what to do, while the market is pricing what it thinks the Fed will do. But forward guidance makes the market less responsive to data, so removing that guidance nevertheless allows markets to respond more freely and flexibly to incoming information. That will likely lead to increased volatility and a related rise in the term premium at the front end of the curve. But it would be positive in the sense that forward guidance has dampened short-rate volatility.

Allison Nathan: Warsh has also argued that this approach will, in turn, enable the Fed to obtain more "direct and unfiltered" information about the economy from markets. But is that true if markets are ultimately noisier as a result?

Stephen Miran: The question is, does removing forward guidance improve the signal coming from markets? Forward guidance generates little volatility and little signal at the front end of the curve, because the Fed's already said what it will do. In my view, reducing guidance generates more volatility but also more signal. Markets will always be volatile. But is that a reason to discard information from them? No. To get any signal from the markets, you must extract it from the noise of volatility, and the Fed has a longer timeline on which to parse it than do levered market participants. The first-best world of signal without volatility doesn't exist. I'd rather have that signal, even if it's messy, than throw the signal out by anchoring market expectations to the Fed's near-term projections.

And it's not just the front end of the curve that contains information. Inflation expectations priced across the yield curve are incredibly helpful when thinking about setting monetary policy. These data also aren't perfect, but they're among the best indicators of where inflation expectations are heading and are less susceptible to the hall of mirrors issue.

Allison Nathan: Are concerns about the Fed "outsourcing" monetary policy to the markets warranted?

Stephen Miran: The Fed shouldn't and likely wouldn't try to outsource policy completely to the markets. It should be the one setting policy. But that doesn't preclude the Fed from taking steps to improve the quality of the signals from markets, and with improved quality, make better use of them.

Allison Nathan: Could less forward guidance that puts greater onus on the market to form its own policy expectations in itself influence Fed behavior?

Stephen Miran: In the short run, the resulting increase in volatility and term premium at the front end of the yield curve would effectively amount to an incremental tightening in financial conditions, which, all else equal, would be contractionary for economic growth on the margin. So, if nothing else about the economy has changed, the Fed would need to offset that tightening by lowering the Fed funds rate. Over the long run, it might be a reason to expect a lower neutral rate, again, all else equal, since any given level of the funds rate would drive tighter financial conditions and therefore a looser funds rate would be required on average. So, reduced forward guidance could affect the level of policy rates consistent with balanced economic growth.

Allison Nathan: Could a sharp rise in market volatility force Warsh to rethink his push to usher in a quieter Fed regime?

Stephen Miran: Absolutely, in an extreme case. Forward guidance, quantitative easing, and related tools can be extraordinarily useful at the zero lower bound. They were appropriate to combat deflation risk in the aftermath of the Global Financial Crisis, but they should have been withdrawn once the emergency had passed. These tools are like antibiotics: they should be used when the economy is sick but overusing them just creates superbugs like the inflation we saw after the pandemic, linked to the housing QE and forward guidance we discussed earlier. The utility of these tools diminishes outside of a deflationary environment, and their preciousness shouldn't be squandered. If volatility ever tightened financial conditions enough to force the Fed to cut back to the zero lower bound, forward guidance could again be warranted. But we're very far from that today.

Allison Nathan: Even if Warsh likely won't be forced to rethink his communication approach anytime soon, won't other Fed officials continue to publicly share their views?

Stephen Miran: If the chairman wants less forward guidance, I would expect the officials that are a part of his leadership group and looped into the decision-making process to similarly communicate less about the Fed's intentions. But dissenting views should still be heard. During my time as a Fed governor, I frequently shared my views publicly because they were out of consensus, so I felt the need to explain myself. If I were in consensus, there'd be little point.

Allison Nathan: I'd be remiss if I didn't take the opportunity to ask you what you think the Fed should do next?

Stephen Miran: Recent too-high PCE inflation is concerning, but monetary policy today can do little about that given the well-documented lags between monetary policy and its impact on the economy. These lags mean that monetary policy today must be set based on where inflation is expected to be in the *second half of 2027*. And so, the question is whether the factors behind today's elevated inflation will repeat well into next year. I don't think that will be the case, as recent high PCE inflation is driven by a combination of energy and deglobalization shocks, as well as measurement error, none of which I forecast to repeat. If they don't, inflation will return toward its underlying levels near 2%. Indeed, core CPI is 2.5%, which historically is fine. If the typical CPI-PCE relationship held at present, PCE would be 2.1%; the nearly 1-point inversion of the CPI-PCE relationship is mostly measurement error, like the portfolio management services we discussed, but also software, which GS' economists have covered too.

At the same time, I view the labor market as still at risk and monetary policy as still modestly restrictive, although the truly astonishing investments in AI have nudged up my expectations of the neutral rate somewhat since I last submitted dots earlier this year. All told, I would still favor slightly easier policy ahead, since inflation is on track to be in a good place after monetary policy lags have elapsed. Hiking rates seems inappropriate, as that would effectively amount to asking people to lose their jobs to compensate for measurement error.

The evolution of Fed communications

1979
Begins publishing a summary of policymakers' individual economic projections in the semiannual Monetary Policy Report to Congress

March 1993
Begins releasing minutes of FOMC meetings (with a 6–8-week lag)

November 1993
Begins releasing transcripts of FOMC meetings (with a 5-year lag)

February 1994
Begins announcing policy actions publicly on the day of the FOMC decision

May 1999
Releases a statement about the FOMC decision even after no change in the Fed funds rate target; Begins announcing a policy "tilt" indicating most likely future interest rate action between then and the next FOMC meeting

January 2000
Commits to publishing a statement after every FOMC meeting; Replaces "tilt" with statement describing "balance of risks" to the economic outlook intended to cover an interval extending beyond the next FOMC meeting

March 2002
Begins releasing votes of individual Committee members and preferred policy choice of any dissenters

August 2003
Begins to issue direct qualitative statements about its future policy inclinations in various verbal formulations

February 2005
Expedites the release of FOMC minutes to make them available before the subsequent FOMC meeting

November 2007
Increases the frequency and expands the content and horizon of its publicly-released inflation and economic activity forecasts

April 2011
Chairman holds first press conference following an FOMC decision

August 2011
Shifts to calendar-based guidance from vague qualitative guidance

January 2012
"Dot plot" is officially added to the Summary of Economic Projections (SEP) and the Statement on Longer-Run Goals and Monetary Policy Strategy debuts

December 2012
Replaces calendar-based guidance with outcome-based guidance

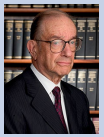
January 2019
Shifts to holding press conferences after every FOMC meeting from only after meetings associated with an SEP

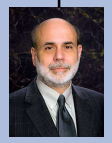
August 2020
Updates its longer-run strategy following its first formal framework review

June 2026
FOMC statement becomes shorter and omits forward guidance, and no SEP projections are submitted by the Chairman

1979 1993 1994 1999 2000 2002 2003 2005 2007 2011 2012 2019 2020 2026


Paul Volcker
1979-1987


Alan Greenspan
1987-2006


Ben Bernanke
2006-2014


Janet Yellen
2014-2018


Jerome Powell
2018-2026


Kevin Warsh
2026-

Source: Federal Reserve, Goldman Sachs GIR.

The Fed's communication toolkit

Tool	Introduction*	Timing/frequency	Issuer/voice	Official purpose/content	Proposed changes
FOMC statement	1994	Released immediately following each scheduled FOMC meeting; eight such meetings are held per year	Collective FOMC	Announces the FOMC's policy decision, vote, and its assessment of economic conditions, inflation, and employment. Prior to Warsh becoming Chairman, the statement included forward-looking language about the likely policy path or conditions for future adjustments	Statement shortened significantly in June 2026 (explicit forward guidance and "balance of risks" content removed)
Chair's press conference	April 2011	Held after each scheduled FOMC meeting	FOMC Chair	Enables the chair to explain the FOMC's policy decision and answer questions from the press about the decision, economic conditions, the future policy path, etc. in real time	Warsh provided limited commentary during his first two press conferences and is considering reducing their frequency
Summary of Economic Projections (SEP)	November 2007	Released following select FOMC meetings, four times per year	Individual FOMC participants, compiled and published by the FOMC	Shows FOMC participants' projections for GDP growth, unemployment, PCE inflation, and interest rates (via the dot plot)	Warsh did not submit an interest rate projection (dot) at his first meeting, and the communications task force is actively reviewing the SEP and the dot plot (see pg. 8)
Publicly-released FOMC minutes	February 1993	Released three weeks after each scheduled FOMC meeting	Collective FOMC	Provide an overview of the discussion that occurred at the FOMC meeting, including a summary of economic and financial conditions, policy issues addressed, and the decisions made by the meeting participants	No active official proposals
Remarks by Governors and Reserve Bank presidents	N/A	Frequent but no set schedule	Individual policymakers	Allow individual policymakers to discuss the economy, regional conditions, research, and their individual policy views	No active official proposals (though Warsh has discussed limiting such remarks)
Implementation note	December 2015	Released alongside the statement following each scheduled FOMC meeting	Collective FOMC	Outlines operational details needed to implement the policy decision, including open market operations and balance sheet instructions	No active official proposals (though broader balance sheet operations are under review by the balance sheet policy task force)
Statement on longer-run goals and monetary policy strategy	January 2012	Updated/reaffirmed annually	Collective FOMC	Articulates the FOMC's monetary policy strategy and its approach to achieving its dual mandate	No active official proposals (though how the Fed understands and responds to inflation is under review by the inflation frameworks task force)
Monetary policy report and congressional testimony	1978	Report is delivered semiannually to the required Senate and House committees along with testimony from the chair	Federal Reserve Board; Chair for testimony	The report explains the conduct of monetary policy, economic developments, and the outlook, in accordance with the Federal Reserve Act that requires the Federal Reserve Board to submit written reports to Congress containing such discussions	No active official proposals
Beige Book	1970 (known initially as the "Red Book" owing to its red cover; public releases began in 1983)	Published eight times per year	Twelve Federal Reserve Banks	Each Federal Reserve Bank gathers anecdotal information on economic conditions in its District through reports from Bank and Branch directors and interviews with business contacts, economists, and market experts. The Beige Book summarizes this information	No active official proposals
FOMC transcripts	1994	Released with a five-year lag	Collective FOMC	Provide a detailed historical record of FOMC meeting proceedings	No active official proposals

**Column shows only the year the tool was introduced; for details on changes to these tools throughout the years, see timeline on pg. 12.*
Source: Federal Reserve, Goldman Sachs GIR.

Transparency: more benefits than costs

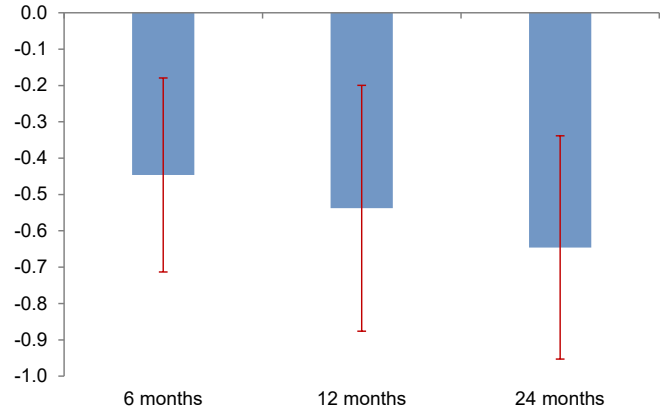
Joseph Briggs and Megan Peters argue that central bank transparency has more benefits than costs

The potential benefits of increased central bank transparency, including a reduction in interest rate volatility and more effective transmission of monetary policy to the real economy, are often weighed against the potential costs of central banks adjusting more slowly to incoming information that conflicts with prior guidance—increasing the risk of policy mistakes—and too much communication generating confusion that amplifies market volatility. In practice, we find strong evidence of the former and little evidence of the latter.

Clear evidence of transparency benefits…

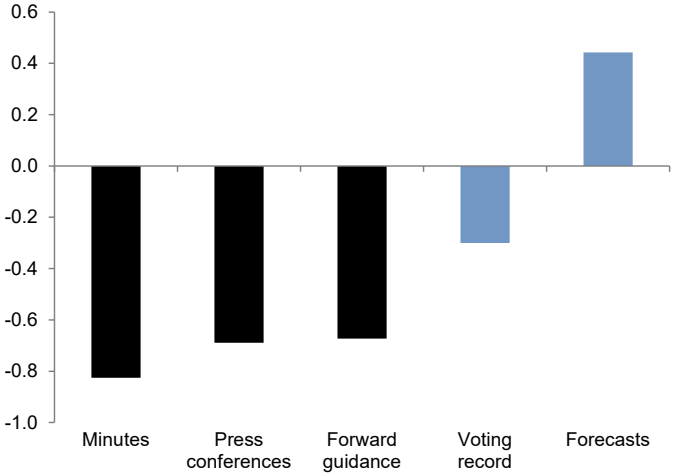
We find evidence that the global shift to greater transparency has yielded benefits through two channels: a reduction in rates volatility and more effective transmission of policy via financial conditions.

Increased information has generally lowered policy volatility…
Changes in realized volatility 6, 12, and 24 months after major reforms to central bank communication, bp/day



Note: Error bars indicate 90% confidence intervals.
Source: GS FICC and Equities, Goldman Sachs GIR.

…with larger effects from minutes, press conferences, and forward guidance
Changes in realized monthly volatility 24 months after major reforms to central bank communication, bp/day



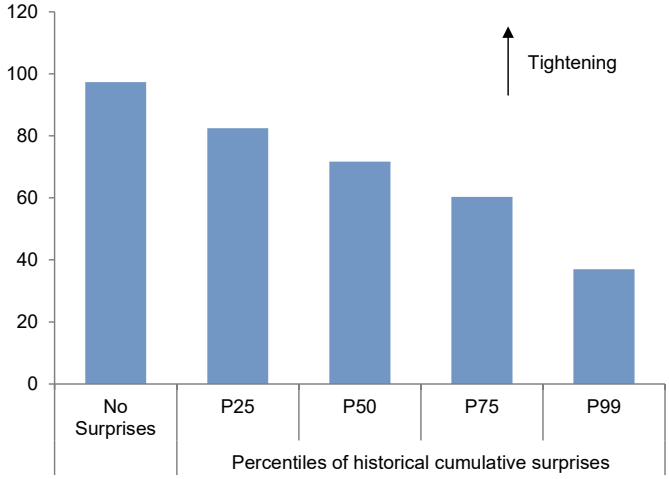
Note: Black bars indicate statistically significant changes at the 10% level.
Source: GS FICC and Equities, Goldman Sachs GIR.

First, after compiling data on central bank communication policy changes over the last 30 years, we find that reforms that increased transparency have historically led to a roughly 10% reduction in daily 1y forward 1y swap rates volatility, with larger effects following the introduction of minutes, press conferences, and forward guidance. While modest, these patterns highlight that increased information has helped lower policy rate uncertainty and volatility.

Second, and more importantly, we find that increased communication and transparency have led to more effective transmission of policy via financial conditions. After conditioning on the average meeting day surprise over the last five years (which we view as a summary measure of communication effectiveness), we find that a 100bp hawkish policy surprise leads to 100bp FCI tightening if a central bank has not surprised recently, but only 70bp for central banks at the median of the historical surprise distribution, 60bp at the 75th percentile, and 40bp at the 99th percentile.

The dampened transmission for more surprise-prone central banks suggests that financial markets may be slow to fully incorporate policy signals if they view them as less credible or persistent, or if they confuse the interpretation of the central bank's reaction function. If markets then discover that the policy surprise is indeed followed by a more persistent policy shift that subsequently tightens (or loosens) financial conditions, the economic impact will arrive later, undesirably extending the lag of monetary policy.

Transparency and credibility have contributed to more effective transmission of policy via financial conditions
Effect of 100bp hawkish policy surprise on GS FCI over next week, by historical tendency to surprise, bp



Source: Bloomberg, Goldman Sachs GIR.

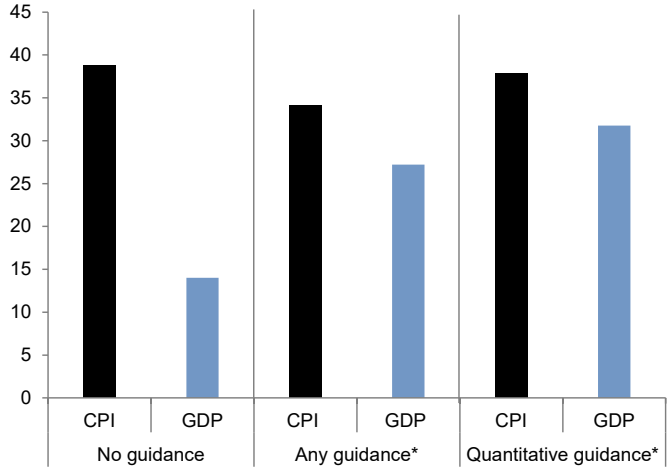
…but little evidence of transparency costs

But what about the potential costs of central banks reacting more slowly to incoming information that goes against guidance and too much communication in itself creating confusion and volatility? We find limited evidence for both.

First, we find[1] that central banks that provide guidance tend to react similarly to inflation surprises as those that do not and react more quickly to GDP surprises. While we are hesitant to over-extrapolate these results given that the differences in reactions are not statistically significant, the estimates push back against the narrative that providing guidance significantly dampens policy responses.

Guidance-providing central banks have, if anything, responded more quickly to forecast surprises

Effect of a 1pp annualized forecast revision on policy rate, by whether forward guidance was used over previous 12m, bp



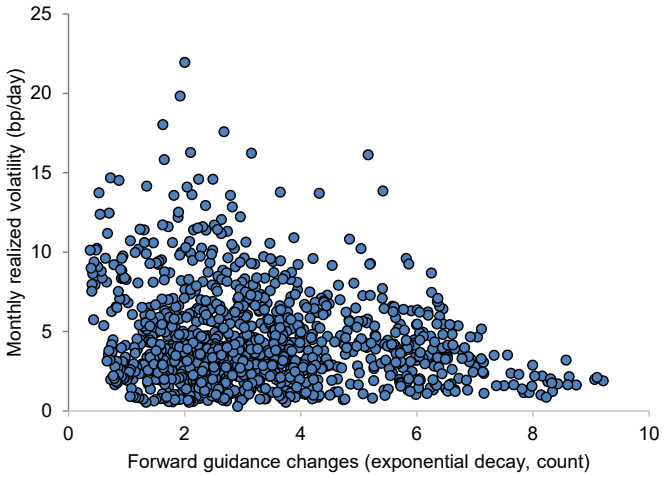
**No statistically significant difference between central bank reaction functions with and without forward guidance.*
Source: Sutherland (2022), Haver Analytics, Goldman Sachs GIR.

Second, we find little evidence that major central banks have reached a point where more information is counterproductive. Specifically, we find that the two-year total change in forward guidance (based on the extended Sutherland (2022) dataset) is associated with a decline in year-ahead policy-rate pricing volatility, although the marginal benefit of more frequent updates appears to diminish rapidly.

We also find an essentially flat relationship between the (normalized) number of central bank speeches and rates volatility. The lack of any visible uptick in volatility at higher communication levels suggests that DM central banks are not at a point where further communication is counterproductive, although the marginal benefits of more communication may be limited.

The marginal benefit of more information diminishes quickly…

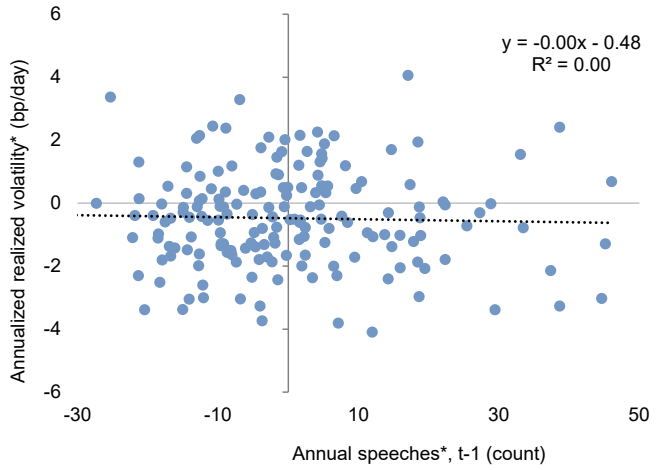
Forward guidance changes (exponential decay, count) vs. monthly realized volatility (bp/day)



Source: GS FICC and Equities, Bloomberg, Sutherland (2022), GS GIR.

…but so far communication has not led to market confusion and increased volatility

Annual speeches (count) vs. annualized realized volatility* (bp/day)



**Controlling for country and time fixed effects.*
Source: GS FICC and Equities, Bloomberg, Sutherland (2022), GS GIR.

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Megan Peters, Global Economist

Email: megan.l.peters@gs.com *Goldman Sachs International*
Tel: 44-20-7051-2058

[1] To assess this, we compile cumulative year-ahead forecast surprises, which we define as the sum of GDP and inflation realizations (vs. benchmark forecasts) and the 2-year ahead forecast revisions (vs. benchmark forecasts). We then regress year-ahead policy outcomes on these same variables while also controlling for lagged policy rates and country fixed effects. We interact our cumulative forecast surprise measures with a general and quantitative guidance indicator (based on the Sutherland (2022) forward guidance dataset) to estimate whether providing guidance leads to a more delayed reaction.

A less transparent Fed and US rates

William Marshall finds that rates have priced in uncertainty related to a less transparent Fed, but whether it persists will depend on the clarity of the Fed's reaction function

US rates markets have already begun to reflect Chairman Kevin Warsh's shift toward a less transparent Fed regime. We find that understanding FOMC actions and the rationale behind them meaningfully improves the rates markets' ability to price future policy outcomes and that even meeting-day surprises don't necessarily equate to structurally higher rates volatility as long as the Fed's reaction function is well understood. Without that understanding, the recent increased uncertainty about the path of policy rates will likely persist, raising rate market volatility and leading investors to demand a higher term premium for holding duration.

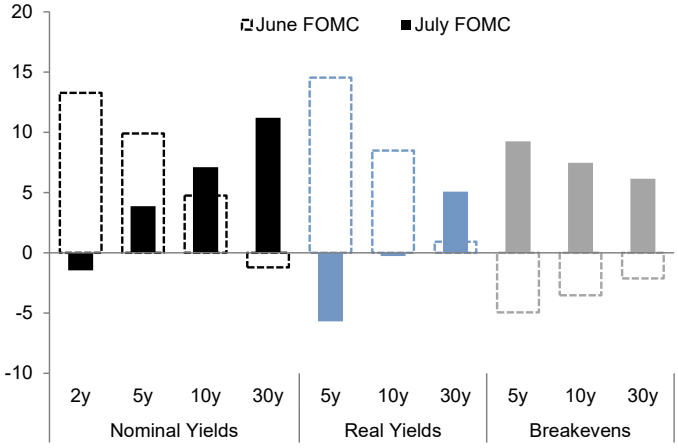
Rates markets already responding

The US rates market has quickly priced in the uncertainty that accompanies less transparent Fed communication. This was most readily visible in the substantial uncertainty implied heading into the July FOMC decision—the rates market priced unusually high ~30% odds of a hike prior to the ultimate decision to keep the policy rate unchanged. While larger "surprises" (measured as the gap between what markets priced ahead of the meeting and the realized outcome) have often accompanied rate cuts, July represented the largest non-cut meeting day "surprise" since the late 1990s.

Rates market moves on meeting days have also reflected the uncertainty inherent in a less transparent regime. The market interpreted the June FOMC as a relatively hawkish surprise, visible in a front-end led and real-yield driven rise in nominal yields. And the July decision to remain on hold despite non-trivial odds of a hike produced a mirror-image move, with long-end yields rising disproportionately and breakeven inflation playing a greater role in the rates selloff.

July's FOMC decision saw more or less the mirror image of the rate move from June

Change on FOMC meeting day, bp



Source: Bloomberg, Goldman Sachs GIR.

On net, market responses to Chairman Warsh's first two FOMC meetings have amounted to relatively higher long-end yields, a steeper curve, and higher volatility in the US compared to other major markets. However, a material reevaluation in the level of traded inflation has not occurred, suggesting that, at least for now, investors are mainly responding to greater uncertainty rather than pricing in diminished credibility.

History as a guide

As the market continues to digest the regime shift, the evolution of Fed communication over the last three decades offers useful takeaways about how this shift may continue to impact rates market dynamics. First, the historical experience suggests that understanding FOMC actions and the rationale behind them meaningfully improves markets' ability to price future policy outcomes. We find that the Fed's efforts to improve policy effectiveness via increased transparency in the mid-2000s compressed the absolute gap between ex-ante market pricing of the Fed funds rate and the eventual policy outcome. While transparency does not necessarily equip markets with perfect foresight, insight into the decision-making process reduces the risk that markets over or underreact to given pieces of information. As a result, we find that even though larger meeting-day surprises tend to correspond to larger yield swings on the day of the decision, they do not necessarily equate to structurally higher rates volatility if the market has a clear understanding of the Fed's reaction function.

Second, we find that, in such environments, Fed press conferences have played a clear role in shaping markets' interpretation of the FOMC statement, though different Fed chairs have used this tool to different ends. Under Bernanke and Yellen, rate moves suggested that press conferences tended to reinforce the statement's message, whereas under Powell yields tended to reverse their initial statement reaction during the press conference, suggesting he mainly used the press conference to add context that altered the rates market's initial read.

That said, we also find that post-meeting press conferences, as well as the publication of the dot plot, have not significantly improved the markets' ability to anticipate future policy rate outcomes. That is not to say that these innovations have not mattered as a rates market driver, but rather that there is stronger evidence that the initial mid-2000s shift towards transparency was more impactful. However, markets do take a signal from the frequency of press conferences—rates markets tended to put lower odds on a move at non-press conference meetings before the shift to holding a press conference after every meeting.

Lesson learned: Reaction function clarity is key

All told, history suggests that a shift toward more meeting-day surprises may change when volatility occurs but need not raise it structurally. But this is conditional on the Fed still providing clarity about its rationale for policy decisions and its reaction function. Markets need to understand which data the Fed thinks matter most and how policy is likely to respond as those data evolve. Less explanation on these fronts would likely amplify uncertainty about the path of policy rates, fueling higher rate market volatility and leading investors to demand greater compensation for holding duration via higher term premium.

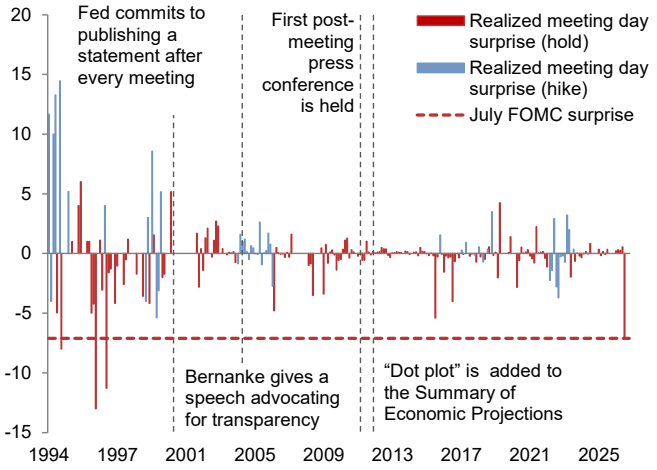
William Marshall, Head of US Rates Strategy

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Tel: 212-357-0413

Fed transparency and markets, in pics

The Fed's move toward greater transparency over the last 30 years has led to a decline in FOMC meeting day surprises…

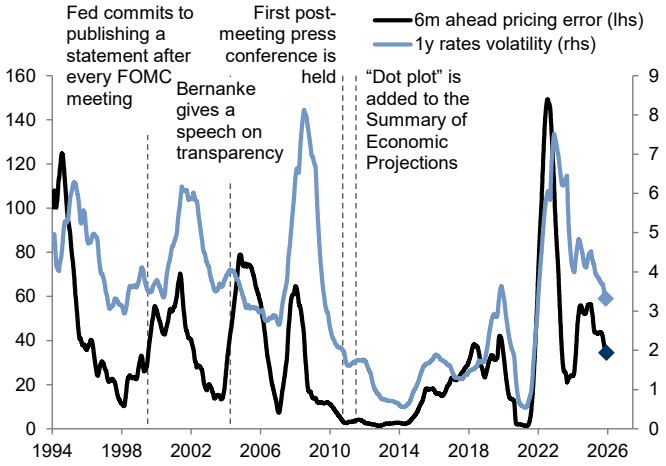
Est. non-cut meeting day "surprise" vs. market pricing, bp



Note: Surprise estimated as meeting day change in weighted 1st or 2nd Fed funds futures prior to 2002, 1m OIS from 2002 to present.
Source: Bloomberg, GS FICC and Equities, Goldman Sachs GIR.

…and more transparency has improved the market's policy path pricing accuracy and reduced volatility, outside of macro shocks

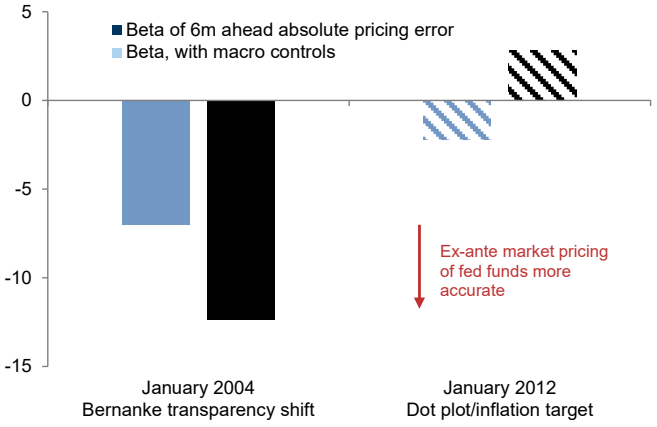
Absolute pricing errors of Fed funds futures (lhs, bp) vs. 1y rates volatility (rhs, bp/day)*



**1y rolling averages.*
Source: GS FICC and Equities, Goldman Sachs GIR.

The shift away from deliberately vague communication under Bernanke drove the largest improvement in accuracy

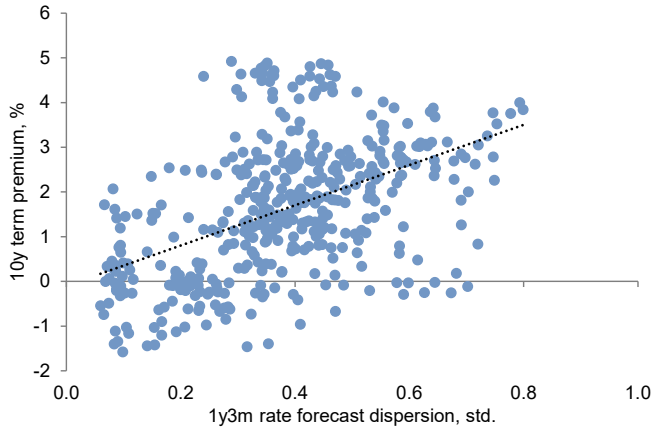
Beta of 6m-ahead policy pricing accuracy to Bernanke transparency speech (2004) and dot plot intro (2012), bp



Note: Shaded bars are beta insignificant.
Source: Goldman Sachs GIR.

Lower uncertainty about the policy path, measured as the dispersion in forecasts, also tends to dampen term premium and reduce borrowing costs

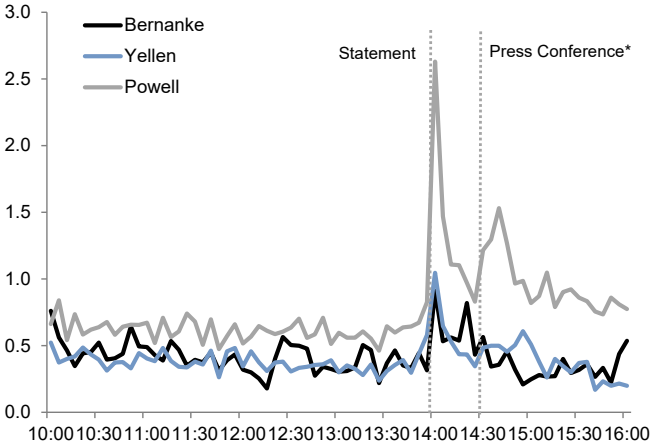
Forecast dispersion of 1y3m rates vs. 10y US term premium, GS estimate



Source: GS FICC and Equities, Consensus Economics, Goldman Sachs GIR.

That said, compared with the Bernanke and Yellen eras, press conferences under Powell generated higher rates volatility…

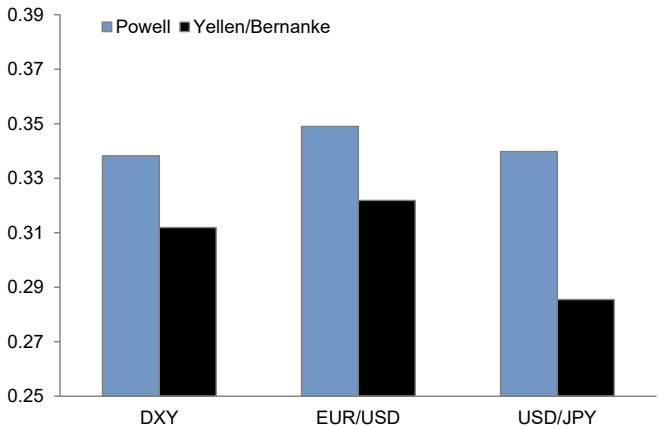
Intraday volatility in 2y Treasury yields on FOMC press conference days, measured in 5-minute intervals, bp



**Press conferences under Chair Bernanke began at 14:15.*
Source: Goldman Sachs GIR.

…as well as FX volatility, partly owing to differences in Powell's communication style, which suggests that transparency alone does not guarantee calmer markets

Std. dev. of FX returns around FOMC press conferences, %



Note: 70-minute window from 10 minutes before to 60 minutes after the beginning of the press conference; we exclude the March 15, 2020 meeting.
Source: USMPD, Goldman Sachs GIR.

Special thanks to rates strategist Friedrich Schaper and FX strategist Lexi Kanter for charts.

Fed communication and FX volatility

Michael Cahill and Lexi Kanter find that global macro conditions remain the biggest driver of Dollar vol, but shifting Fed communication impacts when and how that vol manifests

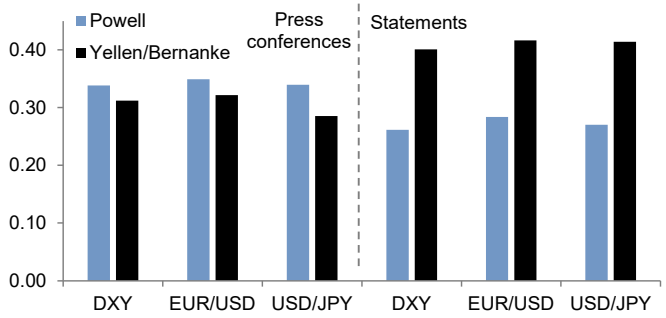
New Fed Chairman Kevin Warsh's push for a less transparent Fed raises the important question of whether this shift would increase Dollar volatility. The answer is more nuanced than just a simple "yes." Macro factors, including the growth and inflation backdrop, relative monetary policy expectations and rate differentials, are key determinants of the overall level of FX volatility. But historical evidence suggests that even small shifts in how—and how much—central banks communicate can change which events and data drive FX volatility and when that volatility is most pronounced.

Looking back: lessons from past Fed communication shifts

History suggests that Fed communication affects FX volatility, but it is more about the timing than the level. We find that intraday FX volatility increased more around the post-FOMC meeting statements under Fed Chairs Bernanke and Yellen, but more around the post-meeting press conference under Chair Powell. This shift is consistent with findings from our rates strategists and the academic literature, which attribute the recent increase in market volatility around press conferences partly to differences between Chair Powell's and his predecessors' communication styles. However, macro conditions and the policy choices they prompted—such as innovations in forward guidance and QE program details released in the FOMC statements—likely also played a role.

Different Fed communication styles have shifted FX volatility

Standard deviation of FX returns around each event, %



Source: USMPD, Goldman Sachs GIR.

Looking up: lessons from Canada

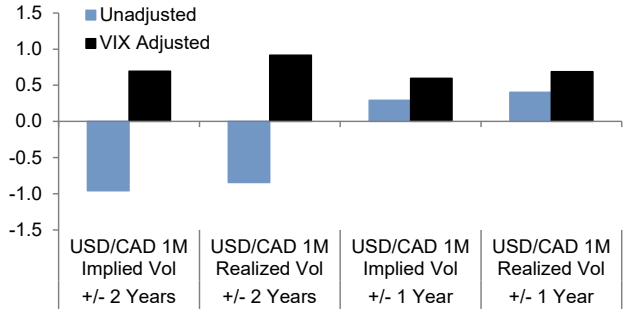
The Bank of Canada's transition from Governor Carney to Governor Poloz in 2013, which marked a significant shift in its communication framework, offers a useful parallel to the current US situation. Poloz deliberately dialed back the detailed, date-based forward guidance pioneered by Carney, arguing that less guidance would encourage the private sector to spend more time assessing Canada's economy and lead markets to better reflect what the Bank *should* do vs. what it *would* do.

We find that USD/CAD volatility was higher during the Poloz era than the global volatility backdrop would have predicted, which we attribute to the joint effect of less forward guidance and a potential shift in the Bank's reaction function. CAD volatility also became more idiosyncratic and less correlated with volatility in other G10 currencies and global factors between the final year of Carney's term and the first year of Poloz's, suggesting that domestic considerations became a more prominent driver of volatility.

The lesson learned is that reduced guidance can make FX volatility more sensitive to domestic developments and less tied to common global drivers. It may not, however, raise the overall level of volatility on its own. The Carney-Poloz transition coincided with a *reduction* in overall volatility, and the relative increase in CAD volatility emerges only after controlling for global factors. Central bank communication style can therefore affect a currency's volatility relative to peers, while macro factors remain far more important in determining the overall volatility regime. This suggests that reduced guidance alone is not necessarily a sign that implied and realized volatility will rise.

A relative rise in CAD vol only after controlling for global factors

Chg in avg USD/CAD implied and realized vol around Carney-Poloz transition



Source: Bloomberg, Goldman Sachs GIR.

Looking ahead: less guidance redistributing USD volatility

The historical Fed and Bank of Canada experiences demonstrate that shifts in central bank communication can have an important effect on FX volatility, often appearing as a redistribution of that volatility—whether from one form of communication to another or to a different set of datapoints—which can be detrimental if markets and policymakers are not in sync (see pgs. 4-5).

The implications are especially important for the Dollar. Given its global role, increased Dollar volatility can have wide-ranging market impacts, which is partly why Fed officials have long sought to suppress policy surprises. When the Fed talks, it also particularly pays to listen given that Fed communication tends to have the largest international spillovers, and, unsurprisingly, FX tends to respond more reliably than other asset classes to policy surprises owing to its relative nature.

While less important than the broader macro backdrop, the evidence shows that even small communication shifts can have a meaningful impact on how FX volatility tends to respond to different events, at different horizons, and even to different data. We expect less transparency from the Fed Chairman to lead to higher front-end volatility, a greater focus on other Fed communication, and potentially more market misfires as investors adapt to a different data framework—all else equal.

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Fed: more transparent than most

The growing recognition that managing market expectations is a key tool in managing the transmission of monetary policy has led to a consistent shift toward more transparent global central banking over the last 25 years. While the Fed was late to adopt many changes that increased transparency, today it is more transparent than most central banks.

Comparing central bank communication strategies

A comparison of global central bank communication strategies reveals that:

- Most major central banks make policy decisions by vote (except for the BoC, Norges Bank, and SNB, which make decisions by consensus) and **provide details of individual voting records** (the outlier is the ECB, which often reveals whether the vote was unanimous in its post-meeting press conference).
- Most DM central banks hold eight meetings per year, with **a press conference after each meeting** (except for the BoE and Norges Bank, which hold press conferences after every other meeting), and all major central banks release **meeting minutes**.
- Most DM central banks publish a quarterly Monetary Policy Report (MPR) with **detailed economic forecasts** that reflect either staff or monetary policy committee consensus (the ECB does not publish an MPR and the Fed shares a semi-annual MPR with Congress), while the Fed and BoJ publish committee members' individual forecasts. Most central banks do not publish policy rate forecasts, and those that do (besides the Fed) usually show a single modal rates path based on their economic projections.
- Nearly all DM central banks have provided some form of **forward guidance** in meetings over the last several years, although most central banks (besides those that publish rates forecasts) have not provided guidance at recent meetings.

Relative to other central banks, the FOMC provides more information about individual forecasts and policy expectations, but is an outlier in not publishing a quarterly monetary policy report and never publishing economic scenarios

	Decision process		Meetings			Forecasts				Forward guidance	
	Decision process	Publishes voting records?	Meetings / year	Press conferences / year	Delay until publication of minutes	Publishes MPR?	Publishes scenarios?	Forecast ownership	Publishes rate forecast?	Forward guidance frequency / year (last 10 years)	Explicit forward guidance today?
Fed	By vote	Attributed	8	8	Three weeks	Semiannually	Never	Individual	Individual	8	Yes
ECB	By vote	No	8	8	Four weeks	No	Sometimes	Staff	No	6	No
BoJ	By vote	Attributed	8	8	One month	Quarterly	Never	Individual	No	7	Yes
BoE	By vote	Attributed	8	4	On meeting date	Quarterly	Sometimes	MPC	No	7	No
BoC	Consensus	N/A	8	8	Two weeks	Quarterly	Sometimes	MPC	No	5	No
RBA	By vote	Unattributed	8	8	Two weeks	Quarterly	Sometimes	Staff	No	5	No
RBNZ	By vote	Attributed	7*	7*	On meeting date	Quarterly	Sometimes	MPC	MPC	7	Yes
Riksbank	By vote	Attributed	8	8	Five days	Quarterly	Sometimes	MPC	MPC	8	Yes
Norges Bank	Consensus	N/A	8	4	On meeting date	Quarterly	Sometimes	MPC	MPC	8	Yes
SNB	Consensus	N/A	4	4	Four weeks	Quarterly	Sometimes	MPC	No	0	No

**8 from 2027 onwards.*
Source: Goldman Sachs GIR.

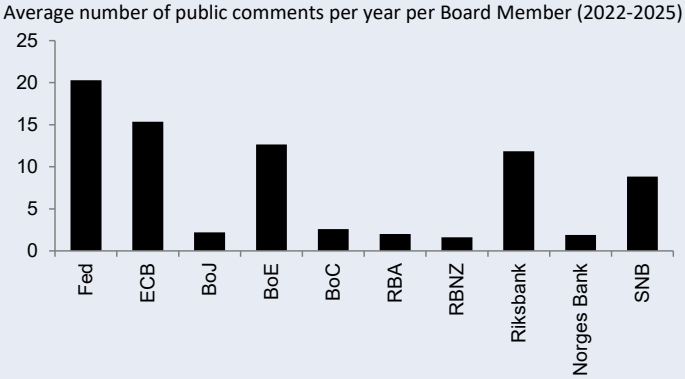
Fed: a more transparent central bank

On net, while the Fed has not always been a leader in providing information—it was relatively late to announce the outcome of policy meetings, adopt press conferences, regularly publish forecasts, and publish meeting statements (see pgs. 12 and 13)—today it is more transparent than most other central banks, mainly because it provides more details on individual FOMC members' economic and policy rate forecasts.

Other patterns support this overarching takeaway, including that the Fed generally provides more information in its meeting minutes and Fed officials tend to provide more public commentary than officials at other central banks. Accordingly, a moderate pullback in communication strategy could reflect a return to cross-country and historical norms, particularly if the changes focus on providing less information around individual forecasts. But that begs the question of whether the Fed's more transparent approach has been an advantage that it would then be giving up (see pgs. 14-15).

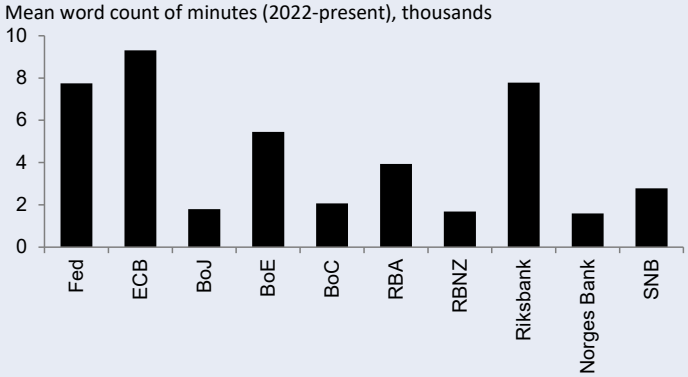
The Fed provides more information than most other DM central banks via more frequent public appearances…

Average number of public comments per year per Board Member (2022-2025)



…and more detailed meeting minutes

Mean word count of minutes (2022-present), thousands



Source: Bloomberg, Federal Reserve, ECB, BoJ, BoE, BoC, RBA, RBNZ, Sveriges Riksbank, Norges Bank, Swiss National Bank.

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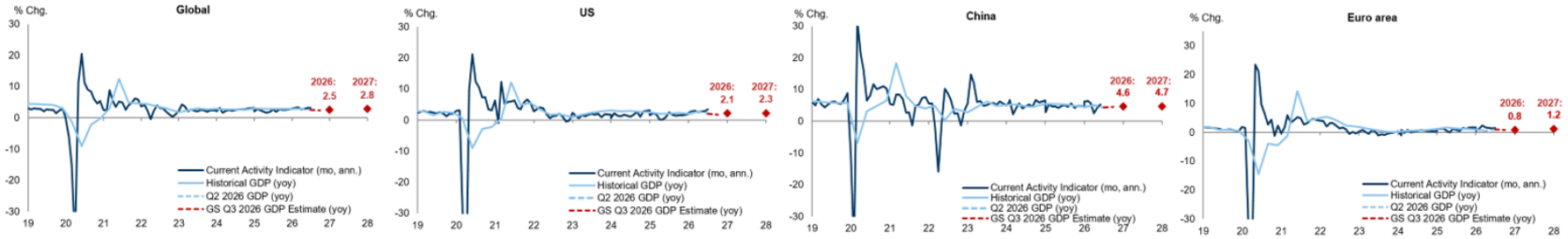
GS GIR: Macro at a glance

Watching

- **Globally,** we expect real GDP growth to slow to 2.5% yoy in 2026 amid headwinds from the rise in energy prices from the conflict in the Middle East. We expect global core inflation to end the year at 2.8%, reflecting a fading tariff boost and further normalization in shelter and wage inflation but a boost from energy price passthrough.
- **In the US,** we expect real GDP growth of 2.1% on a Q4/Q4 basis in 2026, reflecting subdued consumer spending growth but a boost from the AI boom via higher equity wealth as well as strong capex. We expect core PCE inflation of 2.9% in December 2026, boosted by the effects of tariffs, energy price passthrough, and AI demand exaggerated by measurement issues, with it falling closer to 2% in 2027 as those effects subside. We expect the unemployment rate to end 2026 at 4.2%.
- **We expect the Fed** to leave the policy rate unchanged at 3.5-3.75% for the rest of 2026.
- **In the Euro area,** we expect real GDP growth of 0.8% on a Q4/Q4 basis in 2026, reflecting resilient activity data but ongoing headwinds from elevated energy prices as well as subdued consumer confidence. We expect core inflation to rise to a peak of 2.7% yoy in 1Q27, reflecting energy price passthrough, before declining to 2.0% by end-2028.
- **We expect the ECB** to deliver one more 25bp hike in September to a peak policy rate of 2.5%.
- **In China,** we expect real GDP growth of 4.6% yoy in 2026, reflecting strong exports but headwinds from weak domestic demand. We expect CPI/PPI inflation to rise to 1.0%/2.0% yoy this year largely owing to commodity price passthrough.
- **WATCH CONFLICT IN THE MIDDLE EAST.** The situation remains fluid amid swings between de-escalatory and escalatory headlines, and we will be closely watching how oil flows through the region continue to evolve from here.

Goldman Sachs Global Investment Research.

Growth



Source: Haver Analytics, Goldman Sachs Global Investment Research.
Note: GS CAI is a measure of current growth. For more information on the methodology of the CAI please see "Technical Updates to Our Global CAIs," Global Economics Comment, Sep. 01, 2025.

Forecasts

Economics							Markets									Equities								
GDP growth (%)	2026				2027		Interest rates 10Yr (%)	Last	E2026	E2027	FX	Last	3m	12m	S&P 500	E2026		E2027		Returns (%)	12m	YTD	E2026 P/E
	GS (Q4/Q4)	Cons. (Q4/Q4)	GS (CY)	Cons. (CY)	GS (CY)	Cons. (CY)										GS	Cons.	GS	Cons.				
Global	2.4	–	2.5	2.6	2.8	2.6	US	4.71	4.40	4.25	EUR/$	1.16	1.14	1.12	Price	8,000				S&P 500	8	12	22.1x
US	2.1	2.0	2.1	2.2	2.3	2.1	Germany	3.25	3.00	3.00	GBP/$	1.35	1.33	1.28	EPS	$340	$363	$385	$408	MXAPJ	23	22	12.8x
China	4.6	4.6	4.6	4.6	4.7	4.4	Japan	2.94	2.50	2.25	$/JPY	160	162	165	Growth	24%	32%	13%	12%	Topix	9	21	19.9x
Euro area	0.8	0.9	0.8	0.8	1.2	1.2	UK	4.94	4.50	4.35	$/CNY	6.73	6.80	6.50						STOXX 600	7	10	15.6x
Policy rates (%)	2026				2027		Commodities	Last	3Q26	4Q26	Credit (bp)		Last	3Q26	4Q26	Consumer	2026		2027		Wage Tracker 2026 (%)			
	GS	Mkt.			GS	Mkt.											CPI (%, yoy)	Unemp. Rate	CPI (%, yoy)	Unemp. Rate	Q1	Q2	Q3	Q4
US	3.63	3.87			3.13	3.97	Nat Gas, NYMEX ($/mmBtu	2.78	3.50	3.50	USD	IG	81	85	82	US	3.2	4.2	2.3	4.1	3.5	3.6	--	--
Euro area	2.50	2.62			2.00	2.79	Nat Gas, TTF (EUR/MWh)	63.75	60	53		HY	275	305	295	Euro area	2.9	6.3	2.4	6.2	--	--	--	--
China	1.40	1.38			1.30	--	Copper ($/mt)	14,235	13,600	13,700	EUR	IG	90	91	88	China	1.0	--	1.0	--	--	--	--	--
Japan	1.00	1.37			1.50	2.01	Gold ($/troy oz)	4,404	4,600	4,836		HY	255	320	315									

Source: Bloomberg, Goldman Sachs Global Investment Research. For important disclosures, see the Disclosure Appendix or go to www.gs.com/research/hedge.html.

Market pricing as of August 18, 2026

Glossary of GS proprietary indices

Current Activity Indicator (CAI)

GS CAIs measure the growth signal in a broad range of weekly and monthly indicators, offering an alternative to Gross Domestic Product (GDP). GDP is an imperfect guide to current activity: In most countries, it is only available quarterly and is released with a substantial delay, and its initial estimates are often heavily revised. GDP also ignores important measures of real activity, such as employment and the purchasing managers' indexes (PMIs). All of these problems reduce the effectiveness of GDP for investment and policy decisions. Our CAIs aim to address GDP's shortcomings and provide a timelier read on the pace of growth.

For more, see our CAI page and Global Economics Comment: Technical Updates to Our Global CAIs.

Dynamic Equilibrium Exchange Rates (DEER)

The GSDEER framework establishes an equilibrium (or "fair") value of the real exchange rate based on relative productivity and terms-of-trade differentials.

For more, see our GSDEER page, Global Economics Paper No. 227: Finding Fair Value in EM FX, 26 January 2016, and Global Markets Analyst: A Look at Valuation Across G10 FX, 29 June 2017.

Financial Conditions Index (FCI)

GS FCIs gauge the "looseness" or "tightness" of financial conditions across the world's major economies, incorporating variables that directly affect spending on domestically produced goods and services. FCIs can provide valuable information about the economic growth outlook and the direct and indirect effects of monetary policy on real economic activity.

FCIs for the G10 economies are calculated as a weighted average of a policy rate, a long-term risk-free bond yield, a corporate credit spread, an equity price variable, and a trade-weighted exchange rate; the Euro area FCI also includes a sovereign credit spread. The weights mirror the effects of the financial variables on real GDP growth in our models over a one-year horizon. FCIs for emerging markets are calculated as a weighted average of a short-term interest rate, a long-term swap rate, a CDS spread, an equity price variable, a trade-weighted exchange rate, and—in economies with large foreign-currency-denominated debt stocks—a debt-weighted exchange rate index.

For more, see our FCI page, Global Economics Analyst: Our New G10 Financial Conditions Indices, 20 April 2017, and Global Economics Analyst: Tracking EM Financial Conditions – Our New FCIs, 6 October 2017.

Goldman Sachs Analyst Index (GSAI)

The US GSAI is based on a monthly survey of GS equity analysts to obtain their assessments of business conditions in the industries they follow. The results provide timely "bottom-up" information about US economic activity to supplement and cross-check our analysis of "top-down" data. Based on analysts' responses, we create a diffusion index for economic activity comparable to the ISM's indexes for activity in the manufacturing and nonmanufacturing sectors.

Macro-Data Assessment Platform (MAP)

GS MAP scores facilitate rapid interpretation of new data releases for economic indicators worldwide. MAP summarizes the importance of a specific data release (i.e., its historical correlation with GDP) and the degree of surprise relative to the consensus forecast. The sign on the degree of surprise characterizes underperformance with a negative number and outperformance with a positive number. Each of these two components is ranked on a scale from 0 to 5, with the MAP score being the product of the two, i.e., from -25 to +25. For example, a MAP score of +20 (5;+4) would indicate that the data has a very high correlation to GDP (5) and that it came out well above consensus expectations (+4), for a total MAP value of +20.

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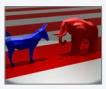
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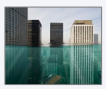
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